

CRF-tutorial

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1 Word Segmentation with Conditional Random Fields (CRF)

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1.2 Date: 25 July 2026

CRFSuiteကိုသုံးပြီး Word Segmentation မော်ဒယ်ဆောက်မယ်။ ပြီးတော့ ဆောက်ထားတဲ့ မော်ဒယ်ကို သုံးပြီး မြန်မာစာ စာလုံးတွေကို ဖြတ်ကြည့်ပြီး ရလဒ် ဘယ်လောက် ကောင်းသလဲ ဆိုတာကို လက်တွေ့ လုပ်ကြည့်ကြရအောင်။

1.3 CRFSuite Installation

Link: <https://github.com/chokkan/crfsuite>

Library link: <https://github.com/chokkan/liblbfgs>

CRFSuite Tutorial link: <https://www.chokkan.org/software/crfsuite/tutorial.html>

```
[1]: %pwd
```

```
[1]: '/home/ye/aif2/word-seg'
```

```
[2]: !mkdir tool
```

```
[3]: %cd tool
```

```
/home/ye/aif2/word-seg/tool
```

```
[4]: !git clone https://github.com/chokkan/crfsuite
```

```
Cloning into 'crfsuite'...
```

```
remote: Enumerating objects: 3314, done.
```

```
remote: Counting objects: 100% (7/7), done.
```

```
remote: Compressing objects: 100% (7/7), done.
```

```
remote: Total 3314 (delta 0), reused 5 (delta 0), pack-reused 3307 (from 1)
```

```
Receiving objects: 100% (3314/3314), 1.04 MiB | 4.05 MiB/s, done.
```

```
Resolving deltas: 100% (2274/2274), done.
```

```
[5]: %cd crfsuite/
```

```
/home/ye/aif2/word-seg/tool/crfsuite
```

```
[6]: !ls
```

AUTHORS	ChangeLog	crfsuite.sln	frontend	INSTALL	README
autogen.sh	configure.in	doc	genbinary.sh.in	lib	swig
bench	COPYING	example	include	Makefile.am	win32

```
[7]: !./autogen.sh
```

```
libtoolize: putting auxiliary files in '.'.
libtoolize: copying file './ltmain.sh'
libtoolize: putting macros in AC_CONFIG_MACRO_DIRS, 'm4'.
libtoolize: copying file 'm4/libtool.m4'
libtoolize: copying file 'm4/ltoptions.m4'
libtoolize: copying file 'm4/ltsugar.m4'
libtoolize: copying file 'm4/ltversion.m4'
libtoolize: copying file 'm4/lt~obsolete.m4'
aclocal: warning: autoconf input should be named 'configure.ac', not
'configure.in'
autoheader: warning: autoconf input should be named 'configure.ac', not
'configure.in'
automake: warning: autoconf input should be named 'configure.ac', not
'configure.in'
configure.in:22: installing './compile'
configure.in:21: installing './config.guess'
configure.in:21: installing './config.sub'
configure.in:30: installing './install-sh'
configure.in:30: installing './missing'
configure.in:140: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or
'*_CPPFLAGS')
configure.in:140: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or
'*_CPPFLAGS')
frontend/Makefile.am: installing './depcomp'
automake: warning: autoconf input should be named 'configure.ac', not
'configure.in'
configure.in:140: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or
'*_CPPFLAGS')
configure.in:140: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or
'*_CPPFLAGS')
lib/cqdb/Makefile.am:9: warning: source file 'src/lookup3.c' is in a
subdirectory,
lib/cqdb/Makefile.am:9: but option 'subdir-objects' is disabled
automake: warning: possible forward-incompatibility.
automake: At least one source file is in a subdirectory, but the 'subdir-
objects'
automake: automake option hasn't been enabled. For now, the corresponding
output
automake: object file(s) will be placed in the top-level directory. However,
this
automake: behavior may change in a future Automake major version, with object
automake: files being placed in the same subdirectory as the corresponding
```

sources.

automake: You are advised to start using 'subdir-objects' option throughout your automake: project, to avoid future incompatibilities.

lib/cqdb/Makefile.am:9: warning: source file 'src/cqdb.c' is in a subdirectory,
lib/cqdb/Makefile.am:9: but option 'subdir-objects' is disabled

configure.in:140: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or '*_CPPFLAGS')

lib/crf/Makefile.am:8: warning: source file 'src/dictionary.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/logging.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/params.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/quark.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/rumavl.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/dataset.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/holdout.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/train_arow.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/train_averaged_perceptron.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/train_l2sgd.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/train_lbfgs.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/train_passive_aggressive.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/crf1d_context.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/crf1d_model.c' is in a subdirectory,

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled

lib/crf/Makefile.am:8: warning: source file 'src/crf1d_feature.c' is in a subdirectory,

```

lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled
lib/crf/Makefile.am:8: warning: source file 'src/crfd_encode.c' is in a
subdirectory,
lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled
lib/crf/Makefile.am:8: warning: source file 'src/crfd_tag.c' is in a
subdirectory,
lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled
lib/crf/Makefile.am:8: warning: source file 'src/crfsuite_train.c' is in a
subdirectory,
lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled
lib/crf/Makefile.am:8: warning: source file 'src/crfsuite.c' is in a
subdirectory,
lib/crf/Makefile.am:8: but option 'subdir-objects' is disabled
configure.in:140: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or
'*_CPPFLAGS')
configure.in:22: warning: The macro `AC_AIX' is obsolete.
configure.in:22: You should run autoupdate.
./lib/autoconf/specific.m4:540: AC_AIX is expanded from...
configure.in:22: the top level
configure.in:23: warning: The macro `AC_MINIX' is obsolete.
configure.in:23: You should run autoupdate.
./lib/autoconf/specific.m4:545: AC_MINIX is expanded from...
configure.in:23: the top level
configure.in:24: warning: The macro `AC_ISC_POSIX' is obsolete.
configure.in:24: You should run autoupdate.
./lib/autoconf/specific.m4:550: AC_ISC_POSIX is expanded from...
configure.in:24: the top level
configure.in:39: warning: The macro `AC_PROG_LIBTOOL' is obsolete.
configure.in:39: You should run autoupdate.
m4/libtool.m4:100: AC_PROG_LIBTOOL is expanded from...
configure.in:39: the top level
configure.in:55: warning: The macro `AC_HEADER_STDC' is obsolete.
configure.in:55: You should run autoupdate.
./lib/autoconf/headers.m4:704: AC_HEADER_STDC is expanded from...
configure.in:55: the top level

```

[8]: `!./configure`

```

checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes

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checking for gcc option to enable C11 features... none needed
checking whether gcc understands -c and -o together... yes
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for inttypes.h... yes
checking for stdint.h... yes
checking for strings.h... yes
checking for sys/stat.h... yes
checking for sys/types.h... yes
checking for unistd.h... yes
checking for wchar.h... yes
checking for minix/config.h... no
checking whether it is safe to define __EXTENSIONS__... yes
checking whether _XOPEN_SOURCE should be defined... no
checking for library containing strerror... none required
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a race-free mkdir -p... /usr/bin/mkdir -p
checking for gawk... gawk
checking whether make sets $(MAKE)... yes
checking whether make supports the include directive... yes (GNU style)
checking whether make supports nested variables... yes
checking dependency style of gcc... gcc3
checking whether to enable maintainer-specific portions of Makefiles... no
checking how to print strings... printf
checking for a sed that does not truncate output... /usr/bin/sed
checking for grep that handles long lines and -e... /usr/bin/grep
checking for egrep... /usr/bin/grep -E
checking for fgrep... /usr/bin/grep -F
checking for ld used by gcc... /usr/bin/ld
checking if the linker (/usr/bin/ld) is GNU ld... yes
checking for BSD- or MS-compatible name lister (nm)... /usr/bin/nm -B
checking the name lister (/usr/bin/nm -B) interface... BSD nm
checking whether ln -s works... yes
checking the maximum length of command line arguments... 1572864
checking how to convert x86_64-pc-linux-gnu file names to x86_64-pc-linux-gnu
format... func_convert_file_noop
checking how to convert x86_64-pc-linux-gnu file names to toolchain format...
func_convert_file_noop
checking for /usr/bin/ld option to reload object files... -r
checking for file... file
checking for objdump... objdump
checking how to recognize dependent libraries... pass_all
checking for dlltool... no
checking how to associate runtime and link libraries... printf %s\n
checking for ar... ar
checking for archiver @FILE support... @

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checking for strip... strip
checking for ranlib... ranlib
checking command to parse /usr/bin/nm -B output from gcc object... ok
checking for sysroot... no
checking for a working dd... /usr/bin/dd
checking how to truncate binary pipes... /usr/bin/dd bs=4096 count=1
checking for mt... mt
checking if mt is a manifest tool... no
checking for dlfcn.h... yes
checking for objdir... .libs
checking if gcc supports -fno-rtti -fno-exceptions... no
checking for gcc option to produce PIC... -fPIC -DPIC
checking if gcc PIC flag -fPIC -DPIC works... yes
checking if gcc static flag -static works... yes
checking if gcc supports -c -o file.o... yes
checking if gcc supports -c -o file.o... (cached) yes
checking whether the gcc linker (/usr/bin/ld -m elf_x86_64) supports shared
libraries... yes
checking whether -lc should be explicitly linked in... no
checking dynamic linker characteristics... GNU/Linux ld.so
checking how to hardcode library paths into programs... immediate
checking whether stripping libraries is possible... yes
checking if libtool supports shared libraries... yes
checking whether to build shared libraries... yes
checking whether to build static libraries... yes
checking whether ln -s works... yes
checking for egrep... (cached) /usr/bin/grep -E
checking for fcntl.h... yes
checking for limits.h... yes
checking for malloc.h... yes
checking for strings.h... (cached) yes
checking for unistd.h... (cached) yes
checking for stdint.h... (cached) yes
checking for an ANSI C-conforming const... yes
checking size of ... 0
checking for size_t... yes
checking whether struct tm is in sys/time.h or time.h... time.h
checking size of short... 2
checking size of unsigned short... 2
checking size of int... 4
checking size of unsigned int... 4
checking size of long... 8
checking size of unsigned long... 8
checking for uint8_t... yes
checking for uint16_t... yes
checking for uint32_t... yes
checking for working alloca.h... yes
checking for alloca... yes

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checking for working memcmp... yes
checking for vprintf... yes
checking for strdup... yes
checking for strerror... yes
checking for strtol... yes
checking for strtoul... yes
checking for rand in -lm... yes
checking for lbfgs in -llbfgs... no
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating Makefile
config.status: creating genbinary.sh
config.status: creating include/Makefile
config.status: creating lib/cqdb/Makefile
config.status: creating lib/crf/Makefile
config.status: creating frontend/Makefile
config.status: creating swig/Makefile
config.status: creating swig/python/setup.py
config.status: creating swig/perl/Makefile.PL
config.status: creating config.h
config.status: executing depfiles commands
config.status: executing libtool commands

```

[9]: `!make`

```

make all-recursive
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite'
Making all in include
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
make[2]: Nothing to be done for 'all'.
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
Making all in lib/cqdb
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../..
-mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline
-std=c99 -MT libcqdb_la-lookup3.lo -MD -MP -MF .deps/libcqdb_la-lookup3.Tpo -c
-o libcqdb_la-lookup3.lo `test -f 'src/lookup3.c' || echo './'`src/lookup3.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../.. -I../..
-I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../..
-mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline
-std=c99 -MT libcqdb_la-lookup3.lo -MD -MP -MF .deps/libcqdb_la-lookup3.Tpo -c
src/lookup3.c -fPIC -DPIC -o .libs/libcqdb_la-lookup3.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../.. -I../..
-I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../.. -I../..
-mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline
-std=c99 -MT libcqdb_la-lookup3.lo -MD -MP -MF .deps/libcqdb_la-lookup3.Tpo -c
src/lookup3.c -o libcqdb_la-lookup3.o
>/dev/null 2>&1

```

```

mv -f .deps/libcqdb_la-lookup3.Tpo .deps/libcqdb_la-lookup3.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I. -I./include
-mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline
-std=c99 -MT libcqdb_la-cqdb.lo -MD -MP -MF .deps/libcqdb_la-cqdb.Tpo -c -o
libcqdb_la-cqdb.lo `test -f 'src/cqdb.c' || echo './`src/cqdb.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I./include -mfpmath=sse -msse2 -DUSE_SSE -O3
-fomit-frame-pointer -ffast-math -Winline -std=c99 -MT libcqdb_la-cqdb.lo -MD
-MP -MF .deps/libcqdb_la-cqdb.Tpo -c src/cqdb.c -fPIC -DPIC -o
.libs/libcqdb_la-cqdb.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I./include -mfpmath=sse -msse2 -DUSE_SSE -O3
-fomit-frame-pointer -ffast-math -Winline -std=c99 -MT libcqdb_la-cqdb.lo -MD
-MP -MF .deps/libcqdb_la-cqdb.Tpo -c src/cqdb.c -o libcqdb_la-cqdb.o >/dev/null
2>&1
mv -f .deps/libcqdb_la-cqdb.Tpo .deps/libcqdb_la-cqdb.Plo
/bin/bash ../../libtool --tag=CC --mode=link gcc -I./include -mfpmath=sse
-msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -no-
undefined -release 0.12 -o libcqdb.la -rpath /usr/local/lib libcqdb_la-
lookup3.lo libcqdb_la-cqdb.lo -lm
libtool: link: gcc -shared -fPIC -DPIC .libs/libcqdb_la-lookup3.o
.libs/libcqdb_la-cqdb.o -lm -mfpmath=sse -msse2 -O3 -Wl,-soname
-Wl,libcqdb-0.12.so -o .libs/libcqdb-0.12.so
libtool: link: (cd ".libs" && rm -f "libcqdb.so" && ln -s "libcqdb-0.12.so"
"libcqdb.so")
libtool: link: ar cr .libs/libcqdb.a libcqdb_la-lookup3.o libcqdb_la-cqdb.o
libtool: link: ranlib .libs/libcqdb.a
libtool: link: ( cd ".libs" && rm -f "libcqdb.la" && ln -s "../libcqdb.la"
"libcqdb.la" )
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
Making all in lib/crf
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-dictionary.lo -MD -MP -MF
.deps/libcrfsuite_la-dictionary.Tpo -c -o libcrfsuite_la-dictionary.lo `test -f
'src/dictionary.c' || echo './`src/dictionary.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-dictionary.lo -MD -MP -MF .deps/libcrfsuite_la-dictionary.Tpo -c
src/dictionary.c -fPIC -DPIC -o .libs/libcrfsuite_la-dictionary.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-dictionary.lo -MD -MP -MF .deps/libcrfsuite_la-dictionary.Tpo -c

```



```

src/dictionary.c -o libcrfsuite_la-dictionary.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-dictionary.Tpo .deps/libcrfsuite_la-dictionary.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-logging.lo -MD -MP -MF
.deps/libcrfsuite_la-logging.Tpo -c -o libcrfsuite_la-logging.lo `test -f
'src/logging.c' || echo './`src/logging.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-logging.lo -MD -MP -MF .deps/libcrfsuite_la-logging.Tpo -c
src/logging.c -fPIC -DPIC -o .libs/libcrfsuite_la-logging.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-logging.lo -MD -MP -MF .deps/libcrfsuite_la-logging.Tpo -c
src/logging.c -o libcrfsuite_la-logging.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-logging.Tpo .deps/libcrfsuite_la-logging.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-params.lo -MD -MP -MF
.deps/libcrfsuite_la-params.Tpo -c -o libcrfsuite_la-params.lo `test -f
'src/params.c' || echo './`src/params.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-params.lo -MD -MP -MF .deps/libcrfsuite_la-params.Tpo -c
src/params.c -fPIC -DPIC -o .libs/libcrfsuite_la-params.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-params.lo -MD -MP -MF .deps/libcrfsuite_la-params.Tpo -c
src/params.c -o libcrfsuite_la-params.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-params.Tpo .deps/libcrfsuite_la-params.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-quark.lo -MD -MP -MF
.deps/libcrfsuite_la-quark.Tpo -c -o libcrfsuite_la-quark.lo `test -f
'src/quark.c' || echo './`src/quark.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-quark.lo -MD -MP -MF .deps/libcrfsuite_la-quark.Tpo -c
src/quark.c -fPIC -DPIC -o .libs/libcrfsuite_la-quark.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.

```

```

-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-quark.lo -MD -MP -MF .deps/libcrfsuite_la-quark.Tpo -c
src/quark.c -o libcrfsuite_la-quark.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-quark.Tpo .deps/libcrfsuite_la-quark.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-rumavl.lo -MD -MP -MF
.deps/libcrfsuite_la-rumavl.Tpo -c -o libcrfsuite_la-rumavl.lo `test -f
'src/rumavl.c' || echo './`src/rumavl.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-rumavl.lo -MD -MP -MF .deps/libcrfsuite_la-rumavl.Tpo -c
src/rumavl.c -fPIC -DPIC -o .libs/libcrfsuite_la-rumavl.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-rumavl.lo -MD -MP -MF .deps/libcrfsuite_la-rumavl.Tpo -c
src/rumavl.c -o libcrfsuite_la-rumavl.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-rumavl.Tpo .deps/libcrfsuite_la-rumavl.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-dataset.lo -MD -MP -MF
.deps/libcrfsuite_la-dataset.Tpo -c -o libcrfsuite_la-dataset.lo `test -f
'src/dataset.c' || echo './`src/dataset.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-dataset.lo -MD -MP -MF .deps/libcrfsuite_la-dataset.Tpo -c
src/dataset.c -fPIC -DPIC -o .libs/libcrfsuite_la-dataset.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-dataset.lo -MD -MP -MF .deps/libcrfsuite_la-dataset.Tpo -c
src/dataset.c -o libcrfsuite_la-dataset.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-dataset.Tpo .deps/libcrfsuite_la-dataset.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-holdout.lo -MD -MP -MF
.deps/libcrfsuite_la-holdout.Tpo -c -o libcrfsuite_la-holdout.lo `test -f
'src/holdout.c' || echo './`src/holdout.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT

```

```

libcrfsuite_la-holdout.lo -MD -MP -MF .deps/libcrfsuite_la-holdout.Tpo -c
src/holdout.c -fPIC -DPIC -o .libs/libcrfsuite_la-holdout.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-holdout.lo -MD -MP -MF .deps/libcrfsuite_la-holdout.Tpo -c
src/holdout.c -o libcrfsuite_la-holdout.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-holdout.Tpo .deps/libcrfsuite_la-holdout.Plo
/bin/bash ../..../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-train_arow.lo -MD -MP -MF
.deps/libcrfsuite_la-train_arow.Tpo -c -o libcrfsuite_la-train_arow.lo `test -f
'src/train_arow.c' || echo './`src/train_arow.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-train_arow.lo -MD -MP -MF .deps/libcrfsuite_la-train_arow.Tpo -c
src/train_arow.c -fPIC -DPIC -o .libs/libcrfsuite_la-train_arow.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-train_arow.lo -MD -MP -MF .deps/libcrfsuite_la-train_arow.Tpo -c
src/train_arow.c -o libcrfsuite_la-train_arow.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-train_arow.Tpo .deps/libcrfsuite_la-train_arow.Plo
/bin/bash ../..../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-train_averaged_perceptron.lo
-MD -MP -MF .deps/libcrfsuite_la-train_averaged_perceptron.Tpo -c -o
libcrfsuite_la-train_averaged_perceptron.lo `test -f
'src/train_averaged_perceptron.c' || echo './`src/train_averaged_perceptron.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-train_averaged_perceptron.lo -MD -MP -MF .deps/libcrfsuite_la-
train_averaged_perceptron.Tpo -c src/train_averaged_perceptron.c -fPIC -DPIC -o
.libs/libcrfsuite_la-train_averaged_perceptron.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-train_averaged_perceptron.lo -MD -MP -MF .deps/libcrfsuite_la-
train_averaged_perceptron.Tpo -c src/train_averaged_perceptron.c -o
libcrfsuite_la-train_averaged_perceptron.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-train_averaged_perceptron.Tpo .deps/libcrfsuite_la-
train_averaged_perceptron.Plo
/bin/bash ../..../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.

```

```

-I../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-train_l2sgd.lo -MD -MP -MF
.deps/libcrfsuite_la-train_l2sgd.Tpo -c -o libcrfsuite_la-train_l2sgd.lo `test
-f 'src/train_l2sgd.c' || echo './`src/train_l2sgd.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../lib/cqdb/include -I.
-I../.. -I../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-train_l2sgd.lo -MD -MP -MF .deps/libcrfsuite_la-train_l2sgd.Tpo
-c src/train_l2sgd.c -fPIC -DPIC -o .libs/libcrfsuite_la-train_l2sgd.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../lib/cqdb/include -I.
-I../.. -I../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-train_l2sgd.lo -MD -MP -MF .deps/libcrfsuite_la-train_l2sgd.Tpo
-c src/train_l2sgd.c -o libcrfsuite_la-train_l2sgd.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-train_l2sgd.Tpo .deps/libcrfsuite_la-train_l2sgd.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../lib/cqdb/include -I. -I../.. -I../lib/cqdb/include -I.
-I../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -MT libcrfsuite_la-train_lbfgs.lo -MD -MP -MF
.deps/libcrfsuite_la-train_lbfgs.Tpo -c -o libcrfsuite_la-train_lbfgs.lo `test
-f 'src/train_lbfgs.c' || echo './`src/train_lbfgs.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../lib/cqdb/include -I.
-I../.. -I../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99 -MT
libcrfsuite_la-train_lbfgs.lo -MD -MP -MF .deps/libcrfsuite_la-train_lbfgs.Tpo
-c src/train_lbfgs.c -fPIC -DPIC -o .libs/libcrfsuite_la-train_lbfgs.o
src/train_lbfgs.c:51:10: fatal error: lbfgs.h: No
such file or directory
   51 | #include <lbfgs.h>
      |         ~~~~~~
compilation terminated.
make[2]: *** [Makefile:596: libcrfsuite_la-train_lbfgs.lo] Error 1
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
make[1]: *** [Makefile:475: all-recursive] Error 1
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite'
make: *** [Makefile:380: all] Error 2

```

1.4 liblbfgs Library Installation

```
[11]: %cd /tmp
```

```
/tmp
```

```
[13]: !git clone https://github.com/chokkan/liblbfgs
```

```

Cloning into 'liblbfgs'...
remote: Enumerating objects: 528, done.
remote: Counting objects: 100% (41/41), done.

```

```
remote: Compressing objects: 100% (18/18), done.
remote: Total 528 (delta 23), reused 31 (delta 21), pack-reused 487 (from 1)
Receiving objects: 100% (528/528), 163.52 KiB | 1.84 MiB/s, done.
Resolving deltas: 100% (326/326), done.
```

```
[15]: %cd liblbfgs/
```

```
/tmp/liblbfgs
```

```
[16]: !./autogen.sh
```

```
libtoolize: putting auxiliary files in '.'.
libtoolize: copying file './ltmain.sh'
libtoolize: putting macros in AC_CONFIG_MACRO_DIRS, 'm4'.
libtoolize: copying file 'm4/libtool.m4'
libtoolize: copying file 'm4/ltoptions.m4'
libtoolize: copying file 'm4/ltsugar.m4'
libtoolize: copying file 'm4/ltversion.m4'
libtoolize: copying file 'm4/lt-obsolete.m4'
configure.ac:30: installing './compile'
configure.ac:30: installing './config.guess'
configure.ac:30: installing './config.sub'
configure.ac:22: installing './install-sh'
configure.ac:22: installing './missing'
configure.ac:102: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or '*_CPPFLAGS')
Makefile.am: installing './INSTALL'
lib/Makefile.am:24: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or '*_CPPFLAGS')
lib/Makefile.am: installing './depcomp'
sample/Makefile.am:15: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or '*_CPPFLAGS')
configure.ac:48: warning: The macro `AC_HEADER_STDC' is obsolete.
configure.ac:48: You should run autoupdate.
./lib/autoconf/headers.m4:704: AC_HEADER_STDC is expanded from...
configure.ac:48: the top level
```

```
[17]: !./configure
```

```
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a race-free mkdir -p... /usr/bin/mkdir -p
checking for gawk... gawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking whether to enable maintainer-specific portions of Makefiles... no
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking how to print strings... printf
```

```

checking whether make supports the include directive... yes (GNU style)
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes
checking for gcc option to enable C11 features... none needed
checking whether gcc understands -c and -o together... yes
checking dependency style of gcc... gcc3
checking for a sed that does not truncate output... /usr/bin/sed
checking for grep that handles long lines and -e... /usr/bin/grep
checking for egrep... /usr/bin/grep -E
checking for fgrep... /usr/bin/grep -F
checking for ld used by gcc... /usr/bin/ld
checking if the linker (/usr/bin/ld) is GNU ld... yes
checking for BSD- or MS-compatible name lister (nm)... /usr/bin/nm -B
checking the name lister (/usr/bin/nm -B) interface... BSD nm
checking whether ln -s works... yes
checking the maximum length of command line arguments... 1572864
checking how to convert x86_64-pc-linux-gnu file names to x86_64-pc-linux-gnu
format... func_convert_file_noop
checking how to convert x86_64-pc-linux-gnu file names to toolchain format...
func_convert_file_noop
checking for /usr/bin/ld option to reload object files... -r
checking for file... file
checking for objdump... objdump
checking how to recognize dependent libraries... pass_all
checking for dlltool... no
checking how to associate runtime and link libraries... printf %s\n
checking for ar... ar
checking for archiver @FILE support... @
checking for strip... strip
checking for ranlib... ranlib
checking command to parse /usr/bin/nm -B output from gcc object... ok
checking for sysroot... no
checking for a working dd... /usr/bin/dd
checking how to truncate binary pipes... /usr/bin/dd bs=4096 count=1
checking for mt... mt
checking if mt is a manifest tool... no
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for inttypes.h... yes
checking for stdint.h... yes
checking for strings.h... yes

```

```

checking for sys/stat.h... yes
checking for sys/types.h... yes
checking for unistd.h... yes
checking for dlfcn.h... yes
checking for objdir... .libs
checking if gcc supports -fno-rtti -fno-exceptions... no
checking for gcc option to produce PIC... -fPIC -DPIC
checking if gcc PIC flag -fPIC -DPIC works... yes
checking if gcc static flag -static works... yes
checking if gcc supports -c -o file.o... yes
checking if gcc supports -c -o file.o... (cached) yes
checking whether the gcc linker (/usr/bin/ld -m elf_x86_64) supports shared
libraries... yes
checking whether -lc should be explicitly linked in... no
checking dynamic linker characteristics... GNU/Linux ld.so
checking how to hardcode library paths into programs... immediate
checking whether stripping libraries is possible... yes
checking if libtool supports shared libraries... yes
checking whether to build shared libraries... yes
checking whether to build static libraries... yes
checking whether ln -s works... yes
checking whether make sets $(MAKE)... (cached) yes
checking for egrep... (cached) /usr/bin/grep -E
checking for xmmintrin.h... yes
checking for emmintrin.h... yes
checking for fabs in -lm... yes
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating Makefile
config.status: creating lib/Makefile
config.status: creating sample/Makefile
config.status: creating config.h
config.status: executing depfiles commands
config.status: executing libtool commands

```

[18]: !make

```

make all-recursive
make[1]: Entering directory '/tmp/liblbfgs'
Making all in lib
make[2]: Entering directory '/tmp/liblbfgs/lib'
/bin/bash ../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I. -I..
-I.. -I../include -O3 -ffast-math -Wall -O3 -ffast-math -Wall -MT lbfgs.lo
-MD -MP -MF .deps/lbfgs.Tpo -c -o lbfgs.lo lbfgs.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -O3 -ffast-
math -Wall -O3 -ffast-math -Wall -MT lbfgs.lo -MD -MP -MF .deps/lbfgs.Tpo -c
lbfgs.c -fPIC -DPIC -o .libs/lbfgs.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -O3 -ffast-

```

```

math -Wall -O3 -ffast-math -Wall -MT lbfgs.lo -MD -MP -MF .deps/lbfgs.Tpo -c
lbfgs.c -o lbfgs.o >/dev/null 2>&1
mv -f .deps/lbfgs.Tpo .deps/lbfgs.Plo
/bin/bash ../libtool --tag=CC --mode=link gcc -O3 -ffast-math -Wall -O3
-ffast-math -Wall -no-undefined -release 1.10 -o liblbfgs.la -rpath
/usr/local/lib lbfgs.lo -lm
libtool: link: gcc -shared -fPIC -DPIC .libs/lbfgs.o -lm -O3 -O3
-Wl,-soname -Wl,liblbfgs-1.10.so -o .libs/liblbfgs-1.10.so
libtool: link: (cd ".libs" && rm -f "liblbfgs.so" && ln -s "liblbfgs-1.10.so"
"liblbfgs.so")
libtool: link: ar cr .libs/liblbfgs.a lbfgs.o
libtool: link: ranlib .libs/liblbfgs.a
libtool: link: ( cd ".libs" && rm -f "liblbfgs.la" && ln -s "../liblbfgs.la"
"liblbfgs.la" )
make[2]: Leaving directory '/tmp/liblbfgs/lib'
Making all in sample
make[2]: Entering directory '/tmp/liblbfgs/sample'
gcc -DHAVE_CONFIG_H -I. -I.. -I../include -O3 -ffast-math -Wall -O3
-ffast-math -Wall -MT sample.o -MD -MP -MF .deps/sample.Tpo -c -o sample.o
sample.c
mv -f .deps/sample.Tpo .deps/sample.Po
/bin/bash ../libtool --tag=CC --mode=link gcc -O3 -ffast-math -Wall -O3
-ffast-math -Wall -o sample sample.o ../lib/liblbfgs.la -lm
libtool: link: gcc -O3 -ffast-math -Wall -O3 -ffast-math -Wall -o .libs/sample
sample.o ../lib/.libs/liblbfgs.so -lm
make[2]: Leaving directory '/tmp/liblbfgs/sample'
make[2]: Entering directory '/tmp/liblbfgs'
make[2]: Leaving directory '/tmp/liblbfgs'
make[1]: Leaving directory '/tmp/liblbfgs'

ye@lst-hpc3090:/tmp/liblbfgs$ sudo make install
[sudo] password for ye:
Making install in lib
make[1]: Entering directory '/tmp/liblbfgs/lib'
make[2]: Entering directory '/tmp/liblbfgs/lib'
  /usr/bin/mkdir -p '/usr/local/lib'
  /bin/bash ../libtool --mode=install /usr/bin/install -c liblbfgs.la '/usr/local/lib'
libtool: install: /usr/bin/install -c .libs/liblbfgs-1.10.so /usr/local/lib/liblbfgs-1.10.so
libtool: install: (cd /usr/local/lib && { ln -s -f liblbfgs-1.10.so liblbfgs.so || { rm -f lib
libtool: install: /usr/bin/install -c .libs/liblbfgs.lai /usr/local/lib/liblbfgs.la
libtool: install: /usr/bin/install -c .libs/liblbfgs.a /usr/local/lib/liblbfgs.a
libtool: install: chmod 644 /usr/local/lib/liblbfgs.a
libtool: install: ranlib /usr/local/lib/liblbfgs.a
libtool: finish: PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin:"
-----
Libraries have been installed in:
  /usr/local/lib

```


If you ever happen to want to link against installed libraries in a given directory, LIBDIR, you must either use libtool, and specify the full pathname of the library, or use the '-LLIBDIR' flag during linking and do at least one of the following:

- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for more information, such as the ld(1) and ld.so(8) manual pages.

```
-----
/usr/bin/mkdir -p '/usr/local/include'
/usr/bin/install -c -m 644 ../include/lbfgs.h '/usr/local/include'
make[2]: Leaving directory '/tmp/liblbfgs/lib'
make[1]: Leaving directory '/tmp/liblbfgs/lib'
Making install in sample
make[1]: Entering directory '/tmp/liblbfgs/sample'
make[2]: Entering directory '/tmp/liblbfgs/sample'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/tmp/liblbfgs/sample'
make[1]: Leaving directory '/tmp/liblbfgs/sample'
make[1]: Entering directory '/tmp/liblbfgs'
make[2]: Entering directory '/tmp/liblbfgs'
make[2]: Nothing to be done for 'install-exec-am'.
/usr/bin/mkdir -p '/usr/local/share/doc/liblbfgs'
/usr/bin/install -c -m 644 README INSTALL COPYING AUTHORS ChangeLog NEWS '/usr/local/share/doc/liblbfgs'
make[2]: Leaving directory '/tmp/liblbfgs'
make[1]: Leaving directory '/tmp/liblbfgs'
ye@lst-hpc3090:/tmp/liblbfgs$
```

1.5 Let's Do CRFSuite Installation Again

```
[24]: %pwd
```

```
[24]: '/home/ye/aif2/word-seg/tool'
```

```
[25]: %cd crfsuite/
```

```
/home/ye/aif2/word-seg/tool/crfsuite
```

```
[27]: !make clean
```

```
Making clean in include
```

```
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
rm -rf .libs _libs
```

```

rm -f *.lo
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
Making clean in lib/cqdb
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
test -z "libcqdb.la" || rm -f libcqdb.la
rm -f ./so_locations
rm -rf .libs _libs
rm -f *.o
rm -f *.lo
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
Making clean in lib/crf
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
test -z "libcrfsuite.la" || rm -f libcrfsuite.la
rm -f ./so_locations
rm -rf .libs _libs
rm -f *.o
rm -f *.lo
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
Making clean in frontend
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/frontend'
rm -f crfsuite
rm -rf .libs _libs
rm -f *.o
rm -f *.lo
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/frontend'
Making clean in swig
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/swig'
rm -rf .libs _libs
rm -f *.lo
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/swig'
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite'
rm -rf .libs _libs
rm -f *.lo
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite'

```

```

[28]: !./configure CFLAGS="-I/usr/local/include -O3" LDFLAGS="-L/usr/local/lib"
↳LIBS="-llbfgs"

```

```

checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes

```

```

checking for gcc option to enable C11 features... none needed
checking whether gcc understands -c and -o together... yes
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for inttypes.h... yes
checking for stdint.h... yes
checking for strings.h... yes
checking for sys/stat.h... yes
checking for sys/types.h... yes
checking for unistd.h... yes
checking for wchar.h... yes
checking for minix/config.h... no
checking whether it is safe to define __EXTENSIONS__... yes
checking whether _XOPEN_SOURCE should be defined... no
checking for library containing strerror... none required
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a race-free mkdir -p... /usr/bin/mkdir -p
checking for gawk... gawk
checking whether make sets $(MAKE)... yes
checking whether make supports the include directive... yes (GNU style)
checking whether make supports nested variables... yes
checking dependency style of gcc... gcc3
checking whether to enable maintainer-specific portions of Makefiles... no
checking how to print strings... printf
checking for a sed that does not truncate output... /usr/bin/sed
checking for grep that handles long lines and -e... /usr/bin/grep
checking for egrep... /usr/bin/grep -E
checking for fgrep... /usr/bin/grep -F
checking for ld used by gcc... /usr/bin/ld
checking if the linker (/usr/bin/ld) is GNU ld... yes
checking for BSD- or MS-compatible name lister (nm)... /usr/bin/nm -B
checking the name lister (/usr/bin/nm -B) interface... BSD nm
checking whether ln -s works... yes
checking the maximum length of command line arguments... 1572864
checking how to convert x86_64-pc-linux-gnu file names to x86_64-pc-linux-gnu
format... func_convert_file_noop
checking how to convert x86_64-pc-linux-gnu file names to toolchain format...
func_convert_file_noop
checking for /usr/bin/ld option to reload object files... -r
checking for file... file
checking for objdump... objdump
checking how to recognize dependent libraries... pass_all
checking for dlltool... no
checking how to associate runtime and link libraries... printf %s\n
checking for ar... ar
checking for archiver @FILE support... @

```

checking for strip... strip
checking for ranlib... ranlib
checking command to parse /usr/bin/nm -B output from gcc object... ok
checking for sysroot... no
checking for a working dd... /usr/bin/dd
checking how to truncate binary pipes... /usr/bin/dd bs=4096 count=1
checking for mt... mt
checking if mt is a manifest tool... no
checking for dlfcn.h... yes
checking for objdir... .libs
checking if gcc supports -fno-rtti -fno-exceptions... no
checking for gcc option to produce PIC... -fPIC -DPIC
checking if gcc PIC flag -fPIC -DPIC works... yes
checking if gcc static flag -static works... yes
checking if gcc supports -c -o file.o... yes
checking if gcc supports -c -o file.o... (cached) yes
checking whether the gcc linker (/usr/bin/ld -m elf_x86_64) supports shared
libraries... yes
checking whether -lc should be explicitly linked in... no
checking dynamic linker characteristics... GNU/Linux ld.so
checking how to hardcode library paths into programs... immediate
checking whether stripping libraries is possible... yes
checking if libtool supports shared libraries... yes
checking whether to build shared libraries... yes
checking whether to build static libraries... yes
checking whether ln -s works... yes
checking for egrep... (cached) /usr/bin/grep -E
checking for fcntl.h... yes
checking for limits.h... yes
checking for malloc.h... yes
checking for strings.h... (cached) yes
checking for unistd.h... (cached) yes
checking for stdint.h... (cached) yes
checking for an ANSI C-conforming const... yes
checking size of ... 0
checking for size_t... yes
checking whether struct tm is in sys/time.h or time.h... time.h
checking size of short... 2
checking size of unsigned short... 2
checking size of int... 4
checking size of unsigned int... 4
checking size of long... 8
checking size of unsigned long... 8
checking for uint8_t... yes
checking for uint16_t... yes
checking for uint32_t... yes
checking for working alloca.h... yes
checking for alloca... yes

```

checking for working memcmp... yes
checking for vprintf... yes
checking for strdup... yes
checking for strerror... yes
checking for strtol... yes
checking for strtoul... yes
checking for rand in -lm... yes
checking for lbfgs in -llbfgs... yes
checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating Makefile
config.status: creating genbinary.sh
config.status: creating include/Makefile
config.status: creating lib/cqdb/Makefile
config.status: creating lib/crf/Makefile
config.status: creating frontend/Makefile
config.status: creating swig/Makefile
config.status: creating swig/python/setup.py
config.status: creating swig/perl/Makefile.PL
config.status: creating config.h
config.status: executing depfiles commands
config.status: executing libtool commands

```

[29]: `!make`

```

make all-recursive
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite'
Making all in include
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
make[2]: Nothing to be done for 'all'.
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
Making all in lib/cqdb
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I. -I./include
-mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline
-std=c99 -I/usr/local/include -O3 -MT libcqdb_la-lookup3.lo -MD -MP -MF
.deps/libcqdb_la-lookup3.Tpo -c -o libcqdb_la-lookup3.lo `test -f
'src/lookup3.c' || echo './'`src/lookup3.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I./include -mfpmath=sse -msse2 -DUSE_SSE -O3
-fomit-frame-pointer -ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT
libcqdb_la-lookup3.lo -MD -MP -MF .deps/libcqdb_la-lookup3.Tpo -c src/lookup3.c
-fPIC -DPIC -o .libs/libcqdb_la-lookup3.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I./include -mfpmath=sse -msse2 -DUSE_SSE -O3
-fomit-frame-pointer -ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT
libcqdb_la-lookup3.lo -MD -MP -MF .deps/libcqdb_la-lookup3.Tpo -c src/lookup3.c

```

```

-o libcrfsuite_la-lookup3.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-lookup3.Tpo .deps/libcrfsuite_la-lookup3.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I. -I./include
-mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline
-std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-cqdb.lo -MD -MP -MF
.deps/libcrfsuite_la-cqdb.Tpo -c -o libcrfsuite_la-cqdb.lo `test -f 'src/cqdb.c' || echo
'./`src/cqdb.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I./include -mfpmath=sse -msse2 -DUSE_SSE -O3
-fomit-frame-pointer -ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT
libcrfsuite_la-cqdb.lo -MD -MP -MF .deps/libcrfsuite_la-cqdb.Tpo -c src/cqdb.c -fPIC
-DPIC -o .libs/libcrfsuite_la-cqdb.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I./include -mfpmath=sse -msse2 -DUSE_SSE -O3
-fomit-frame-pointer -ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT
libcrfsuite_la-cqdb.lo -MD -MP -MF .deps/libcrfsuite_la-cqdb.Tpo -c src/cqdb.c -o
libcrfsuite_la-cqdb.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-cqdb.Tpo .deps/libcrfsuite_la-cqdb.Plo
/bin/bash ../../libtool --tag=CC --mode=link gcc -I./include -mfpmath=sse
-msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -no-undefined -release 0.12 -o libcrfsuite_la
/usr/local/lib libcrfsuite_la-lookup3.lo libcrfsuite_la-cqdb.lo -llbfgs -lm -llbfgs
libtool: link: gcc -shared -fPIC -DPIC .libs/libcrfsuite_la-lookup3.o
.libs/libcrfsuite_la-cqdb.o -lm /usr/local/lib/liblbfgs.so -mfpmath=sse -msse2
-O3 -O3 -Wl,-soname -Wl,libcrfsuite-0.12.so -o .libs/libcrfsuite-0.12.so
libtool: link: (cd ".libs" && rm -f "libcrfsuite.so" && ln -s "libcrfsuite-0.12.so"
"libcrfsuite.so")
libtool: link: ar cr .libs/libcrfsuite.a libcrfsuite_la-lookup3.o libcrfsuite_la-cqdb.o
libtool: link: ranlib .libs/libcrfsuite.a
libtool: link: ( cd ".libs" && rm -f "libcrfsuite.la" && ln -s "../libcrfsuite.la"
"libcrfsuite.la" )
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
Making all in lib/crf
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
dictionary.lo -MD -MP -MF .deps/libcrfsuite_la-dictionary.Tpo -c -o
libcrfsuite_la-dictionary.lo `test -f 'src/dictionary.c' || echo
'./`src/dictionary.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-dictionary.lo -MD -MP -MF
.deps/libcrfsuite_la-dictionary.Tpo -c src/dictionary.c -fPIC -DPIC -o
.libs/libcrfsuite_la-dictionary.o

```

```

libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-dictionary.lo -MD -MP -MF
.deps/libcrfsuite_la-dictionary.Tpo -c src/dictionary.c -o libcrfsuite_la-
dictionary.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-dictionary.Tpo .deps/libcrfsuite_la-dictionary.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
logging.lo -MD -MP -MF .deps/libcrfsuite_la-logging.Tpo -c -o libcrfsuite_la-
logging.lo `test -f 'src/logging.c' || echo './`src/logging.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-logging.lo -MD -MP -MF
.deps/libcrfsuite_la-logging.Tpo -c src/logging.c -fPIC -DPIC -o
.libs/libcrfsuite_la-logging.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-logging.lo -MD -MP -MF
.deps/libcrfsuite_la-logging.Tpo -c src/logging.c -o libcrfsuite_la-logging.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-logging.Tpo .deps/libcrfsuite_la-logging.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
params.lo -MD -MP -MF .deps/libcrfsuite_la-params.Tpo -c -o libcrfsuite_la-
params.lo `test -f 'src/params.c' || echo './`src/params.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-params.lo -MD -MP -MF
.deps/libcrfsuite_la-params.Tpo -c src/params.c -fPIC -DPIC -o
.libs/libcrfsuite_la-params.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-params.lo -MD -MP -MF
.deps/libcrfsuite_la-params.Tpo -c src/params.c -o libcrfsuite_la-params.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-params.Tpo .deps/libcrfsuite_la-params.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer

```

```

-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
quark.lo -MD -MP -MF .deps/libcrfsuite_la-quark.Tpo -c -o libcrfsuite_la-
quark.lo `test -f 'src/quark.c' || echo './`src/quark.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-quark.lo -MD -MP -MF
.deps/libcrfsuite_la-quark.Tpo -c src/quark.c -fPIC -DPIC -o
.libs/libcrfsuite_la-quark.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-quark.lo -MD -MP -MF
.deps/libcrfsuite_la-quark.Tpo -c src/quark.c -o libcrfsuite_la-quark.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-quark.Tpo .deps/libcrfsuite_la-quark.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
rumavl.lo -MD -MP -MF .deps/libcrfsuite_la-rumavl.Tpo -c -o libcrfsuite_la-
rumavl.lo `test -f 'src/rumavl.c' || echo './`src/rumavl.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-rumavl.lo -MD -MP -MF
.deps/libcrfsuite_la-rumavl.Tpo -c src/rumavl.c -fPIC -DPIC -o
.libs/libcrfsuite_la-rumavl.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-rumavl.lo -MD -MP -MF
.deps/libcrfsuite_la-rumavl.Tpo -c src/rumavl.c -o libcrfsuite_la-rumavl.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-rumavl.Tpo .deps/libcrfsuite_la-rumavl.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
dataset.lo -MD -MP -MF .deps/libcrfsuite_la-dataset.Tpo -c -o libcrfsuite_la-
dataset.lo `test -f 'src/dataset.c' || echo './`src/dataset.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-dataset.lo -MD -MP -MF
.deps/libcrfsuite_la-dataset.Tpo -c src/dataset.c -fPIC -DPIC -o
.libs/libcrfsuite_la-dataset.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.

```



```

-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-dataset.lo -MD -MP -MF
.deps/libcrfsuite_la-dataset.Tpo -c src/dataset.c -o libcrfsuite_la-dataset.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-dataset.Tpo .deps/libcrfsuite_la-dataset.Plo
/bin/bash ../..../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
holdout.lo -MD -MP -MF .deps/libcrfsuite_la-holdout.Tpo -c -o libcrfsuite_la-
holdout.lo `test -f 'src/holdout.c' || echo './`src/holdout.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-holdout.lo -MD -MP -MF
.deps/libcrfsuite_la-holdout.Tpo -c src/holdout.c -fPIC -DPIC -o
.libs/libcrfsuite_la-holdout.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-holdout.lo -MD -MP -MF
.deps/libcrfsuite_la-holdout.Tpo -c src/holdout.c -o libcrfsuite_la-holdout.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-holdout.Tpo .deps/libcrfsuite_la-holdout.Plo
/bin/bash ../..../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
train_arow.lo -MD -MP -MF .deps/libcrfsuite_la-train_arow.Tpo -c -o
libcrfsuite_la-train_arow.lo `test -f 'src/train_arow.c' || echo
'./`src/train_arow.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_arow.lo -MD -MP -MF
.deps/libcrfsuite_la-train_arow.Tpo -c src/train_arow.c -fPIC -DPIC -o
.libs/libcrfsuite_la-train_arow.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_arow.lo -MD -MP -MF
.deps/libcrfsuite_la-train_arow.Tpo -c src/train_arow.c -o libcrfsuite_la-
train_arow.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-train_arow.Tpo .deps/libcrfsuite_la-train_arow.Plo
/bin/bash ../..../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer

```

```

-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
train_averaged_perceptron.lo -MD -MP -MF .deps/libcrfsuite_la-
train_averaged_perceptron.Tpo -c -o libcrfsuite_la-train_averaged_perceptron.lo
`test -f 'src/train_averaged_perceptron.c' || echo
'./'`src/train_averaged_perceptron.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../../include -I.
-I../.. -I../../include -I. -I../../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_averaged_perceptron.lo -MD -MP
-MF .deps/libcrfsuite_la-train_averaged_perceptron.Tpo -c
src/train_averaged_perceptron.c -fPIC -DPIC -o .libs/libcrfsuite_la-
train_averaged_perceptron.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../../include -I.
-I../.. -I../../include -I. -I../../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_averaged_perceptron.lo -MD -MP
-MF .deps/libcrfsuite_la-train_averaged_perceptron.Tpo -c
src/train_averaged_perceptron.c -o libcrfsuite_la-train_averaged_perceptron.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-train_averaged_perceptron.Tpo .deps/libcrfsuite_la-
train_averaged_perceptron.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../../include -I. -I../.. -I../../include -I.
-I../../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
train_l2sgd.lo -MD -MP -MF .deps/libcrfsuite_la-train_l2sgd.Tpo -c -o
libcrfsuite_la-train_l2sgd.lo `test -f 'src/train_l2sgd.c' || echo
'./'`src/train_l2sgd.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../../include -I.
-I../.. -I../../include -I. -I../../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_l2sgd.lo -MD -MP -MF
.deps/libcrfsuite_la-train_l2sgd.Tpo -c src/train_l2sgd.c -fPIC -DPIC -o
.libs/libcrfsuite_la-train_l2sgd.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../../include -I.
-I../.. -I../../include -I. -I../../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_l2sgd.lo -MD -MP -MF
.deps/libcrfsuite_la-train_l2sgd.Tpo -c src/train_l2sgd.c -o libcrfsuite_la-
train_l2sgd.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-train_l2sgd.Tpo .deps/libcrfsuite_la-train_l2sgd.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../../include -I. -I../.. -I../../include -I.
-I../../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
train_lbfgs.lo -MD -MP -MF .deps/libcrfsuite_la-train_lbfgs.Tpo -c -o
libcrfsuite_la-train_lbfgs.lo `test -f 'src/train_lbfgs.c' || echo
'./'`src/train_lbfgs.c

```

```

libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_lbfgs.lo -MD -MP -MF
.deps/libcrfsuite_la-train_lbfgs.Tpo -c src/train_lbfgs.c -fPIC -DPIC -o
.libs/libcrfsuite_la-train_lbfgs.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_lbfgs.lo -MD -MP -MF
.deps/libcrfsuite_la-train_lbfgs.Tpo -c src/train_lbfgs.c -o libcrfsuite_la-
train_lbfgs.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-train_lbfgs.Tpo .deps/libcrfsuite_la-train_lbfgs.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
train_passive_aggressive.lo -MD -MP -MF .deps/libcrfsuite_la-
train_passive_aggressive.Tpo -c -o libcrfsuite_la-train_passive_aggressive.lo
`test -f 'src/train_passive_aggressive.c' || echo
'./'`src/train_passive_aggressive.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_passive_aggressive.lo -MD -MP
-MF .deps/libcrfsuite_la-train_passive_aggressive.Tpo -c
src/train_passive_aggressive.c -fPIC -DPIC -o .libs/libcrfsuite_la-
train_passive_aggressive.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-train_passive_aggressive.lo -MD -MP
-MF .deps/libcrfsuite_la-train_passive_aggressive.Tpo -c
src/train_passive_aggressive.c -o libcrfsuite_la-train_passive_aggressive.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-train_passive_aggressive.Tpo .deps/libcrfsuite_la-
train_passive_aggressive.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
crfld_context.lo -MD -MP -MF .deps/libcrfsuite_la-crfld_context.Tpo -c -o
libcrfsuite_la-crfld_context.lo `test -f 'src/crfld_context.c' || echo
'./'`src/crfld_context.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfld_context.lo -MD -MP -MF

```

```

.deps/libcrfsuite_la-crf1d_context.Tpo -c src/crf1d_context.c -fPIC -DPIC -o
.libs/libcrfsuite_la-crf1d_context.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crf1d_context.lo -MD -MP -MF
.deps/libcrfsuite_la-crf1d_context.Tpo -c src/crf1d_context.c -o libcrfsuite_la-
crf1d_context.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-crf1d_context.Tpo .deps/libcrfsuite_la-
crf1d_context.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
crf1d_model.lo -MD -MP -MF .deps/libcrfsuite_la-crf1d_model.Tpo -c -o
libcrfsuite_la-crf1d_model.lo `test -f 'src/crf1d_model.c' || echo
'./'`src/crf1d_model.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crf1d_model.lo -MD -MP -MF
.deps/libcrfsuite_la-crf1d_model.Tpo -c src/crf1d_model.c -fPIC -DPIC -o
.libs/libcrfsuite_la-crf1d_model.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crf1d_model.lo -MD -MP -MF
.deps/libcrfsuite_la-crf1d_model.Tpo -c src/crf1d_model.c -o libcrfsuite_la-
crf1d_model.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-crf1d_model.Tpo .deps/libcrfsuite_la-crf1d_model.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
crf1d_feature.lo -MD -MP -MF .deps/libcrfsuite_la-crf1d_feature.Tpo -c -o
libcrfsuite_la-crf1d_feature.lo `test -f 'src/crf1d_feature.c' || echo
'./'`src/crf1d_feature.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crf1d_feature.lo -MD -MP -MF
.deps/libcrfsuite_la-crf1d_feature.Tpo -c src/crf1d_feature.c -fPIC -DPIC -o
.libs/libcrfsuite_la-crf1d_feature.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crf1d_feature.lo -MD -MP -MF
.deps/libcrfsuite_la-crf1d_feature.Tpo -c src/crf1d_feature.c -o libcrfsuite_la-

```

```

crfld_feature.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-crfld_feature.Tpo .deps/libcrfsuite_la-
crfld_feature.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
crfld_encode.lo -MD -MP -MF .deps/libcrfsuite_la-crfld_encode.Tpo -c -o
libcrfsuite_la-crfld_encode.lo `test -f 'src/crfld_encode.c' || echo
'./'`src/crfld_encode.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfld_encode.lo -MD -MP -MF
.deps/libcrfsuite_la-crfld_encode.Tpo -c src/crfld_encode.c -fPIC -DPIC -o
.libs/libcrfsuite_la-crfld_encode.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfld_encode.lo -MD -MP -MF
.deps/libcrfsuite_la-crfld_encode.Tpo -c src/crfld_encode.c -o libcrfsuite_la-
crfld_encode.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-crfld_encode.Tpo .deps/libcrfsuite_la-
crfld_encode.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
crfld_tag.lo -MD -MP -MF .deps/libcrfsuite_la-crfld_tag.Tpo -c -o
libcrfsuite_la-crfld_tag.lo `test -f 'src/crfld_tag.c' || echo
'./'`src/crfld_tag.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfld_tag.lo -MD -MP -MF
.deps/libcrfsuite_la-crfld_tag.Tpo -c src/crfld_tag.c -fPIC -DPIC -o
.libs/libcrfsuite_la-crfld_tag.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfld_tag.lo -MD -MP -MF
.deps/libcrfsuite_la-crfld_tag.Tpo -c src/crfld_tag.c -o libcrfsuite_la-
crfld_tag.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-crfld_tag.Tpo .deps/libcrfsuite_la-crfld_tag.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-

```

```

crfsuite_train.lo -MD -MP -MF .deps/libcrfsuite_la-crfsuite_train.Tpo -c -o
libcrfsuite_la-crfsuite_train.lo `test -f 'src/crfsuite_train.c' || echo
'./'`src/crfsuite_train.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfsuite_train.lo -MD -MP -MF
.deps/libcrfsuite_la-crfsuite_train.Tpo -c src/crfsuite_train.c -fPIC -DPIC -o
.libs/libcrfsuite_la-crfsuite_train.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfsuite_train.lo -MD -MP -MF
.deps/libcrfsuite_la-crfsuite_train.Tpo -c src/crfsuite_train.c -o
libcrfsuite_la-crfsuite_train.o >/dev/null 2>&1
mv -f .deps/libcrfsuite_la-crfsuite_train.Tpo .deps/libcrfsuite_la-
crfsuite_train.Plo
/bin/bash ../../libtool --tag=CC --mode=compile gcc -DHAVE_CONFIG_H -I.
-I../.. -I../.. -I../..../include -I. -I../.. -I../..../include -I.
-I../..../lib/cqdb/include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer
-ffast-math -Winline -std=c99 -I/usr/local/include -O3 -MT libcrfsuite_la-
crfsuite.lo -MD -MP -MF .deps/libcrfsuite_la-crfsuite.Tpo -c -o libcrfsuite_la-
crfsuite.lo `test -f 'src/crfsuite.c' || echo './'`src/crfsuite.c
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfsuite.lo -MD -MP -MF
.deps/libcrfsuite_la-crfsuite.Tpo -c src/crfsuite.c -fPIC -DPIC -o
.libs/libcrfsuite_la-crfsuite.o
libtool: compile: gcc -DHAVE_CONFIG_H -I. -I../.. -I../.. -I../..../include -I.
-I../.. -I../..../include -I. -I../..../lib/cqdb/include -mfpmath=sse -msse2
-DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -MT libcrfsuite_la-crfsuite.lo -MD -MP -MF
.deps/libcrfsuite_la-crfsuite.Tpo -c src/crfsuite.c -o libcrfsuite_la-crfsuite.o
>/dev/null 2>&1
mv -f .deps/libcrfsuite_la-crfsuite.Tpo .deps/libcrfsuite_la-crfsuite.Plo
/bin/bash ../../libtool --tag=CC --mode=link gcc -I../..../lib/cqdb/include
-mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline
-std=c99 -I/usr/local/include -O3 -no-undefined -release 0.12 -o libcrfsuite.la
-rpath /usr/local/lib libcrfsuite_la-dictionary.lo libcrfsuite_la-logging.lo
libcrfsuite_la-params.lo libcrfsuite_la-quark.lo libcrfsuite_la-rumavl.lo
libcrfsuite_la-dataset.lo libcrfsuite_la-holdout.lo libcrfsuite_la-train_arow.lo
libcrfsuite_la-train_averaged_perceptron.lo libcrfsuite_la-train_l2sgd.lo
libcrfsuite_la-train_lbfgs.lo libcrfsuite_la-train_passive_aggressive.lo
libcrfsuite_la-crfd_context.lo libcrfsuite_la-crfd_model.lo libcrfsuite_la-
crfd_feature.lo libcrfsuite_la-crfd_encode.lo libcrfsuite_la-crfd_tag.lo
libcrfsuite_la-crfsuite_train.lo libcrfsuite_la-crfsuite.lo
../..../lib/cqdb/libcqdb.la -llbfgs -lm -llbfgs

```

```

libtool: link: gcc -shared -fPIC -DPIC .libs/libcrfsuite_la-dictionary.o
.libs/libcrfsuite_la-logging.o .libs/libcrfsuite_la-params.o
.libs/libcrfsuite_la-quark.o .libs/libcrfsuite_la-rumavl.o .libs/libcrfsuite_la-
dataset.o .libs/libcrfsuite_la-holdout.o .libs/libcrfsuite_la-train_arow.o
.libs/libcrfsuite_la-train_averaged_perceptron.o .libs/libcrfsuite_la-
train_l2sgd.o .libs/libcrfsuite_la-train_lbfgs.o .libs/libcrfsuite_la-
train_passive_aggressive.o .libs/libcrfsuite_la-crfd_context.o
.libs/libcrfsuite_la-crfd_model.o .libs/libcrfsuite_la-crfd_feature.o
.libs/libcrfsuite_la-crfd_encode.o .libs/libcrfsuite_la-crfd_tag.o
.libs/libcrfsuite_la-crfsuite_train.o .libs/libcrfsuite_la-crfsuite.o
-Wl,-rpath -Wl,/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb/.libs
../../lib/cqdb/.libs/libcqdb.so -lm /usr/local/lib/liblbfgs.so -mfpmath=sse
-msse2 -O3 -O3 -Wl,-soname -Wl,libcrfsuite-0.12.so -o
.libs/libcrfsuite-0.12.so
libtool: link: (cd ".libs" && rm -f "libcrfsuite.so" && ln -s
"libcrfsuite-0.12.so" "libcrfsuite.so")
libtool: link: ar cr .libs/libcrfsuite.a libcrfsuite_la-dictionary.o
libcrfsuite_la-logging.o libcrfsuite_la-params.o libcrfsuite_la-quark.o
libcrfsuite_la-rumavl.o libcrfsuite_la-dataset.o libcrfsuite_la-holdout.o
libcrfsuite_la-train_arow.o libcrfsuite_la-train_averaged_perceptron.o
libcrfsuite_la-train_l2sgd.o libcrfsuite_la-train_lbfgs.o libcrfsuite_la-
train_passive_aggressive.o libcrfsuite_la-crfd_context.o libcrfsuite_la-
crfd_model.o libcrfsuite_la-crfd_feature.o libcrfsuite_la-crfd_encode.o
libcrfsuite_la-crfd_tag.o libcrfsuite_la-crfsuite_train.o libcrfsuite_la-
crfsuite.o
libtool: link: ranlib .libs/libcrfsuite.a
libtool: link: ( cd ".libs" && rm -f "libcrfsuite.la" && ln -s
"./libcrfsuite.la" "libcrfsuite.la" )
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
Making all in frontend
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/frontend'
gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -I. -I.. -I../include -I.
-I../include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math
-Winline -std=c99 -I/usr/local/include -O3 -MT crfsuite-iwa.o -MD -MP -MF
.deps/crfsuite-iwa.Tpo -c -o crfsuite-iwa.o `test -f 'iwa.c' || echo './'`iwa.c
mv -f .deps/crfsuite-iwa.Tpo .deps/crfsuite-iwa.Po
gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -I. -I.. -I../include -I.
-I../include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math
-Winline -std=c99 -I/usr/local/include -O3 -MT crfsuite-option.o -MD -MP -MF
.deps/crfsuite-option.Tpo -c -o crfsuite-option.o `test -f 'option.c' || echo
'./'`option.c
mv -f .deps/crfsuite-option.Tpo .deps/crfsuite-option.Po
gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -I. -I.. -I../include -I.
-I../include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math
-Winline -std=c99 -I/usr/local/include -O3 -MT crfsuite-reader.o -MD -MP -MF
.deps/crfsuite-reader.Tpo -c -o crfsuite-reader.o `test -f 'reader.c' || echo
'./'`reader.c
mv -f .deps/crfsuite-reader.Tpo .deps/crfsuite-reader.Po

```

```

gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -I. -I.. -I../include -I.
-I../include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math
-Winline -std=c99 -I/usr/local/include -O3 -MT crfsuite-learn.o -MD -MP -MF
.deps/crfsuite-learn.Tpo -c -o crfsuite-learn.o `test -f 'learn.c' || echo
'./'`learn.c
mv -f .deps/crfsuite-learn.Tpo .deps/crfsuite-learn.Po
gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -I. -I.. -I../include -I.
-I../include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math
-Winline -std=c99 -I/usr/local/include -O3 -MT crfsuite-tag.o -MD -MP -MF
.deps/crfsuite-tag.Tpo -c -o crfsuite-tag.o `test -f 'tag.c' || echo './'`tag.c
mv -f .deps/crfsuite-tag.Tpo .deps/crfsuite-tag.Po
gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -I. -I.. -I../include -I.
-I../include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math
-Winline -std=c99 -I/usr/local/include -O3 -MT crfsuite-dump.o -MD -MP -MF
.deps/crfsuite-dump.Tpo -c -o crfsuite-dump.o `test -f 'dump.c' || echo
'./'`dump.c
mv -f .deps/crfsuite-dump.Tpo .deps/crfsuite-dump.Po
gcc -DHAVE_CONFIG_H -I. -I.. -I.. -I../include -I. -I.. -I../include -I.
-I../include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math
-Winline -std=c99 -I/usr/local/include -O3 -MT crfsuite-main.o -MD -MP -MF
.deps/crfsuite-main.Tpo -c -o crfsuite-main.o `test -f 'main.c' || echo
'./'`main.c
mv -f .deps/crfsuite-main.Tpo .deps/crfsuite-main.Po
/bin/bash ../libtool --tag=CC --mode=link gcc -I../include -mfpmath=sse
-msse2 -DUSE_SSE -O3 -fomit-frame-pointer -ffast-math -Winline -std=c99
-I/usr/local/include -O3 -o crfsuite crfsuite-iwa.o crfsuite-option.o
crfsuite-reader.o crfsuite-learn.o crfsuite-tag.o crfsuite-dump.o crfsuite-
main.o ../lib/crf/libcrfsuite.la -llbfgs -lm -llbfgs
libtool: link: gcc -I../include -mfpmath=sse -msse2 -DUSE_SSE -O3 -fomit-frame-
pointer -ffast-math -Winline -std=c99 -I/usr/local/include -O3 -o .libs/crfsuite
crfsuite-iwa.o crfsuite-option.o crfsuite-reader.o crfsuite-learn.o crfsuite-
tag.o crfsuite-dump.o crfsuite-main.o ../lib/crf/.libs/libcrfsuite.so -lm
/usr/local/lib/liblbfgs.so
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/frontend'
Making all in swig
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/swig'
make[2]: Nothing to be done for 'all'.
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/swig'
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite'
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite'
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite'

make install run ပို့က sudo right ရှိရမယ်။

ye@lst-hpc3090:~/aif2/word-seg/tool/crfsuite$ sudo make install
Making install in include
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/include'

```



```

make[2]: Nothing to be done for 'install-exec-am'.
  /usr/bin/mkdir -p '/usr/local/include'
  /usr/bin/install -c -m 644 crfsuite.h crfsuite_api.hpp crfsuite.hpp '/usr/local/include'
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/include'
Making install in lib/cqdb
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
  /usr/bin/mkdir -p '/usr/local/lib'
  /bin/bash ../../libtool  --mode=install /usr/bin/install -c  libcqdb.la '/usr/local/lib'
libtool: install: /usr/bin/install -c .libs/libcqdb-0.12.so /usr/local/lib/libcqdb-0.12.so
libtool: install: (cd /usr/local/lib && { ln -s -f libcqdb-0.12.so libcqdb.so || { rm -f libcqdb.so;
libtool: install: /usr/bin/install -c .libs/libcqdb.lai /usr/local/lib/libcqdb.la
libtool: install: /usr/bin/install -c .libs/libcqdb.a /usr/local/lib/libcqdb.a
libtool: install: chmod 644 /usr/local/lib/libcqdb.a
libtool: install: ranlib /usr/local/lib/libcqdb.a
libtool: finish: PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin:"
-----

```

Libraries have been installed in:
 /usr/local/lib

If you ever happen to want to link against installed libraries in a given directory, LIBDIR, you must either use libtool, and specify the full pathname of the library, or use the '-LLIBDIR' flag during linking and do at least one of the following:

- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for more information, such as the ld(1) and ld.so(8) manual pages.

```

-----
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/cqdb'
Making install in lib/crf
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
  /usr/bin/mkdir -p '/usr/local/lib'
  /bin/bash ../../libtool  --mode=install /usr/bin/install -c  libcrfsuite.la '/usr/local/lib'
libtool: warning: relinking 'libcrfsuite.la'
libtool: install: (cd /home/ye/aif2/word-seg/tool/crfsuite/lib/crf; /bin/bash "/home/ye/aif2/w
libtool: relink: gcc -shared -fPIC -DPIC  .libs/libcrfsuite_la-dictionary.o .libs/libcrfsuite
libtool: install: /usr/bin/install -c .libs/libcrfsuite-0.12.soT /usr/local/lib/libcrfsuite-0.
libtool: install: (cd /usr/local/lib && { ln -s -f libcrfsuite-0.12.so libcrfsuite.so || { rm

```

```

libtool: install: /usr/bin/install -c .libs/libcrfsuite.lai /usr/local/lib/libcrfsuite.la
libtool: install: /usr/bin/install -c .libs/libcrfsuite.a /usr/local/lib/libcrfsuite.a
libtool: install: chmod 644 /usr/local/lib/libcrfsuite.a
libtool: install: ranlib /usr/local/lib/libcrfsuite.a
libtool: finish: PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin:"
-----

```

Libraries have been installed in:
 /usr/local/lib

If you ever happen to want to link against installed libraries in a given directory, LIBDIR, you must either use libtool, and specify the full pathname of the library, or use the '-LLIBDIR' flag during linking and do at least one of the following:

- add LIBDIR to the 'LD_LIBRARY_PATH' environment variable during execution
- add LIBDIR to the 'LD_RUN_PATH' environment variable during linking
- use the '-Wl,-rpath -Wl,LIBDIR' linker flag
- have your system administrator add LIBDIR to '/etc/ld.so.conf'

See any operating system documentation about shared libraries for more information, such as the ld(1) and ld.so(8) manual pages.

```

-----
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/lib/crf'
Making install in frontend
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/frontend'
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/frontend'
  /usr/bin/mkdir -p '/usr/local/bin'
  /bin/bash ../libtool  --mode=install /usr/bin/install -c crfsuite '/usr/local/bin'
libtool: install: /usr/bin/install -c .libs/crfsuite /usr/local/bin/crfsuite
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/frontend'
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/frontend'
Making install in swig
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/swig'
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite/swig'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/swig'
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite/swig'
make[1]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite'
make[2]: Entering directory '/home/ye/aif2/word-seg/tool/crfsuite'
make[2]: Nothing to be done for 'install-exec-am'.
  /usr/bin/mkdir -p '/usr/local/share/doc/crfsuite'
  /usr/bin/install -c -m 644 README INSTALL COPYING AUTHORS ChangeLog '/usr/local/share/doc/crfsuite'
make[2]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite'

```

```
make[1]: Leaving directory '/home/ye/aif2/word-seg/tool/crfsuite'
ye@lst-hpc3090:~/aif2/word-seg/tool/crfsuite$
```

```
[30]: !crfsuite --help
```

```
CRFSuite 0.12.2 Copyright (c) 2007-2013 Naoaki Okazaki
```

```
USAGE: crfsuite <COMMAND> [OPTIONS]
```

```
COMMAND      Command name to specify the processing
OPTIONS      Arguments for the command (optional; command-specific)
```

```
COMMAND:
```

```
learn        Obtain a model from a training set of instances
tag          Assign suitable labels to given instances by using a model
dump         Output a model in a plain-text format
```

For the usage of each command, specify -h option in the command argument.

```
[31]: !crfsuite learn --help
```

```
CRFSuite 0.12.2 Copyright (c) 2007-2013 Naoaki Okazaki
```

```
USAGE: crfsuite learn [OPTIONS] [DATA1] [DATA2] ...
```

Trains a model using training data set(s).

```
DATA      file(s) corresponding to data set(s) for training; if multiple N files
          are specified, this utility assigns a group number (1...N) to the
          instances in each file; if a file name is '-', the utility reads a
          data set from STDIN
```

```
OPTIONS:
```

```
-t, --type=TYPE      specify a graphical model (DEFAULT='1d'):
                     (this option is reserved for the future use)
                     1d          1st-order Markov CRF with state and transition
                               features; transition features are not conditioned
                               on observations
-a, --algorithm=NAME specify a training algorithm (DEFAULT='lbfgs')
                     lbfgs      L-BFGS with L1/L2 regularization
                     l2sgd      SGD with L2-regularization
                     ap         Averaged Perceptron
                     pa         Passive Aggressive
                     arow       Adaptive Regularization of Weights (AROW)
-p, --set=NAME=VALUE set the algorithm-specific parameter NAME to VALUE;
                     use '-H' or '--help-parameters' with the algorithm name
                     specified by '-a' or '--algorithm' and the graphical
                     model specified by '-t' or '--type' to see the list of
                     algorithm-specific parameters
-m, --model=FILE     store the model to FILE (DEFAULT=''); if the value is
                     empty, this utility does not store the model
```

```

-g, --split=N          split the instances into N groups; this option is
                        useful for holdout evaluation and cross validation
-e, --holdout=M        use the M-th data for holdout evaluation and the rest
                        for training
-x, --cross-validate   repeat holdout evaluations for #i in {1, ..., N} groups
                        (N-fold cross validation)
-l, --log-to-file      write the training log to a file instead of to STDOUT;
                        The filename is determined automatically by the training
                        algorithm, parameters, and source files
-L, --logbase=BASE     set the base name for a log file (used with -l option)
-h, --help             show the usage of this command and exit
-H, --help-parameters show the help message of algorithm-specific parameters;
                        specify an algorithm with '-a' or '--algorithm' option,
                        and specify a graphical model with '-t' or '--type'

```

option

```
[32]: !crfsuite tag --help
```

CRFSuite 0.12.2 Copyright (c) 2007-2013 Naoaki Okazaki

USAGE: crfsuite tag [OPTIONS] [DATA]

Assign suitable labels to the instances in the data set given by a file (DATA).

If the argument DATA is omitted or '-', this utility reads a data from STDIN.

Evaluate the performance of the model on labeled instances (with -t option).

OPTIONS:

```

-m, --model=MODEL      Read a model from a file (MODEL)
-t, --test             Report the performance of the model on the data
-r, --reference        Output the reference labels in the input data
-p, --probability      Output the probability of the label sequences
-i, --marginal         Output the marginal probability of items for their
predicted label
-l, --marginal-all    Output the marginal probabilities of items for all
labels
-q, --quiet            Suppress tagging results (useful for test mode)
-h, --help             Show the usage of this command and exit

```

1.6 Data Preparation

Github repository တစ်ခုလုံးကို download မလုပ်ပဲ လိုချင်တဲ့ ဖိုင်တစ်ဖိုင်ထဲကိုပဲ ကိုယ့်စက်ထဲကို ကော်ပီကူးချင်ရင် raw format အဖြစ်ပြောင်းကြည့်ပြီး ပုံမှန်အတိုင်း copy/paste လုပ်ယူလည်း ရတယ်။ သို့မဟုတ် အဲဒီ raw link ကို ယူလိုက်ပြီး wget command နဲ့လည်း ကော်ပီကူးယူလို့ ရပါတယ်။

```
[34]: %cd ../../data/myPOS/
```

/home/ye/aif2/word-seg/data/myPOS

```
[36]: !wget https://raw.githubusercontent.com/ye-kyaw-thu/myPOS/refs/heads/master/
↳corpus-ver-3.0/corpus/mypos-ver.3.0.shuf.notag.nopunc.txt
```

```
--2026-07-25 15:16:17-- https://raw.githubusercontent.com/ye-kyaw-thu/myPOS/refs/heads/master/corpus-ver-3.0/corpus/mypos-ver.3.0.shuf.notag.nopunc.txt
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.110.133, 185.199.109.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.110.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 7303817 (7.0M) [application/octet-stream]
Saving to: 'mypos-ver.3.0.shuf.notag.nopunc.txt'

mypos-ver.3.0.shuf. 100%[=====>] 6.96M --.-KB/s in 0.07s

2026-07-25 15:16:17 (103 MB/s) - 'mypos-ver.3.0.shuf.notag.nopunc.txt' saved
[7303817/7303817]
```

```
[37]: !head mypos-ver.3.0.shuf.notag.nopunc.txt
```

၁၉၆၂ ခုနှစ် ခန့်မှန်း သန်းခေါင်စာရင်း အရ လူဦးရေ ၁၁၅၉၃၁ ယောက် ရှိ သည်
 လူ တိုင်း တွင် သင့်မြတ် လျော်ကန် စွာ ကန့်သတ် ထား သည့် အလုပ် လုပ် ချိန် အပြင် လစာ
 နှင့်တကွ အခါ ကာလ အားလျော်စွာ သတ်မှတ် ထား သည့် အလုပ် အားလပ်ရက် များ ပါဝင် သည့်
 အနားယူခွင့် နှင့် အားလပ်ခွင့် ခံစားပိုင်ခွင့် ရှိ သည်
 ဤ နည်း ကို စစ်ယူ သော နည်း ဟု ခေါ် သည်
 စာပြန်ပွဲ ဆို တာ က အာဂုံဆောင် အလွတ်ကျက် ထား တဲ့ ပိဋကတ်သုံးပုံ စာပေ တွေ ကို စာစစ်
 သံဃာတော်ကြီး တွေ ရဲ့ ရှေ့ မှာ အလွတ် ပြန် ပြီး ရွတ်ပြ ရ တာ ပေါ့
 ဒီ မှာ ကျွန်တော့် သက်သေခံကတ် ပါ
 ၂၀ ရာစု မြန်မာ့ သမိုင်း သန်းဝင်းလှိုင် ၂၀၀၉ ခု မေ လ ကံကော်ဝတ်ရည် စာပေ
 ကျွန်တော် မျက်မှန် တစ် လက် လုပ် ချင် ပါ တယ်
 ကျွန်တော် တို့ က ဒီ အမှု ရဲ့ ကြံရာပါ ကို ဖမ်းမိ ဖို့ ကြိုးစား ခဲ့ တယ်
 ကလေး မီးဖွား ဖို့ ခန့်မှန်း ရက် က ဘယ်တော့ ပါ လဲ
 အရိုးရှင်းဆုံး ကာဗိုဟိုက်ဒရိတ် မှာ ဂလူးကိုစ် ဂလက်တို့စ် ဖရပ်တို့စ် စသည့်
 မိုနိုဆက်ကရိုက် များ ဖြစ် သည်

```
[38]: !wc mypos-ver.3.0.shuf.notag.nopunc.txt
```

```
43196 510644 7303817 mypos-ver.3.0.shuf.notag.nopunc.txt
```

```
[39]: !wget https://raw.githubusercontent.com/ye-kyaw-thu/myPOS/refs/heads/master/
      ↪corpus-ver-3.0/corpus/otest.1k.nopipe.txt
```

```
--2026-07-25 15:18:11-- https://raw.githubusercontent.com/ye-kyaw-thu/myPOS/refs/heads/master/corpus-ver-3.0/corpus/otest.1k.nopipe.txt
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.110.133, 185.199.111.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.110.133|:443... connected.
```

```
HTTP request sent, awaiting response... 200 OK
Length: 229758 (224K) [text/plain]
Saving to: 'otest.1k.nopipe.txt.1'
```

```
otest.1k.nopipe.txt 100%[=====>] 224.37K  --.-KB/s    in 0.006s
```

```
2026-07-25 15:18:11 (36.9 MB/s) - 'otest.1k.nopipe.txt.1' saved [229758/229758]
```

```
[40]: !head otest.1k.nopipe.txt
```

```
တစ်/tn ကိုက်/n ကို/ppm ဝမ်/n ခုနှစ်ထောင်/tn ပါ/part //punc
မနှစ်/n က/ppm သူ/pron ကျွန်မ/pron ကို/ppm သင်/v ဝေး/part တယ်/ppm //punc
ကျွန်တော့်/pron ခုံ/n သွား/v ရှာ/v မလို/part //punc
အတန်း/n စ/v တာ/part ကြာ/v ပြီ/ppm လား/part //punc
ဆေး/n နည်းနည်း/adv စား/v လိုက်/part ၊/punc သုံး/tn လေး/tn ရက်/n လောက်/part
အနားယူ/v လိုက်/part ရင်/conj ပျောက်/v သွား/part မှာ/ppm ပါ/part //punc
အေးချမ်း/v မှု/part နဲ့/conj စည်းကမ်း/n ကို/ppm တည်မြဲ/v အောင်/part ထိန်းသိမ်း/v
သည်/ppm //punc
ဇွန်း/n ကို/ppm လိုအပ်/v တယ်/ppm //punc
ဘွဲ့/n ရ/v ရင်/conj ဘာ/n လုပ်/v မ/part လို့/part လဲ/part //punc
ကျွန်တော်/pron ချောင်းဆိုး/v ခြင်း/part အတွက်/ppm တစ်/tn ခု/part ခု/part လို/v
ချင်/part တယ်/ppm //punc
အသီးအနှံ/n တို့/part မှ/ppm လွဲ/v လျှင်/conj လူ/n တို့/part ၏/ppm အဓိက/n
အစားအစာ/n မှာ/ppm ငါး/n ဖြစ်/v သည်/ppm //punc
```

```
[41]: !wc otest.1k.nopipe.txt
```

```
1000 13468 229758 otest.1k.nopipe.txt
```

```
[43]: !perl ./mk-wordtag.pl ./otest.1k.nopipe.txt "\/" w > ./otest.txt
```

```
[44]: !head ./otest.txt
```

```
တစ် ကိုက် ကို ဝမ် ခုနှစ်ထောင် ပါ ။
မနှစ် က သူ ကျွန်မ ကို သင် ဝေး တယ် ။
ကျွန်တော့် ခုံ သွား ရှာ မလို ။
အတန်း စ တာ ကြာ ပြီ လား ။
ဆေး နည်းနည်း စား လိုက် ၊ သုံး လေး ရက် လောက် အနားယူ လိုက် ရင် ပျောက် သွား မှာ ပါ
။
အေးချမ်း မှု နဲ့ စည်းကမ်း ကို တည်မြဲ အောင် ထိန်းသိမ်း သည် ။
ဇွန်း ကို လိုအပ် တယ် ။
ဘွဲ့ ရ ရင် ဘာ လုပ် မ လို့ လဲ ။
ကျွန်တော် ချောင်းဆိုး ခြင်း အတွက် တစ် ခု ခု လို ချင် တယ် ။
အသီးအနှံ တို့ မှ လွဲ လျှင် လူ တို့ ၏ အဓိက အစားအစာ မှာ ငါး ဖြစ် သည် ။
```

1.7 Tag Preparation for Word Segmentation

Based on paper: Win Pa Pa, Ye Kyaw Thu, Andrew Finch, Eiichiro Sumita, “Word Boundary Identification for Myanmar Text Using Conditional Random Fields”, In Proceedings of the Ninth International Conference on Genetic and Evolutionary Computing (ICGEC 2015), August 26-28, 2015, Yangon, Myanmar, pp. 447-456.

[46]: `!cat /home/ye/aif2/word-seg/util/myWord_tagger.py`

```
# myWordTagger.py
# Author: Thura Aung
# 9th December, 2023
# Data preparation for word segmentation CRF model training
# Ref: Win Pa Pa, Ye Kyaw Thu, Andrew Finch, Eiichiro Sumita, "Word Boundary
Identification for Myanmar Text Using Conditional Random Fields", In Proceedings
of the Ninth International Conference on Genetic and Evolutionary Computing
(ICGEC 2015), August 26-28, 2015, Yangon, Myanmar, pp. 447-456.

import re
import argparse

parser = argparse.ArgumentParser(description='Data preparation for word
segmentation')
parser.add_argument('-i', '--input', type=str, help='input file', required=True)
parser.add_argument('-m', '--mode', type=str, default=r's', help='s for syllable
and c for character tagging', required=False)
parser.add_argument('-n', '--nTag', type=int, default=4, help='Number of Tags
{2, 3, 4}', required=False)
args = parser.parse_args()

inputFile = getattr(args, 'input')
mode = getattr(args, 'mode')
n = getattr(args, 'nTag')

myConsonant = r"က-အ"
enChar = r"a-zA-Z0-9"
otherChar = r"ဣတ္ထိဉ္ဇီဒြဏ္ဍေသဉ္စုၤၤဝ-၉ၣ်!-/:-@[-`{-~\s"
ssSymbol = r'့'
aThat = r'်'

# Regular expression pattern for Myanmar syllable breaking
# A consonant not after a subscript symbol AND a consonant is not followed by
a-That character or a subscript symbol
BreakPattern = re.compile(r"((?<!" + ssSymbol + r")[" + myConsonant + r"])(?![" +
aThat + ssSymbol + r"])" + r"[" + enChar + otherChar + r"])"

# Function to segment words into syllables according to the provided rules
def segment_into_syllables(text):
```

```

data = ""
line = re.sub(BreakPattern, " "+r"\1",text)
data += line
return data.split()

def make_word_ls(line, chk='s'):
    word_ls = []
    if chk == 's':
        words = line.split()
        for word in words:
            word_ls.append(list(segment_into_syllables(word)))
    else:
        word_ls = [list(c) for c in line.split()]
    return word_ls

def two_tagger(word_ls):
    for word in word_ls:
        n_of_element = len(word)
        if n_of_element == 1:
            print(f"{word[0]}\t|")
        else:
            for i in range(0, n_of_element-1):
                print(f"{word[i]}\t-")
            print(f"{word[n_of_element-1]}\t|")

def three_tagger(word_ls):
    for word in word_ls:
        n_of_element = len(word)
        if n_of_element >= 3:
            print(f"{word[0]}\t<")
            for i in range(1, n_of_element-1):
                print(f"{word[i]}\t-")
            print(f"{word[n_of_element-1]}\t|")
        elif n_of_element == 2:
            print(f"{word[0]}\t<")
            print(f"{word[1]}\t|")
        else:
            print(f"{word[0]}\t|")

def four_tagger(word_ls):
    for word in word_ls:
        n_of_element = len(word)
        if n_of_element >= 4:
            print(f"{word[0]}\t<")
            for i in range(1, n_of_element-2):
                print(f"{word[i]}\t-")
            print(f"{word[n_of_element-2]}\t>")
            print(f"{word[n_of_element-1]}\t|")

```



```

        elif n_of_element == 3:
            print(f"{word[0]}\t<")
            print(f"{word[1]}\t>")
            print(f"{word[2]}\t|")
        elif n_of_element == 2:
            print(f"{word[0]}\t<")
            print(f"{word[1]}\t|")
        else:
            print(f"{word[0]}\t|")

try:
    with open(inputFile, 'r') as file:
        if n == 2:
            for line in file:
                two_tagger(make_word_ls(line, mode))
                print()
        if n == 3:
            for line in file:
                three_tagger(make_word_ls(line, mode))
                print()
        if n == 4:
            for line in file:
                four_tagger(make_word_ls(line, mode))
                print()
        else:
            print("Only 2 to 4 accepted.")
except FileNotFoundError:
    print(f"File '{file_path}' not found.")
except Exception as e:
    print(f"An error occurred: {e}")

```

1.8 -help Calling

```
[47]: !python /home/ye/aif2/word-seg/util/myWord_tagger.py --help
```

```
usage: myWord_tagger.py [-h] -i INPUT [-m MODE] [-n NTAG]
```

Data preparation for word segmentation

options:

```

-h, --help            show this help message and exit
-i INPUT, --input INPUT
                        input file
-m MODE, --mode MODE  s for syllable and c for character tagging
-n NTAG, --ntag NTAG  Number of Tags {2, 3, 4}

```

```
[48]: %cd /home/ye/aif2/word-seg/data/myPOS
```

```
/home/ye/aif2/word-seg/data/myPOS
```

```
[49]: !mkdir tag
```

```
[50]: %cd tag
```

```
/home/ye/aif2/word-seg/data/myPOS/tag
```

```
[55]: !python /home/ye/aif2/word-seg/util/myWord_tagger.py --input /home/ye/aif2/  
↪word-seg/data/myPOS/mypos-ver.3.0.shuf.notag.nopunc.txt --mode s -n 2 > /  
↪home/ye/aif2/word-seg/data/myPOS/tag/train.tag2.txt
```

```
[56]: !head -n 30 /home/ye/aif2/word-seg/data/myPOS/tag/train.tag2.txt
```

```
၁ -  
၉ -  
၉ -  
၂ |  
၃ -  
နှစ် |  
ခန့် -  
မှန်း |  
သန်း -  
ခေါင် -  
စာ -  
ရင်း |  
အ -  
ရ |  
လူ -  
ဦး -  
ရေ |  
၁ -  
၁ -  
၅ -  
၉ -  
၃ -  
၁ |  
ယောက် |  
ရှိ |  
သည် |  
  
လူ |  
တိုင်း |  
တွင် |
```

1.9 Syllable, 3 tag Format

```
[57]: !python /home/ye/aif2/word-seg/util/myWord_tagger.py --input /home/ye/aif2/  
      ↪word-seg/data/myPOS/mypos-ver.3.0.shuf.notag.nopunc.txt --mode s -n 3 > /  
      ↪home/ye/aif2/word-seg/data/myPOS/tag/train.tag3.txt
```

```
[58]: !head -n 50 /home/ye/aif2/word-seg/data/myPOS/tag/train.tag3.txt
```

```
၁      <  
၉      -  
၉      -  
၂      |  
ခု      <  
နှစ်   |  
ခန့်    <  
မှန်း  |  
သန်း  <  
ခေါင်  -  
စာ      -  
ရင်း   |  
အ      <  
ရ      |  
လူ     <  
ဦး    -  
ရေ     |  
၁      <  
၁      -  
၅      -  
၉      -  
၃      -  
၁      |  
ယောက် |  
ရှိ     |  
သည်  |  
  
လူ     |  
တိုင်း |  
တွင်   |  
သင့်    <  
မြတ်   |  
လျော်  <  
ကန်    |  
စွာ     |  
ကန်    <
```

သတ်	
ထား	
သည့်	
အ	<
လုပ်	
လုပ်	
ချိန်	
အ	<
ပြင်	
လ	<
စာ	
နှင့်	<
တ	-
ကွ	

1.10 Tagging for Test Data

```
[59]: !python /home/ye/aif2/word-seg/util/myWord_tagger.py --input /home/ye/aif2/
      ↪word-seg/data/myPOS/otest.txt --mode s -n 3 > /home/ye/aif2/word-seg/data/
      ↪myPOS/tag/test.tag3.txt
```

```
[61]: !head -n 100 /home/ye/aif2/word-seg/data/myPOS/tag/test.tag3.txt
```

တစ်	
ကိုက်	
ကို	
ဝမ်	
ခု	<
နှစ်	-
ထောင်	
ပါ	
။	
မ	<
နှစ်	
က	
သူ	
ကျွန်	<
မ	
ကို	
သင်	
ပေး	
တယ်	
။	

ကျွန်	<
တော့	
ခုံ	
သွား	
ရှာ	
မ	<
လို့	
။	

အ	<
တန်း	
စ	
တာ	
ကြာ	
ပြီ	
လာ	
။	

ဆေး	
နည်း	<
နည်း	
စား	
လိုက်	
၊	
သုံး	
လေး	
ရက်	
လောက်	
အ	<
နား	-
ယူ	
လိုက်	
ရင်	
ပျောက်	
သွား	
မှာ	
ပါ	
။	

အေး	<
ချမ်း	
မှု	
နဲ့	
စည်း	<

ကမ်း: |
 ဘို: |
 တည်: <
 မြဲ: |
 အောင်: |
 ထိန်း: <
 သိမ်း: |
 သည်: |
 ။: |

စွန်း: |
 ဘို: |
 လို: <
 အပ်: |
 တယ်: |
 ။: |

သွဲ: |
 ရ: |
 ရင်: |
 ဘာ: |
 လုပ်: |
 မ: |
 လို့: |
 လဲ: |
 ။: |

ကျွန်: <
 တော်: |
 ချောင်း: <
 ဆိုး: |
 ခြင်း: |
 အ: <
 တွက်: |

1.11 Change into CRFSuite Data Format

```
[62]: %cd /home/ye/aif2/word-seg/tool/crfsuite/example
```

```
/home/ye/aif2/word-seg/tool/crfsuite/example
```

```
[64]: !ls
```

```
chunking.py crfutils.py ner.py pos.py template.py
```

for Word segmentation, we need to use chunking.py code.

```
[77]: !cat /home/ye/aif2/word-seg/data/myPOS/tag/debug.tag3.txt | python ./chunking.  
      ↪py > debug.crfsuite.txt
```

```
[78]: !cat debug.crfsuite.txt
```

```
['w', 'y']  
|      w[0]=တစ်      w[1]=ကိုက်      w[2]=ကို      w[0]|w[1]=တစ်|ကိုက်  
__BOS__  
|      w[-1]=တစ်      w[0]=ကိုက်      w[1]=ကို      w[2]=ဝမ်  
w[-1]|w[0]=တစ်|ကိုက်      w[0]|w[1]=ကိုက်|ကို  
|      w[-2]=တစ်      w[-1]=ကိုက်      w[0]=ကို      w[1]=ဝမ်      w[2]=ခု  
w[-1]|w[0]=ကိုက်|ကို      w[0]|w[1]=ကို|ဝမ်  
|      w[-2]=ကိုက်      w[-1]=ကို      w[0]=ဝမ်      w[1]=ခု      w[2]=နှစ်  
w[-1]|w[0]=ကို|ဝမ်      w[0]|w[1]=ဝမ်|ခု  
<      w[-2]=ကို      w[-1]=ဝမ်      w[0]=ခု      w[1]=နှစ်      w[2]=ထောင်  
w[-1]|w[0]=ဝမ်|ခု      w[0]|w[1]=ခု|နှစ်  
-      w[-2]=ဝမ်      w[-1]=ခု      w[0]=နှစ်      w[1]=ထောင်      w[2]=ပါ  
w[-1]|w[0]=ခု|နှစ်      w[0]|w[1]=နှစ်|ထောင်  
|      w[-2]=ခု      w[-1]=နှစ်      w[0]=ထောင်      w[1]=ပါ      w[2]=။  
w[-1]|w[0]=နှစ်|ထောင်      w[0]|w[1]=ထောင်|ပါ  
|      w[-2]=နှစ်      w[-1]=ထောင်      w[0]=ပါ      w[1]=။      w[-1]|w[0]=ထောင်|ပါ  
w[0]|w[1]=ပါ|။  
|      w[-2]=ထောင်      w[-1]=ပါ      w[0]=။      w[-1]|w[0]=ပါ|။      __EOS__  
  
<      w[0]=မ      w[1]=နှစ်      w[2]=က      w[0]|w[1]=မ|နှစ်      __BOS__  
|      w[-1]=မ      w[0]=နှစ်      w[1]=က      w[2]=သူ      w[-1]|w[0]=မ|နှစ်  
w[0]|w[1]=နှစ်|က  
|      w[-2]=မ      w[-1]=နှစ်      w[0]=က      w[1]=သူ      w[2]=ကျွန်  
w[-1]|w[0]=နှစ်|က      w[0]|w[1]=က|သူ  
|      w[-2]=နှစ်      w[-1]=က      w[0]=သူ      w[1]=ကျွန်      w[2]=မ      w[-1]|w[0]=က|သူ  
w[0]|w[1]=သူ|ကျွန်  
<      w[-2]=က      w[-1]=သူ      w[0]=ကျွန်      w[1]=မ      w[2]=ကို  
w[-1]|w[0]=သူ|ကျွန်      w[0]|w[1]=ကျွန်|မ  
|      w[-2]=သူ      w[-1]=ကျွန်      w[0]=မ      w[1]=ကို      w[2]=သင်  
w[-1]|w[0]=ကျွန်|မ      w[0]|w[1]=မ|ကို  
|      w[-2]=ကျွန်      w[-1]=မ      w[0]=ကို      w[1]=သင်      w[2]=ပေး  
w[-1]|w[0]=မ|ကို      w[0]|w[1]=ကို|သင်  
|      w[-2]=မ      w[-1]=ကို      w[0]=သင်      w[1]=ပေး      w[2]=တယ်  
w[-1]|w[0]=ကို|သင်      w[0]|w[1]=သင်|ပေး  
|      w[-2]=ကို      w[-1]=သင်      w[0]=ပေး      w[1]=တယ်      w[2]=။  
w[-1]|w[0]=သင်|ပေး      w[0]|w[1]=ပေး|တယ်
```

```

|          w[-2]=သင်          w[-1]=ပေး          w[0]=တယ်          w[1]=။
w[-1]|w[0]=ပေး|တယ်          w[0]|w[1]=တယ်|။
|          w[-2]=ပေး          w[-1]=တယ်          w[0]=။  w[-1]|w[0]=တယ်|။
↳ __EOS__

<          w[0]=ကျွန်          w[1]=တော့          w[2]=ခုံ          w[0]|w[1]=ကျွန်|တော့
__BOS__
|          w[-1]=ကျွန်          w[0]=တော့          w[1]=ခုံ          w[2]=သွား
w[-1]|w[0]=ကျွန်|တော့          w[0]|w[1]=တော့|ခုံ
|          w[-2]=ကျွန်          w[-1]=တော့          w[0]=ခုံ          w[1]=သွား          w[2]=ရှာ
w[-1]|w[0]=တော့|ခုံ          w[0]|w[1]=ခုံ|သွား
|          w[-2]=တော့          w[-1]=ခုံ          w[0]=သွား          w[1]=ရှာ          w[2]=မ
w[-1]|w[0]=ခုံ|သွား          w[0]|w[1]=သွား|ရှာ
|          w[-2]=ခုံ          w[-1]=သွား          w[0]=ရှာ          w[1]=မ          w[2]=လို့
w[-1]|w[0]=သွား|ရှာ          w[0]|w[1]=ရှာ|မ
<          w[-2]=သွား          w[-1]=ရှာ          w[0]=မ          w[1]=လို့          w[2]=။
w[-1]|w[0]=ရှာ|မ          w[0]|w[1]=မ|လို့
|          w[-2]=ရှာ          w[-1]=မ          w[0]=လို့          w[1]=။  w[-1]|w[0]=မ|လို့
w[0]|w[1]=လို့|။
|          w[-2]=မ          w[-1]=လို့          w[0]=။  w[-1]|w[0]=လို့|။  __EOS__

```

1.12 We Found Error! :)

```

{\burmesefont } <
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{\burmesefont } |

```

Only 2 to 4 accepted.

1.13 Backup Original Code

```
[80]: !mkdir bk
```

```
mkdir: cannot create directory 'bk': File exists
```



```
[81]: !cp chunking.py ./bk/
```

```
[82]: !ls ./bk/
```

chunking.py

1.14 Code Updating

```
[83]: %pwd
```

```
[83]: '/home/ye/aif2/word-seg/tool/crfsuite/example'
```

```
[84]: !cat ./chunking.py
```

```
#!/usr/bin/env python
```

```
"""
```

```
A feature extractor for chunking.  
Copyright 2010,2011 Naoaki Okazaki.  
"""
```

```
# Separator of field values.  
separator = '\t'
```

```
# Field names of the input data.  
fields = 'w y'
```

```
# Attribute templates.  
templates = (  
    (('w', -2), ),  
    (('w', -1), ),  
    (('w', 0), ),  
    (('w', 1), ),  
    (('w', 2), ),  
    (('w', -1), ('w', 0)),  
    (('w', 0), ('w', 1)),  
)
```

```
import crfutils
```

```
def feature_extractor(X):  
    # Apply attribute templates to obtain features (in fact, attributes)  
    crfutils.apply_templates(X, templates)  
    if X:  
        # Append BOS and EOS features manually  
        X[0]['F'].append('__BOS__')    # BOS feature  
        X[-1]['F'].append('__EOS__')   # EOS feature
```

```
if __name__ == '__main__':
    crfutils.main(feature_extractor, fields=fields, sep=separator)
```

[]:

```
[72]: !cat /home/ye/aif2/word-seg/data/myPOS/tag/train.tag3.txt | python ./chunking.
      ↪py > train.crfsuite.txt
```

Traceback (most recent call last):

File "/home/ye/aif2/word-seg/tool/crfsuite/example/./chunking.py", line 60, in
<module>

```
    crfutils.main(feature_extractor, fields=fields, sep=separator)
```

File "/home/ye/aif2/word-seg/tool/crfsuite/example/crfutils.py", line 160, in
main

```
    for X in readiter(fi, F, options.separator):
```

File "/home/ye/aif2/word-seg/tool/crfsuite/example/crfutils.py", line 64, in
readiter

```
    raise ValueError(
```

ValueError: Too few fields (1) for ['w', 'y']

```
    <
    <
```

cat: write error: Broken pipe

1.15 Change into CRFSuite Data Format Again

```
[85]: !cat /home/ye/aif2/word-seg/data/myPOS/tag/test.tag3.txt | python ./chunking.py
      ↪ /home/ye/aif2/word-seg/data/myPOS/tag/test.crfsuite.txt
```

```
[87]: !head -n 30 ./test.crfsuite.txt
```

```
['w', 'y']
|      w[0]=တစ်      w[1]=ကိုက်      w[2]=ကို      w[0]|w[1]=တစ်|ကိုက်
__BOS__
|      w[-1]=တစ်      w[0]=ကိုက်      w[1]=ကို      w[2]=ဝမ်
w[-1]|w[0]=တစ်|ကိုက်      w[0]|w[1]=ကိုက်|ကို
|      w[-2]=တစ်      w[-1]=ကိုက်      w[0]=ကို      w[1]=ဝမ်      w[2]=ခု
w[-1]|w[0]=ကိုက်|ကို      w[0]|w[1]=ကို|ဝမ်
|      w[-2]=ကိုက်      w[-1]=ကို      w[0]=ဝမ်      w[1]=ခု      w[2]=နှစ်
w[-1]|w[0]=ကို|ဝမ်      w[0]|w[1]=ဝမ်|ခု
<      w[-2]=ကို      w[-1]=ဝမ်      w[0]=ခု      w[1]=နှစ်      w[2]=ထောင်
w[-1]|w[0]=ဝမ်|ခု      w[0]|w[1]=ခု|နှစ်
-      w[-2]=ဝမ်      w[-1]=ခု      w[0]=နှစ်      w[1]=ထောင်      w[2]=ပါ
w[-1]|w[0]=ခု|နှစ်      w[0]|w[1]=နှစ်|ထောင်
|      w[-2]=ခု      w[-1]=နှစ်      w[0]=ထောင်      w[1]=ပါ      w[2]=။
w[-1]|w[0]=နှစ်|ထောင်      w[0]|w[1]=ထောင်|ပါ
|      w[-2]=နှစ်      w[-1]=ထောင်      w[0]=ပါ      w[1]=။      w[-1]|w[0]=ထောင်|ပါ
```

$w[0] | w[1] = \text{ပါ} | \parallel$
 $| \quad w[-2] = \text{ထောင်} \quad w[-1] = \text{ပါ} \quad w[0] = \parallel \quad w[-1] | w[0] = \text{ပါ} | \parallel \quad _ _ \text{EOS} _ _$

$< \quad w[0] = \text{မ} \quad w[1] = \text{နှစ်} \quad w[2] = \text{က} \quad w[0] | w[1] = \text{မ} | \text{နှစ်} \quad _ _ \text{BOS} _ _$
 $| \quad w[-1] = \text{မ} \quad w[0] = \text{နှစ်} \quad w[1] = \text{က} \quad w[2] = \text{သူ} \quad w[-1] | w[0] = \text{မ} | \text{နှစ်}$
 $w[0] | w[1] = \text{နှစ်} | \text{က}$
 $| \quad w[-2] = \text{မ} \quad w[-1] = \text{နှစ်} \quad w[0] = \text{က} \quad w[1] = \text{သူ} \quad w[2] = \text{ကျွန်}$
 $w[-1] | w[0] = \text{နှစ်} | \text{က} \quad w[0] | w[1] = \text{က} | \text{သူ}$
 $| \quad w[-2] = \text{နှစ်} \quad w[-1] = \text{က} \quad w[0] = \text{သူ} \quad w[1] = \text{ကျွန်} \quad w[2] = \text{မ} \quad w[-1] | w[0] = \text{က} | \text{သူ}$
 $w[0] | w[1] = \text{သူ} | \text{ကျွန်}$
 $< \quad w[-2] = \text{က} \quad w[-1] = \text{သူ} \quad w[0] = \text{ကျွန်} \quad w[1] = \text{မ} \quad w[2] = \text{ကို}$
 $w[-1] | w[0] = \text{သူ} | \text{ကျွန်} \quad w[0] | w[1] = \text{ကျွန်} | \text{မ}$
 $| \quad w[-2] = \text{သူ} \quad w[-1] = \text{ကျွန်} \quad w[0] = \text{မ} \quad w[1] = \text{ကို} \quad w[2] = \text{သင်}$
 $w[-1] | w[0] = \text{ကျွန်} | \text{မ} \quad w[0] | w[1] = \text{မ} | \text{ကို}$
 $| \quad w[-2] = \text{ကျွန်} \quad w[-1] = \text{မ} \quad w[0] = \text{ကို} \quad w[1] = \text{သင်} \quad w[2] = \text{ပေး}$
 $w[-1] | w[0] = \text{မ} | \text{ကို} \quad w[0] | w[1] = \text{ကို} | \text{သင်}$
 $| \quad w[-2] = \text{မ} \quad w[-1] = \text{ကို} \quad w[0] = \text{သင်} \quad w[1] = \text{ပေး} \quad w[2] = \text{တယ်}$
 $w[-1] | w[0] = \text{ကို} | \text{သင်} \quad w[0] | w[1] = \text{သင်} | \text{ပေး}$
 $| \quad w[-2] = \text{ကို} \quad w[-1] = \text{သင်} \quad w[0] = \text{ပေး} \quad w[1] = \text{တယ်} \quad w[2] = \parallel$
 $w[-1] | w[0] = \text{သင်} | \text{ပေး} \quad w[0] | w[1] = \text{ပေး} | \text{တယ်}$
 $| \quad w[-2] = \text{သင်} \quad w[-1] = \text{ပေး} \quad w[0] = \text{တယ်} \quad w[1] = \parallel$
 $w[-1] | w[0] = \text{ပေး} | \text{တယ်} \quad w[0] | w[1] = \text{တယ်} | \parallel$
 $| \quad w[-2] = \text{ပေး} \quad w[-1] = \text{တယ်} \quad w[0] = \parallel \quad w[-1] | w[0] = \text{တယ်} | \parallel \quad \color{red}{\square}$
 $\color{red}{\hookrightarrow} _ _ \text{EOS} _ _$

$< \quad w[0] = \text{ကျွန်} \quad w[1] = \text{တော့} \quad w[2] = \text{ခု} \quad w[0] | w[1] = \text{ကျွန်} | \text{တော့}$
 $_ _ \text{BOS} _ _$
 $| \quad w[-1] = \text{ကျွန်} \quad w[0] = \text{တော့} \quad w[1] = \text{ခု} \quad w[2] = \text{သွား}$
 $w[-1] | w[0] = \text{ကျွန်} | \text{တော့} \quad w[0] | w[1] = \text{တော့} | \text{ခု}$
 $| \quad w[-2] = \text{ကျွန်} \quad w[-1] = \text{တော့} \quad w[0] = \text{ခု} \quad w[1] = \text{သွား} \quad w[2] = \text{ရှာ}$
 $w[-1] | w[0] = \text{တော့} | \text{ခု} \quad w[0] | w[1] = \text{ခု} | \text{သွား}$
 $| \quad w[-2] = \text{တော့} \quad w[-1] = \text{ခု} \quad w[0] = \text{သွား} \quad w[1] = \text{ရှာ} \quad w[2] = \text{မ}$
 $w[-1] | w[0] = \text{ခု} | \text{သွား} \quad w[0] | w[1] = \text{သွား} | \text{ရှာ}$
 $| \quad w[-2] = \text{ခု} \quad w[-1] = \text{သွား} \quad w[0] = \text{ရှာ} \quad w[1] = \text{မ} \quad w[2] = \text{လို့}$
 $w[-1] | w[0] = \text{သွား} | \text{ရှာ} \quad w[0] | w[1] = \text{ရှာ} | \text{မ}$
 $< \quad w[-2] = \text{သွား} \quad w[-1] = \text{ရှာ} \quad w[0] = \text{မ} \quad w[1] = \text{လို့} \quad w[2] = \parallel$
 $w[-1] | w[0] = \text{ရှာ} | \text{မ} \quad w[0] | w[1] = \text{မ} | \text{လို့}$
 $| \quad w[-2] = \text{ရှာ} \quad w[-1] = \text{မ} \quad w[0] = \text{လို့} \quad w[1] = \parallel \quad w[-1] | w[0] = \text{မ} | \text{လို့}$
 $w[0] | w[1] = \text{လို့} | \parallel$

Format changing for training data:

```
[88]: !cat /home/ye/aif2/word-seg/data/myPOS/tag/train.tag3.txt | python ./chunking.  
      ↪py > /home/ye/aif2/word-seg/data/myPOS/tag/train.crfsuite.txt
```

```
[89]: !head -n 30 /home/ye/aif2/word-seg/data/myPOS/tag/train.crfsuite.txt
```

```
['w', 'y']
<      w[0]=၁  w[1]=၉  w[2]=၆  w[0]|w[1]=၁|၉  __BOS__
-      w[-1]=၁ w[0]=၉  w[1]=၆  w[2]=၂  w[-1]|w[0]=၁|၉  w[0]|w[1]=၉|၆
-      w[-2]=၁ w[-1]=၉ w[0]=၆  w[1]=၂  w[2]=၃ w[-1]|w[0]=၉|၆  w[0]|w[1]=၆|၂
|      w[-2]=၉ w[-1]=၆ w[0]=၂  w[1]=၃ w[2]=နှစ်  w[-1]|w[0]=၆|၂
w[0]|w[1]=၂|၃
<      w[-2]=၆ w[-1]=၂ w[0]=၃ w[1]=နှစ်  w[2]=ခန့်  w[-1]|w[0]=၂|၃
w[0]|w[1]=၃|နှစ်
|      w[-2]=၂ w[-1]=၃  w[0]=နှစ်  w[1]=ခန့်  w[2]=မှန်း
w[-1]|w[0]=၃|နှစ်  w[0]|w[1]=နှစ်|ခန့်
<      w[-2]=၃  w[-1]=နှစ်  w[0]=ခန့်  w[1]=မှန်း
w[2]=သန်း  w[-1]|w[0]=နှစ်|ခန့်  w[0]|w[1]=ခန့်|မှန်း
|      w[-2]=နှစ်  w[-1]=ခန့်  w[0]=မှန်း  w[1]=သန်း
w[2]=ခေါင်  w[-1]|w[0]=ခန့်|မှန်း  w[0]|w[1]=မှန်း|သန်း
<      w[-2]=ခန့်  w[-1]=မှန်း  w[0]=သန်း  w[1]=ခေါင်  w[2]=စာ
w[-1]|w[0]=မှန်း|သန်း  w[0]|w[1]=သန်း|ခေါင်
-      w[-2]=မှန်း  w[-1]=သန်း  w[0]=ခေါင်  w[1]=စာ w[2]=ရင်း
w[-1]|w[0]=သန်း|ခေါင်  w[0]|w[1]=ခေါင်|စာ
-      w[-2]=သန်း  w[-1]=ခေါင်  w[0]=စာ w[1]=ရင်း  w[2]=အ
w[-1]|w[0]=ခေါင်|စာ  w[0]|w[1]=စာ|ရင်း
|      w[-2]=ခေါင်  w[-1]=စာ  w[0]=ရင်း  w[1]=အ  w[2]=ရ
w[-1]|w[0]=စာ|ရင်း  w[0]|w[1]=ရင်း|အ
<      w[-2]=စာ  w[-1]=ရင်း  w[0]=အ  w[1]=ရ  w[2]=လူ
w[-1]|w[0]=ရင်း|အ  w[0]|w[1]=အ|ရ
|      w[-2]=ရင်း  w[-1]=အ w[0]=ရ  w[1]=လူ w[2]=ဦး w[-1]|w[0]=အ|ရ
w[0]|w[1]=ရ|လူ
<      w[-2]=အ w[-1]=ရ w[0]=လူ w[1]=ဦး w[2]=ရေ w[-1]|w[0]=ရ|လူ w[0]|w[1]=  
      ↪လူ|ဦး
-      w[-2]=ရ w[-1]=လူ  w[0]=ဦး w[1]=ရေ w[2]=၁  w[-1]|w[0]=လူ|ဦး
w[0]|w[1]=ဦး|ရေ
|      w[-2]=လူ  w[-1]=ဦး  w[0]=ရေ w[1]=၁  w[2]=၁  w[-1]|w[0]=  
      ↪ဦး|ရေ
w[0]|w[1]=ရေ|၁
<      w[-2]=ဦး  w[-1]=ရေ  w[0]=၁  w[1]=၁  w[2]=၅  w[-1]|w[0]=  
      ↪ရေ|၁
w[0]|w[1]=၁|၁
-      w[-2]=ရေ  w[-1]=၁ w[0]=၁  w[1]=၅  w[2]=၉  w[-1]|w[0]=၁|၁
w[0]|w[1]=၁|၅
```

```

-      w[-2]=၁ w[-1]=၁ w[0]=၅ w[1]=၉ w[2]=၃ w[-1]|w[0]=၁|၅ w[0]|w[1]=၁
  ၅|၉
-      w[-2]=၁ w[-1]=၅ w[0]=၉ w[1]=၃ w[2]=၁ w[-1]|w[0]=၅|၉ w[0]|w[1]=၁
  ၉|၃
-      w[-2]=၅ w[-1]=၉ w[0]=၃ w[1]=၁ w[2]=ယောက် w[-1]|w[0]=၉|၃
w[0]|w[1]=၃|၁
|      w[-2]=၉ w[-1]=၃ w[0]=၁ w[1]=ယောက် w[2]=ရှိ w[-1]|w[0]=၃|၁
w[0]|w[1]=၁|ယောက်
|      w[-2]=၃ w[-1]=၁ w[0]=ယောက် w[1]=ရှိ w[2]=သည်
w[-1]|w[0]=၁|ယောက် w[0]|w[1]=ယောက်|ရှိ
|      w[-2]=၁ w[-1]=ယောက် w[0]=ရှိ w[1]=သည်
w[-1]|w[0]=ယောက်|ရှိ w[0]|w[1]=ရှိ|သည်
|      w[-2]=ယောက် w[-1]=ရှိ w[0]=သည် w[-1]|w[0]=ရှိ|သည်
__EOS__

|      w[0]=လူ w[1]=တိုင်း w[2]=တွင် w[0]|w[1]=လူ|တိုင်း: __BOS__
|      w[-1]=လူ w[0]=တိုင်း w[1]=တွင် w[2]=သင့်
w[-1]|w[0]=လူ|တိုင်း: w[0]|w[1]=တိုင်း|တွင်

```

1.16 Training CRF Model for Word Segmentation

```
[90]: !time crfsuite learn -m ws_tag3.model /home/ye/aif2/word-seg/data/myPOS/tag/
      ↪train.crfsuite.txt
```

CRFSuite 0.12.2 Copyright (c) 2007-2013 Naoaki Okazaki

Start time of the training: 2026-07-25T10:06:36Z

Reading the data set(s)

[1] /home/ye/aif2/word-seg/data/myPOS/tag/train.crfsuite.txt

0...1...2...3...4...5...6...7...8...9...10

Number of instances: 43196

Seconds required: 1.797

Statistics the data set(s)

Number of data sets (groups): 1

Number of instances: 43195

Number of items: 776176

Number of attributes: 229827

Number of labels: 4

Feature generation

type: CRF1d

feature.minfreq: 0.000000

feature.possible_states: 0

feature.possible_transitions: 0

0...1...2...3...4...5...6...7...8...9...10
Number of features: 272898
Seconds required: 0.644

L-BFGS optimization
c1: 0.000000
c2: 1.000000
num_memories: 6
max_iterations: 2147483647
epsilon: 0.000010
stop: 10
delta: 0.000010
linesearch: MoreThuente
linesearch.max_iterations: 20

***** Iteration #1 *****
Loss: 807954.639519
Feature norm: 1.000000
Error norm: 201980.900695
Active features: 270313
Line search trials: 1
Line search step: 0.000003
Seconds required for this iteration: 0.226

***** Iteration #2 *****
Loss: 674306.276580
Feature norm: 3.052376
Error norm: 250346.062864
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.110

***** Iteration #3 *****
Loss: 573621.847172
Feature norm: 3.439057
Error norm: 88998.972011
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.111

***** Iteration #4 *****
Loss: 538665.549400
Feature norm: 4.046224
Error norm: 86766.406881
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.112

***** Iteration #5 *****
Loss: 488246.329969
Feature norm: 5.097064
Error norm: 109321.710836
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.113

***** Iteration #6 *****
Loss: 431280.143056
Feature norm: 7.201213
Error norm: 127163.558420
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.113

***** Iteration #7 *****
Loss: 387467.946512
Feature norm: 8.453381
Error norm: 81184.252196
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129

***** Iteration #8 *****
Loss: 336255.015964
Feature norm: 11.675881
Error norm: 26148.245092
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129

***** Iteration #9 *****
Loss: 319636.408741
Feature norm: 13.758644
Error norm: 30025.412165
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.125

***** Iteration #10 *****
Loss: 288994.994613
Feature norm: 18.262237
Error norm: 30469.256709
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #11 *****
Loss: 255861.948368
Feature norm: 25.310903
Error norm: 20690.241078
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #12 *****
Loss: 250680.891134
Feature norm: 33.751359
Error norm: 74091.856371
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #13 *****
Loss: 231192.177337
Feature norm: 35.958347
Error norm: 15901.350460
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #14 *****
Loss: 228883.643656
Feature norm: 36.730348
Error norm: 12019.610737
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #15 *****
Loss: 225229.360735
Feature norm: 38.095753

Error norm: 19480.574810
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #16 *****
Loss: 219480.596835
Feature norm: 39.924495
Error norm: 27088.421981
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #17 *****
Loss: 210655.711024
Feature norm: 42.326182
Error norm: 28789.599037
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #18 *****
Loss: 207832.669306
Feature norm: 42.408634
Error norm: 29122.677092
Active features: 272898
Line search trials: 2
Line search step: 0.139763
Seconds required for this iteration: 0.224

***** Iteration #19 *****
Loss: 198118.364383
Feature norm: 43.296004
Error norm: 15896.657432
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #20 *****
Loss: 192277.539272
Feature norm: 43.097504
Error norm: 9946.438066
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #21 *****
Loss: 187072.017965
Feature norm: 42.883251
Error norm: 14576.436281
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #22 *****
Loss: 181345.221306
Feature norm: 43.374597
Error norm: 14739.523175
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #23 *****
Loss: 176901.901153
Feature norm: 43.564653
Error norm: 24512.413104
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #24 *****
Loss: 173704.264693
Feature norm: 43.909640
Error norm: 9614.808849
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #25 *****
Loss: 171123.083385
Feature norm: 44.743312
Error norm: 10406.688652
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #26 *****
Loss: 169105.620242
Feature norm: 45.661141
Error norm: 15887.587086
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #27 *****
Loss: 164411.934288
Feature norm: 47.994988
Error norm: 21491.889190
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #28 *****
Loss: 157168.476238
Feature norm: 51.875468
Error norm: 21617.378679
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #29 *****
Loss: 154945.844896
Feature norm: 53.896165
Error norm: 24744.712883
Active features: 272898
Line search trials: 2
Line search step: 0.205627
Seconds required for this iteration: 0.222

***** Iteration #30 *****
Loss: 149615.461314
Feature norm: 57.230446
Error norm: 13131.200335
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #31 *****
Loss: 147080.793666
Feature norm: 59.072227

Error norm: 6165.783863
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #32 *****
Loss: 145751.834951
Feature norm: 60.324682
Error norm: 8404.740880
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #33 *****
Loss: 143387.530871
Feature norm: 61.980885
Error norm: 8596.149006
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #34 *****
Loss: 142990.708797
Feature norm: 61.981760
Error norm: 36417.328174
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #35 *****
Loss: 137027.815479
Feature norm: 64.400482
Error norm: 11358.972520
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #36 *****
Loss: 135084.649876
Feature norm: 64.582757
Error norm: 5220.622002
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #37 *****
Loss: 133193.519316
Feature norm: 64.391811
Error norm: 8480.129345
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #38 *****
Loss: 131982.868836
Feature norm: 64.527621
Error norm: 9151.050792
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #39 *****
Loss: 130733.346339
Feature norm: 67.176121
Error norm: 19720.533676
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #40 *****
Loss: 127394.566397
Feature norm: 68.654514
Error norm: 6969.156125
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #41 *****
Loss: 125920.563685
Feature norm: 70.688915
Error norm: 4159.377104
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #42 *****
Loss: 124804.635370
Feature norm: 73.354892
Error norm: 4853.857126
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #43 *****
Loss: 123175.858008
Feature norm: 76.310438
Error norm: 4250.425232
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #44 *****
Loss: 122167.852788
Feature norm: 78.481947
Error norm: 10024.454151
Active features: 272898
Line search trials: 2
Line search step: 0.223603
Seconds required for this iteration: 0.223

***** Iteration #45 *****
Loss: 120004.457281
Feature norm: 82.057666
Error norm: 4478.246175
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #46 *****
Loss: 118585.066497
Feature norm: 84.004939
Error norm: 3760.670348
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #47 *****
Loss: 117238.914900
Feature norm: 85.430172

Error norm: 3671.055276
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #48 *****
Loss: 116560.030483
Feature norm: 85.625631
Error norm: 9665.437058
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #49 *****
Loss: 115342.057162
Feature norm: 85.558156
Error norm: 3886.603964
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #50 *****
Loss: 114197.935212
Feature norm: 84.631484
Error norm: 4521.853491
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.122

***** Iteration #51 *****
Loss: 113296.389665
Feature norm: 84.218659
Error norm: 6227.349057
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.128

***** Iteration #52 *****
Loss: 111972.945950
Feature norm: 83.566505
Error norm: 6977.700676
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #53 *****
Loss: 110545.562117
Feature norm: 83.838307
Error norm: 6660.067498
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #54 *****
Loss: 108983.849238
Feature norm: 84.333879
Error norm: 4736.940710
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #55 *****
Loss: 106868.935318
Feature norm: 85.557969
Error norm: 5251.413045
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #56 *****
Loss: 105916.231624
Feature norm: 86.804251
Error norm: 4792.726938
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #57 *****
Loss: 105329.727758
Feature norm: 87.473240
Error norm: 3731.768447
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #58 *****
Loss: 103691.210920
Feature norm: 89.265085
Error norm: 2653.794004
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #59 *****
Loss: 102185.072972
Feature norm: 91.008212
Error norm: 3200.760622
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #60 *****
Loss: 101535.117703
Feature norm: 92.297764
Error norm: 6890.959999
Active features: 272898
Line search trials: 2
Line search step: 0.374907
Seconds required for this iteration: 0.223

***** Iteration #61 *****
Loss: 100338.159716
Feature norm: 93.546240
Error norm: 3946.829093
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #62 *****
Loss: 99014.886459
Feature norm: 94.983082
Error norm: 2983.784274
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #63 *****
Loss: 97384.639518
Feature norm: 96.946004

Error norm: 3437.439691
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #64 *****
Loss: 96993.773811
Feature norm: 97.424415
Error norm: 5135.963562
Active features: 272898
Line search trials: 2
Line search step: 0.349783
Seconds required for this iteration: 0.222

***** Iteration #65 *****
Loss: 96332.410080
Feature norm: 98.052008
Error norm: 2464.673341
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #66 *****
Loss: 95925.141502
Feature norm: 98.182630
Error norm: 1978.754959
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #67 *****
Loss: 95010.128717
Feature norm: 98.664730
Error norm: 2431.211710
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #68 *****
Loss: 94057.323730
Feature norm: 100.373775
Error norm: 7071.314554
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #69 *****
Loss: 92750.007978
Feature norm: 102.901024
Error norm: 3824.900184
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #70 *****
Loss: 91764.964226
Feature norm: 104.785537
Error norm: 2540.328142
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #71 *****
Loss: 90928.593084
Feature norm: 106.607837
Error norm: 2451.407805
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #72 *****
Loss: 90397.219544
Feature norm: 107.902109
Error norm: 3846.552774
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #73 *****
Loss: 89912.773813
Feature norm: 108.897595
Error norm: 2860.300966
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #74 *****
Loss: 89429.226957
Feature norm: 109.371609
Error norm: 2449.749358
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #75 *****
Loss: 88257.232800
Feature norm: 111.336406
Error norm: 2258.050518
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #76 *****
Loss: 87805.507112
Feature norm: 112.311833
Error norm: 4756.926299
Active features: 272898
Line search trials: 2
Line search step: 0.278980
Seconds required for this iteration: 0.222

***** Iteration #77 *****
Loss: 87098.452094
Feature norm: 114.149920
Error norm: 1868.631726
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #78 *****
Loss: 86853.903797
Feature norm: 114.700484
Error norm: 1618.714042
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #79 *****
Loss: 86292.806043
Feature norm: 115.840523

Error norm: 2011.630817
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #80 *****
Loss: 86012.044530
Feature norm: 116.742264
Error norm: 3541.052181
Active features: 272898
Line search trials: 2
Line search step: 0.504240
Seconds required for this iteration: 0.228

***** Iteration #81 *****
Loss: 85558.386508
Feature norm: 117.355194
Error norm: 1923.410027
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #82 *****
Loss: 85179.778319
Feature norm: 117.584084
Error norm: 2001.226724
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #83 *****
Loss: 84444.134565
Feature norm: 117.853890
Error norm: 2635.391507
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #84 *****
Loss: 84208.767289
Feature norm: 118.197856
Error norm: 3457.022225
Active features: 272898
Line search trials: 2

Line search step: 0.254324
Seconds required for this iteration: 0.222

***** Iteration #85 *****
Loss: 83901.489461
Feature norm: 118.384217
Error norm: 2227.013812
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #86 *****
Loss: 83463.526997
Feature norm: 119.082626
Error norm: 1495.192395
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #87 *****
Loss: 83116.815759
Feature norm: 119.811631
Error norm: 1715.236621
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #88 *****
Loss: 82512.965575
Feature norm: 121.116803
Error norm: 3057.452539
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #89 *****
Loss: 82174.499694
Feature norm: 122.139620
Error norm: 3412.156959
Active features: 272898
Line search trials: 2
Line search step: 0.432026
Seconds required for this iteration: 0.223

***** Iteration #90 *****
Loss: 81768.757612
Feature norm: 122.674336
Error norm: 1720.964293
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.130

***** Iteration #91 *****
Loss: 81516.072350
Feature norm: 122.804627
Error norm: 1359.505578
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #92 *****
Loss: 81304.036556
Feature norm: 122.857110
Error norm: 1565.048067
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #93 *****
Loss: 81070.985115
Feature norm: 123.142380
Error norm: 3879.466639
Active features: 272898
Line search trials: 2
Line search step: 0.457220
Seconds required for this iteration: 0.221

***** Iteration #94 *****
Loss: 80699.745143
Feature norm: 123.484059
Error norm: 2504.343030
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #95 *****
Loss: 80159.278579
Feature norm: 124.484316

Error norm: 1544.265684
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #96 *****
Loss: 79906.508029
Feature norm: 125.126094
Error norm: 1542.621961
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #97 *****
Loss: 79655.484169
Feature norm: 126.256488
Error norm: 4873.872110
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #98 *****
Loss: 79304.255735
Feature norm: 126.818111
Error norm: 1229.027637
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #99 *****
Loss: 79184.054358
Feature norm: 126.691368
Error norm: 1120.534500
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #100 *****
Loss: 78888.967198
Feature norm: 126.705127
Error norm: 1341.250945
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #101 *****
Loss: 78751.097789
Feature norm: 126.937322
Error norm: 3770.144528
Active features: 272898
Line search trials: 2
Line search step: 0.286679
Seconds required for this iteration: 0.222

***** Iteration #102 *****
Loss: 78393.267664
Feature norm: 127.318113
Error norm: 2774.631040
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #103 *****
Loss: 77880.802929
Feature norm: 128.259512
Error norm: 1505.922057
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #104 *****
Loss: 77633.072097
Feature norm: 129.045734
Error norm: 1144.025529
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #105 *****
Loss: 77467.675620
Feature norm: 129.769725
Error norm: 2815.826812
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #106 *****
Loss: 77246.605502
Feature norm: 130.297923
Error norm: 1339.659435
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #107 *****
Loss: 77081.790709
Feature norm: 130.331063
Error norm: 1388.719510
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #108 *****
Loss: 76661.061283
Feature norm: 130.815668
Error norm: 2528.243526
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #109 *****
Loss: 76618.165495
Feature norm: 131.051800
Error norm: 3713.082397
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #110 *****
Loss: 76320.321279
Feature norm: 131.234501
Error norm: 1277.939503
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #111 *****
Loss: 76213.438389
Feature norm: 131.463152

Error norm: 1376.023574
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #112 *****
Loss: 75974.665460
Feature norm: 132.009770
Error norm: 1382.650371
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #113 *****
Loss: 75877.109271
Feature norm: 132.972987
Error norm: 5783.634353
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #114 *****
Loss: 75465.930329
Feature norm: 133.201568
Error norm: 1635.201319
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.121

***** Iteration #115 *****
Loss: 75316.626670
Feature norm: 133.245079
Error norm: 1161.773767
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.137

***** Iteration #116 *****
Loss: 75179.986323
Feature norm: 133.422986
Error norm: 1282.414878
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.135

***** Iteration #117 *****
Loss: 74922.716133
Feature norm: 133.776563
Error norm: 2252.202068
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.135

***** Iteration #118 *****
Loss: 74785.536481
Feature norm: 134.821339
Error norm: 3541.597848
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.128

***** Iteration #119 *****
Loss: 74577.964790
Feature norm: 134.690171
Error norm: 913.389287
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.124

***** Iteration #120 *****
Loss: 74538.961306
Feature norm: 134.693933
Error norm: 1018.627196
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120

***** Iteration #121 *****
Loss: 74422.134443
Feature norm: 134.869691
Error norm: 1105.515091
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.126

***** Iteration #122 *****
Loss: 74350.217620
Feature norm: 135.177782
Error norm: 2749.285982
Active features: 272898
Line search trials: 2
Line search step: 0.217210
Seconds required for this iteration: 0.236

***** Iteration #123 *****
Loss: 74149.441427
Feature norm: 135.671025
Error norm: 1733.484961
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #124 *****
Loss: 73976.637198
Feature norm: 136.166543
Error norm: 981.672860
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #125 *****
Loss: 73807.926025
Feature norm: 136.725128
Error norm: 1062.802012
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #126 *****
Loss: 73637.706280
Feature norm: 137.237897
Error norm: 1449.679992
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #127 *****
Loss: 73547.139449
Feature norm: 137.675937

Error norm: 2162.561913
Active features: 272898
Line search trials: 2
Line search step: 0.414991
Seconds required for this iteration: 0.225

***** Iteration #128 *****
Loss: 73379.908610
Feature norm: 137.999856
Error norm: 1104.144636
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #129 *****
Loss: 73290.236912
Feature norm: 138.153010
Error norm: 894.708881
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #130 *****
Loss: 73193.886409
Feature norm: 138.157833
Error norm: 982.277889
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.126

***** Iteration #131 *****
Loss: 73095.467847
Feature norm: 138.446662
Error norm: 3275.001372
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.125

***** Iteration #132 *****
Loss: 72947.675734
Feature norm: 138.528199
Error norm: 1871.383217
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.131

***** Iteration #133 *****
Loss: 72759.810059
Feature norm: 138.907541
Error norm: 982.169747
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #134 *****
Loss: 72703.862140
Feature norm: 139.113805
Error norm: 1115.169702
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #135 *****
Loss: 72614.565236
Feature norm: 139.852096
Error norm: 2259.858177
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #136 *****
Loss: 72501.570214
Feature norm: 139.950081
Error norm: 1007.017894
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #137 *****
Loss: 72441.566461
Feature norm: 139.899894
Error norm: 1043.864795
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120

***** Iteration #138 *****
Loss: 72289.077506
Feature norm: 140.020565
Error norm: 1185.189058
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #139 *****
Loss: 72056.297697
Feature norm: 140.402037
Error norm: 1077.889721
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #140 *****
Loss: 71977.946705
Feature norm: 140.578403
Error norm: 2812.449277
Active features: 272898
Line search trials: 2
Line search step: 0.347117
Seconds required for this iteration: 0.224

***** Iteration #141 *****
Loss: 71790.568613
Feature norm: 140.934402
Error norm: 1329.320759
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #142 *****
Loss: 71690.736622
Feature norm: 141.081831
Error norm: 693.814695
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #143 *****
Loss: 71596.857563
Feature norm: 141.121842

Error norm: 782.820864
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #144 *****
Loss: 71511.391602
Feature norm: 141.178298
Error norm: 1520.340756
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #145 *****
Loss: 71398.649708
Feature norm: 141.151202
Error norm: 1166.552038
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #146 *****
Loss: 71302.894513
Feature norm: 141.156821
Error norm: 768.817814
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #147 *****
Loss: 71180.569547
Feature norm: 141.271261
Error norm: 790.004652
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #148 *****
Loss: 71138.203488
Feature norm: 141.281484
Error norm: 1210.185918
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #149 *****
Loss: 71079.413345
Feature norm: 141.319508
Error norm: 839.462755
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #150 *****
Loss: 70900.420390
Feature norm: 141.445231
Error norm: 987.756192
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #151 *****
Loss: 70795.831748
Feature norm: 141.533661
Error norm: 1237.906302
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.132

***** Iteration #152 *****
Loss: 70755.033130
Feature norm: 141.562988
Error norm: 1285.131687
Active features: 272898
Line search trials: 2
Line search step: 0.362763
Seconds required for this iteration: 0.262

***** Iteration #153 *****
Loss: 70711.374238
Feature norm: 141.574412
Error norm: 724.565348
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.134

***** Iteration #154 *****
Loss: 70669.947483
Feature norm: 141.569986
Error norm: 685.145303
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.126

***** Iteration #155 *****
Loss: 70588.453893
Feature norm: 141.579218
Error norm: 986.941546
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #156 *****
Loss: 70495.121105
Feature norm: 141.610867
Error norm: 1085.434116
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #157 *****
Loss: 70459.531768
Feature norm: 141.652895
Error norm: 2376.376521
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #158 *****
Loss: 70362.455755
Feature norm: 141.661650
Error norm: 746.717459
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #159 *****
Loss: 70317.996890
Feature norm: 141.676279

Error norm: 785.115932
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #160 *****
Loss: 70273.430869
Feature norm: 141.696250
Error norm: 1009.490597
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #161 *****
Loss: 70194.960496
Feature norm: 141.745102
Error norm: 1322.632250
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #162 *****
Loss: 70153.497713
Feature norm: 141.782072
Error norm: 1090.880836
Active features: 272898
Line search trials: 2
Line search step: 0.431657
Seconds required for this iteration: 0.223

***** Iteration #163 *****
Loss: 70122.027971
Feature norm: 141.808708
Error norm: 545.583571
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #164 *****
Loss: 70101.798709
Feature norm: 141.827755
Error norm: 528.877001
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #165 *****
Loss: 70072.974596
Feature norm: 141.848270
Error norm: 479.049139
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #166 *****
Loss: 70041.829688
Feature norm: 141.856716
Error norm: 1559.833735
Active features: 272898
Line search trials: 2
Line search step: 0.269188
Seconds required for this iteration: 0.223

***** Iteration #167 *****
Loss: 69963.852417
Feature norm: 141.907314
Error norm: 916.470007
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #168 *****
Loss: 69874.782991
Feature norm: 141.944296
Error norm: 586.763451
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #169 *****
Loss: 69799.878314
Feature norm: 141.946951
Error norm: 769.191871
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #170 *****
Loss: 69717.310231
Feature norm: 141.957653
Error norm: 702.822004
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #171 *****
Loss: 69698.358270
Feature norm: 141.946073
Error norm: 919.523951
Active features: 272898
Line search trials: 2
Line search step: 0.224848
Seconds required for this iteration: 0.222

***** Iteration #172 *****
Loss: 69665.247289
Feature norm: 141.926279
Error norm: 639.493280
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #173 *****
Loss: 69632.676296
Feature norm: 141.878381
Error norm: 565.856710
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129

***** Iteration #174 *****
Loss: 69590.987454
Feature norm: 141.816683
Error norm: 1299.858011
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120

***** Iteration #175 *****
Loss: 69541.953208
Feature norm: 141.789700

Error norm: 701.405415
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #176 *****
Loss: 69479.818799
Feature norm: 141.768916
Error norm: 487.052624
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #177 *****
Loss: 69447.483079
Feature norm: 141.767205
Error norm: 524.519143
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #178 *****
Loss: 69371.313560
Feature norm: 141.757897
Error norm: 692.612583
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #179 *****
Loss: 69349.090000
Feature norm: 141.773362
Error norm: 1045.877272
Active features: 272898
Line search trials: 2
Line search step: 0.234804
Seconds required for this iteration: 0.220

***** Iteration #180 *****
Loss: 69297.357704
Feature norm: 141.785678
Error norm: 496.187757
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #181 *****
Loss: 69265.600777
Feature norm: 141.791082
Error norm: 473.735939
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #182 *****
Loss: 69214.020290
Feature norm: 141.803188
Error norm: 616.530706
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #183 *****
Loss: 69195.658546
Feature norm: 141.809618
Error norm: 797.270963
Active features: 272898
Line search trials: 2
Line search step: 0.336597
Seconds required for this iteration: 0.221

***** Iteration #184 *****
Loss: 69161.373624
Feature norm: 141.816097
Error norm: 516.252165
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #185 *****
Loss: 69125.036666
Feature norm: 141.820263
Error norm: 466.681981
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #186 *****
Loss: 69115.820255
Feature norm: 141.859766
Error norm: 2509.222400
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #187 *****
Loss: 69040.707890
Feature norm: 141.852795
Error norm: 718.783274
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.130

***** Iteration #188 *****
Loss: 69028.166553
Feature norm: 141.843940
Error norm: 438.683561
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #189 *****
Loss: 69010.036399
Feature norm: 141.845599
Error norm: 389.000378
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #190 *****
Loss: 68981.758091
Feature norm: 141.854798
Error norm: 695.023225
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #191 *****
Loss: 68953.011894
Feature norm: 141.865018

Error norm: 656.441958
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #192 *****
Loss: 68927.667194
Feature norm: 141.890621
Error norm: 784.082452
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #193 *****
Loss: 68910.632698
Feature norm: 141.888653
Error norm: 494.000623
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #194 *****
Loss: 68887.424176
Feature norm: 141.881298
Error norm: 422.320623
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #195 *****
Loss: 68875.401557
Feature norm: 141.873661
Error norm: 780.290764
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #196 *****
Loss: 68854.526640
Feature norm: 141.863536
Error norm: 771.469022
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #197 *****
Loss: 68811.503542
Feature norm: 141.836990
Error norm: 691.298627
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #198 *****
Loss: 68777.081936
Feature norm: 141.819034
Error norm: 547.488060
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #199 *****
Loss: 68755.257196
Feature norm: 141.808514
Error norm: 550.051425
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #200 *****
Loss: 68726.608938
Feature norm: 141.791749
Error norm: 957.754521
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #201 *****
Loss: 68703.058011
Feature norm: 141.770932
Error norm: 591.628863
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #202 *****
Loss: 68684.661033
Feature norm: 141.771621
Error norm: 362.931611
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #203 *****
Loss: 68666.547704
Feature norm: 141.766030
Error norm: 318.645532
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #204 *****
Loss: 68644.105824
Feature norm: 141.752824
Error norm: 344.653890
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #205 *****
Loss: 68622.161518
Feature norm: 141.733777
Error norm: 1179.414885
Active features: 272898
Line search trials: 2
Line search step: 0.467461
Seconds required for this iteration: 0.224

***** Iteration #206 *****
Loss: 68586.834751
Feature norm: 141.704300
Error norm: 356.397706
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #207 *****
Loss: 68576.773840
Feature norm: 141.699004

Error norm: 293.291471
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #208 *****
Loss: 68555.560953
Feature norm: 141.684674
Error norm: 467.704713
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #209 *****
Loss: 68548.231118
Feature norm: 141.667613
Error norm: 1069.085120
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #210 *****
Loss: 68531.495427
Feature norm: 141.666133
Error norm: 336.519262
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #211 *****
Loss: 68525.509771
Feature norm: 141.663408
Error norm: 329.277995
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #212 *****
Loss: 68514.457741
Feature norm: 141.654556
Error norm: 373.536559
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #213 *****
Loss: 68503.565380
Feature norm: 141.638962
Error norm: 912.842932
Active features: 272898
Line search trials: 2
Line search step: 0.516084
Seconds required for this iteration: 0.227

***** Iteration #214 *****
Loss: 68485.450102
Feature norm: 141.620873
Error norm: 389.942279
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.130

***** Iteration #215 *****
Loss: 68475.986694
Feature norm: 141.611815
Error norm: 240.709884
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129

***** Iteration #216 *****
Loss: 68468.022158
Feature norm: 141.604871
Error norm: 280.638804
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.121

***** Iteration #217 *****
Loss: 68454.655515
Feature norm: 141.595201
Error norm: 354.173949
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #218 *****
Loss: 68443.037409
Feature norm: 141.580787
Error norm: 547.217489
Active features: 272898
Line search trials: 2
Line search step: 0.465573
Seconds required for this iteration: 0.222

***** Iteration #219 *****
Loss: 68427.353723
Feature norm: 141.572163
Error norm: 329.498672
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #220 *****
Loss: 68420.007927
Feature norm: 141.568296
Error norm: 299.658004
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #221 *****
Loss: 68415.056990
Feature norm: 141.566570
Error norm: 384.844876
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #222 *****
Loss: 68409.840576
Feature norm: 141.565336
Error norm: 296.167094
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #223 *****
Loss: 68402.331474
Feature norm: 141.562724

Error norm: 390.995524
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #224 *****
Loss: 68386.816714
Feature norm: 141.554425
Error norm: 450.327364
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #225 *****
Loss: 68369.548688
Feature norm: 141.544300
Error norm: 677.437660
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #226 *****
Loss: 68360.474916
Feature norm: 141.538761
Error norm: 477.289348
Active features: 272898
Line search trials: 2
Line search step: 0.489101
Seconds required for this iteration: 0.224

***** Iteration #227 *****
Loss: 68354.221765
Feature norm: 141.537631
Error norm: 234.842171
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #228 *****
Loss: 68349.363878
Feature norm: 141.537205
Error norm: 256.000751
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #229 *****
Loss: 68344.371355
Feature norm: 141.536850
Error norm: 252.654133
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119

***** Iteration #230 *****
Loss: 68340.333552
Feature norm: 141.537313
Error norm: 733.967383
Active features: 272898
Line search trials: 2
Line search step: 0.281614
Seconds required for this iteration: 0.223

***** Iteration #231 *****
Loss: 68331.037477
Feature norm: 141.537130
Error norm: 508.991563
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #232 *****
Loss: 68318.772490
Feature norm: 141.538224
Error norm: 261.529675
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #233 *****
Loss: 68311.165061
Feature norm: 141.539818
Error norm: 255.306335
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #234 *****
Loss: 68304.955171
Feature norm: 141.542426
Error norm: 379.534251
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #235 *****
Loss: 68302.270434
Feature norm: 141.546036
Error norm: 529.485349
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #236 *****
Loss: 68297.587324
Feature norm: 141.545234
Error norm: 186.624759
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #237 *****
Loss: 68295.238488
Feature norm: 141.545606
Error norm: 178.277582
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #238 *****
Loss: 68291.688637
Feature norm: 141.547270
Error norm: 255.992470
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #239 *****
Loss: 68285.451103
Feature norm: 141.552745

Error norm: 603.278252
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #240 *****
Loss: 68275.856691
Feature norm: 141.561525
Error norm: 433.181477
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #241 *****
Loss: 68267.302418
Feature norm: 141.571465
Error norm: 239.305860
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #242 *****
Loss: 68261.737663
Feature norm: 141.579145
Error norm: 160.236356
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119

***** Iteration #243 *****
Loss: 68257.976660
Feature norm: 141.584400
Error norm: 224.041215
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #244 *****
Loss: 68252.318782
Feature norm: 141.596134
Error norm: 183.226073
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #245 *****
Loss: 68249.905847
Feature norm: 141.607952
Error norm: 736.378651
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #246 *****
Loss: 68242.412054
Feature norm: 141.611061
Error norm: 290.609568
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #247 *****
Loss: 68238.267612
Feature norm: 141.615576
Error norm: 170.229604
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #248 *****
Loss: 68236.120517
Feature norm: 141.620058
Error norm: 220.721567
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #249 *****
Loss: 68235.178102
Feature norm: 141.635144
Error norm: 657.571665
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #250 *****
Loss: 68230.372432
Feature norm: 141.636121
Error norm: 215.846255
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #251 *****
Loss: 68227.252119
Feature norm: 141.639776
Error norm: 195.531563
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #252 *****
Loss: 68223.407803
Feature norm: 141.649122
Error norm: 336.981945
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #253 *****
Loss: 68218.619918
Feature norm: 141.663811
Error norm: 323.150164
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #254 *****
Loss: 68214.038445
Feature norm: 141.689297
Error norm: 358.961129
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #255 *****
Loss: 68208.902530
Feature norm: 141.698741

Error norm: 218.922838
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119

***** Iteration #256 *****
Loss: 68202.749805
Feature norm: 141.711842
Error norm: 262.201281
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #257 *****
Loss: 68199.328868
Feature norm: 141.727698
Error norm: 219.607245
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129

***** Iteration #258 *****
Loss: 68196.279058
Feature norm: 141.734894
Error norm: 172.033177
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.124

***** Iteration #259 *****
Loss: 68191.196856
Feature norm: 141.751742
Error norm: 255.680049
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #260 *****
Loss: 68183.640187
Feature norm: 141.781974
Error norm: 198.452703
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #261 *****
Loss: 68178.307703
Feature norm: 141.809436
Error norm: 294.798038
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119

***** Iteration #262 *****
Loss: 68174.855407
Feature norm: 141.845139
Error norm: 393.502290
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #263 *****
Loss: 68171.837879
Feature norm: 141.838641
Error norm: 151.272210
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #264 *****
Loss: 68170.380415
Feature norm: 141.836894
Error norm: 139.657501
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #265 *****
Loss: 68166.892247
Feature norm: 141.854064
Error norm: 325.313196
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #266 *****
Loss: 68163.589972
Feature norm: 141.882805
Error norm: 313.856414
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #267 *****
Loss: 68161.088450
Feature norm: 141.894360
Error norm: 178.587952
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.123

***** Iteration #268 *****
Loss: 68158.324443
Feature norm: 141.914311
Error norm: 147.911342
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #269 *****
Loss: 68155.780803
Feature norm: 141.929003
Error norm: 196.794852
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #270 *****
Loss: 68154.514330
Feature norm: 141.963202
Error norm: 365.626264
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #271 *****
Loss: 68151.883533
Feature norm: 141.958825

Error norm: 134.804853
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #272 *****
Loss: 68150.967911
Feature norm: 141.954872
Error norm: 136.813524
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #273 *****
Loss: 68147.214712
Feature norm: 141.964045
Error norm: 248.187513
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119

***** Iteration #274 *****
Loss: 68144.986805
Feature norm: 141.985634
Error norm: 367.249143
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #275 *****
Loss: 68141.376045
Feature norm: 142.008265
Error norm: 157.377740
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #276 *****
Loss: 68140.305855
Feature norm: 142.017202
Error norm: 130.629022
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #277 *****
Loss: 68137.923725
Feature norm: 142.037754
Error norm: 94.853403
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #278 *****
Loss: 68137.421075
Feature norm: 142.063830
Error norm: 548.334762
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #279 *****
Loss: 68134.053788
Feature norm: 142.065580
Error norm: 166.390787
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #280 *****
Loss: 68132.828232
Feature norm: 142.063642
Error norm: 111.556670
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #281 *****
Loss: 68131.575956
Feature norm: 142.065278
Error norm: 122.838956
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #282 *****
Loss: 68130.603854
Feature norm: 142.068029
Error norm: 246.055218
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #283 *****
Loss: 68129.082426
Feature norm: 142.080919
Error norm: 100.987922
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #284 *****
Loss: 68127.712600
Feature norm: 142.093016
Error norm: 120.599112
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #285 *****
Loss: 68126.050815
Feature norm: 142.110049
Error norm: 155.831507
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #286 *****
Loss: 68122.358881
Feature norm: 142.146974
Error norm: 202.639684
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #287 *****
Loss: 68121.428995
Feature norm: 142.159502

Error norm: 286.553022
Active features: 272898
Line search trials: 2
Line search step: 0.195042
Seconds required for this iteration: 0.222

***** Iteration #288 *****
Loss: 68118.649067
Feature norm: 142.194422
Error norm: 139.264494
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #289 *****
Loss: 68117.255076
Feature norm: 142.202946
Error norm: 89.335116
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #290 *****
Loss: 68115.179078
Feature norm: 142.212979
Error norm: 102.241501
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #291 *****
Loss: 68114.501955
Feature norm: 142.216069
Error norm: 255.000259
Active features: 272898
Line search trials: 2
Line search step: 0.271573
Seconds required for this iteration: 0.230

***** Iteration #292 *****
Loss: 68113.191656
Feature norm: 142.221574
Error norm: 197.782043
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #293 *****
Loss: 68110.543666
Feature norm: 142.242770
Error norm: 87.631979
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #294 *****
Loss: 68109.361493
Feature norm: 142.259120
Error norm: 108.145104
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #295 *****
Loss: 68108.738683
Feature norm: 142.280168
Error norm: 260.417850
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #296 *****
Loss: 68107.459602
Feature norm: 142.285942
Error norm: 118.344139
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #297 *****
Loss: 68106.556226
Feature norm: 142.288436
Error norm: 130.848005
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #298 *****
Loss: 68105.199366
Feature norm: 142.298660
Error norm: 162.597129
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #299 *****
Loss: 68104.118412
Feature norm: 142.313385
Error norm: 213.771279
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #300 *****
Loss: 68102.982684
Feature norm: 142.333159
Error norm: 132.651080
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129

***** Iteration #301 *****
Loss: 68102.379500
Feature norm: 142.335668
Error norm: 119.279215
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129

***** Iteration #302 *****
Loss: 68101.329071
Feature norm: 142.347834
Error norm: 92.390765
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.125

***** Iteration #303 *****
Loss: 68100.506553
Feature norm: 142.361404

Error norm: 72.846543
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #304 *****
Loss: 68098.872669
Feature norm: 142.394334
Error norm: 157.894246
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #305 *****
Loss: 68097.706789
Feature norm: 142.426539
Error norm: 234.018075
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #306 *****
Loss: 68096.625212
Feature norm: 142.427772
Error norm: 137.868397
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #307 *****
Loss: 68095.512913
Feature norm: 142.429756
Error norm: 87.000860
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #308 *****
Loss: 68094.977643
Feature norm: 142.432459
Error norm: 79.885317
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #309 *****
Loss: 68094.343480
Feature norm: 142.465717
Error norm: 304.959386
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #310 *****
Loss: 68093.225486
Feature norm: 142.463074
Error norm: 74.564349
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #311 *****
Loss: 68092.937807
Feature norm: 142.467174
Error norm: 59.983555
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #312 *****
Loss: 68092.676734
Feature norm: 142.473621
Error norm: 68.764567
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #313 *****
Loss: 68091.594605
Feature norm: 142.493606
Error norm: 101.003782
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #314 *****
Loss: 68090.928205
Feature norm: 142.530766
Error norm: 279.924424
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #315 *****
Loss: 68089.208981
Feature norm: 142.555510
Error norm: 98.057337
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #316 *****
Loss: 68088.677786
Feature norm: 142.549113
Error norm: 66.534084
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119

***** Iteration #317 *****
Loss: 68087.967884
Feature norm: 142.550358
Error norm: 56.285948
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #318 *****
Loss: 68087.353883
Feature norm: 142.552021
Error norm: 144.684048
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #319 *****
Loss: 68086.462257
Feature norm: 142.555582

Error norm: 103.317686
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #320 *****
Loss: 68085.551156
Feature norm: 142.562835
Error norm: 111.216908
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #321 *****
Loss: 68083.974059
Feature norm: 142.573118
Error norm: 78.184542
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #322 *****
Loss: 68082.837450
Feature norm: 142.579998
Error norm: 67.671415
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #323 *****
Loss: 68082.584394
Feature norm: 142.581817
Error norm: 134.896542
Active features: 272898
Line search trials: 2
Line search step: 0.225136
Seconds required for this iteration: 0.222

***** Iteration #324 *****
Loss: 68081.990458
Feature norm: 142.583533
Error norm: 83.213364
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #325 *****
Loss: 68081.396587
Feature norm: 142.584754
Error norm: 60.992673
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #326 *****
Loss: 68080.710854
Feature norm: 142.587096
Error norm: 132.450236
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #327 *****
Loss: 68080.275338
Feature norm: 142.590267
Error norm: 144.502443
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #328 *****
Loss: 68079.934173
Feature norm: 142.590499
Error norm: 77.678997
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #329 *****
Loss: 68079.579671
Feature norm: 142.592035
Error norm: 57.707746
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #330 *****
Loss: 68079.340158
Feature norm: 142.593866
Error norm: 74.986423
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #331 *****
Loss: 68078.945168
Feature norm: 142.598094
Error norm: 84.566282
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #332 *****
Loss: 68078.588977
Feature norm: 142.601675
Error norm: 101.799613
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #333 *****
Loss: 68078.232112
Feature norm: 142.601732
Error norm: 67.979459
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #334 *****
Loss: 68077.276999
Feature norm: 142.602538
Error norm: 99.795583
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #335 *****
Loss: 68076.709858
Feature norm: 142.604507

Error norm: 97.904149
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #336 *****
Loss: 68076.100532
Feature norm: 142.607346
Error norm: 137.124982
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #337 *****
Loss: 68075.499280
Feature norm: 142.609922
Error norm: 70.148164
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #338 *****
Loss: 68075.148586
Feature norm: 142.611527
Error norm: 68.938427
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #339 *****
Loss: 68074.763276
Feature norm: 142.613163
Error norm: 64.132495
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #340 *****
Loss: 68074.399069
Feature norm: 142.615131
Error norm: 166.336866
Active features: 272898
Line search trials: 2

Line search step: 0.288264
Seconds required for this iteration: 0.223

***** Iteration #341 *****
Loss: 68073.635300
Feature norm: 142.618611
Error norm: 100.213586
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #342 *****
Loss: 68072.958851
Feature norm: 142.622300
Error norm: 49.772027
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #343 *****
Loss: 68072.584523
Feature norm: 142.624404
Error norm: 57.053226
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.127

***** Iteration #344 *****
Loss: 68072.390821
Feature norm: 142.626112
Error norm: 106.809368
Active features: 272898
Line search trials: 2
Line search step: 0.501428
Seconds required for this iteration: 0.245

***** Iteration #345 *****
Loss: 68072.084785
Feature norm: 142.627140
Error norm: 67.329365
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #346 *****
Loss: 68071.670043
Feature norm: 142.628504
Error norm: 54.186887
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #347 *****
Loss: 68071.163148
Feature norm: 142.631174
Error norm: 67.768541
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #348 *****
Loss: 68070.824060
Feature norm: 142.634049
Error norm: 133.148415
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #349 *****
Loss: 68070.393222
Feature norm: 142.636167
Error norm: 54.617993
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #350 *****
Loss: 68070.181033
Feature norm: 142.636544
Error norm: 57.915089
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #351 *****
Loss: 68069.674832
Feature norm: 142.638784

Error norm: 50.498498
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #352 *****
Loss: 68069.529051
Feature norm: 142.640139
Error norm: 125.369144
Active features: 272898
Line search trials: 2
Line search step: 0.161222
Seconds required for this iteration: 0.225

***** Iteration #353 *****
Loss: 68069.154562
Feature norm: 142.642826
Error norm: 86.407114
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #354 *****
Loss: 68068.677657
Feature norm: 142.647414
Error norm: 44.398793
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #355 *****
Loss: 68068.392717
Feature norm: 142.650172
Error norm: 42.828133
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #356 *****
Loss: 68068.008381
Feature norm: 142.653896
Error norm: 55.870375
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #357 *****
Loss: 68067.835735
Feature norm: 142.656719
Error norm: 90.344749
Active features: 272898
Line search trials: 2
Line search step: 0.273721
Seconds required for this iteration: 0.223

***** Iteration #358 *****
Loss: 68067.492664
Feature norm: 142.659528
Error norm: 48.070728
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #359 *****
Loss: 68067.242386
Feature norm: 142.661276
Error norm: 39.218841
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #360 *****
Loss: 68066.965691
Feature norm: 142.663141
Error norm: 76.865872
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #361 *****
Loss: 68066.849581
Feature norm: 142.667054
Error norm: 181.818000
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #362 *****
Loss: 68066.476944
Feature norm: 142.666237
Error norm: 67.111521
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #363 *****
Loss: 68066.228757
Feature norm: 142.666718
Error norm: 38.775596
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #364 *****
Loss: 68066.049031
Feature norm: 142.668532
Error norm: 44.195853
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #365 *****
Loss: 68065.846831
Feature norm: 142.672779
Error norm: 77.459612
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #366 *****
Loss: 68065.564600
Feature norm: 142.679517
Error norm: 47.352748
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #367 *****
Loss: 68065.359930
Feature norm: 142.682168

Error norm: 44.734004
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #368 *****
Loss: 68064.894809
Feature norm: 142.690248
Error norm: 41.025878
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #369 *****
Loss: 68064.749809
Feature norm: 142.693275
Error norm: 126.057526
Active features: 272898
Line search trials: 2
Line search step: 0.253726
Seconds required for this iteration: 0.222

***** Iteration #370 *****
Loss: 68064.423955
Feature norm: 142.700182
Error norm: 74.757972
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #371 *****
Loss: 68064.181600
Feature norm: 142.705740
Error norm: 32.301501
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #372 *****
Loss: 68064.073981
Feature norm: 142.707303
Error norm: 32.387120
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #373 *****
Loss: 68063.918858
Feature norm: 142.709493
Error norm: 50.658451
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #374 *****
Loss: 68063.804447
Feature norm: 142.716065
Error norm: 125.536297
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #375 *****
Loss: 68063.552056
Feature norm: 142.716121
Error norm: 50.208497
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #376 *****
Loss: 68063.397481
Feature norm: 142.718249
Error norm: 31.672530
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #377 *****
Loss: 68063.316651
Feature norm: 142.720712
Error norm: 34.109032
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #378 *****
Loss: 68063.256293
Feature norm: 142.723305
Error norm: 81.250721
Active features: 272898
Line search trials: 2
Line search step: 0.291201
Seconds required for this iteration: 0.222

***** Iteration #379 *****
Loss: 68063.128354
Feature norm: 142.726580
Error norm: 65.855095
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #380 *****
Loss: 68062.795869
Feature norm: 142.736905
Error norm: 34.275831
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #381 *****
Loss: 68062.611849
Feature norm: 142.743046
Error norm: 34.170252
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #382 *****
Loss: 68062.461469
Feature norm: 142.749758
Error norm: 42.474571
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

***** Iteration #383 *****
Loss: 68062.358259
Feature norm: 142.752548

Error norm: 56.375861
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #384 *****
Loss: 68062.262162
Feature norm: 142.753227
Error norm: 30.264393
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #385 *****
Loss: 68062.157692
Feature norm: 142.755282
Error norm: 30.046160
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129

***** Iteration #386 *****
Loss: 68062.098487
Feature norm: 142.756886
Error norm: 36.321762
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #387 *****
Loss: 68061.993822
Feature norm: 142.761548
Error norm: 36.117598
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #388 *****
Loss: 68061.815052
Feature norm: 142.770827
Error norm: 36.374475
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #389 *****
Loss: 68061.622790
Feature norm: 142.783567
Error norm: 45.444694
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #390 *****
Loss: 68061.489105
Feature norm: 142.791022
Error norm: 36.082361
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #391 *****
Loss: 68061.406160
Feature norm: 142.794592
Error norm: 33.783946
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #392 *****
Loss: 68061.326555
Feature norm: 142.795553
Error norm: 28.222472
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #393 *****
Loss: 68061.202671
Feature norm: 142.800340
Error norm: 26.309925
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #394 *****
Loss: 68061.025583
Feature norm: 142.810934
Error norm: 41.152793
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118

***** Iteration #395 *****
Loss: 68060.974004
Feature norm: 142.814793
Error norm: 42.071557
Active features: 272898
Line search trials: 2
Line search step: 0.256415
Seconds required for this iteration: 0.221

***** Iteration #396 *****
Loss: 68060.912199
Feature norm: 142.819336
Error norm: 22.529407
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #397 *****
Loss: 68060.866786
Feature norm: 142.821915
Error norm: 20.818644
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #398 *****
Loss: 68060.759265
Feature norm: 142.828672
Error norm: 36.058399
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #399 *****
Loss: 68060.660560
Feature norm: 142.839669

Error norm: 53.622593
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #400 *****
Loss: 68060.558542
Feature norm: 142.843771
Error norm: 34.938868
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #401 *****
Loss: 68060.499828
Feature norm: 142.845371
Error norm: 20.903432
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #402 *****
Loss: 68060.458321
Feature norm: 142.847303
Error norm: 19.080213
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #403 *****
Loss: 68060.369869
Feature norm: 142.853299
Error norm: 30.149118
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #404 *****
Loss: 68060.272008
Feature norm: 142.863373
Error norm: 39.061599
Active features: 272898
Line search trials: 1

Line search step: 1.000000
Seconds required for this iteration: 0.114

***** Iteration #405 *****
Loss: 68060.229720
Feature norm: 142.878866
Error norm: 84.547530
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

***** Iteration #406 *****
Loss: 68060.141685
Feature norm: 142.875460
Error norm: 26.243049
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

***** Iteration #407 *****
Loss: 68060.111293
Feature norm: 142.875790
Error norm: 20.366081
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.123

***** Iteration #408 *****
Loss: 68060.087704
Feature norm: 142.877666
Error norm: 24.899780
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

L-BFGS terminated with the stopping criteria
Total seconds required for training: 51.371

Storing the model
Number of active features: 272898 (272898)
Number of active attributes: 229827 (229827)
Number of active labels: 4 (4)
Writing labels
Writing attributes

```
Writing feature references for transitions
Writing feature references for attributes
Seconds required: 0.050
```

End time of the training: 2026-07-25T10:07:30Z

```
real    0m53.922s
user    0m53.862s
sys     0m0.056s
```

```
[91]: !time crfsuite learn -e2 /home/ye/aif2/word-seg/data/myPOS/tag/train.crfsuite.
      ↪txt /home/ye/aif2/word-seg/data/myPOS/tag/test.crfsuite.txt
```

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Start time of the training: 2026-07-25T10:12:53Z

```
Reading the data set(s)
[1] /home/ye/aif2/word-seg/data/myPOS/tag/train.crfsuite.txt
0...1...2...3...4...5...6...7...8...9...10
Number of instances: 43196
Seconds required: 1.806
[2] /home/ye/aif2/word-seg/data/myPOS/tag/test.crfsuite.txt
0...1...2...3...4...5...6...7...8...9...10
Number of instances: 1000
Seconds required: 0.048
```

```
Statistics the data set(s)
Number of data sets (groups): 2
Number of instances: 44194
Number of items: 796165
Number of attributes: 230564
Number of labels: 4
```

Holdout group: 2

```
Feature generation
type: CRF1d
feature.minfreq: 0.000000
feature.possible_states: 0
feature.possible_transitions: 0
0...1...2...3...4...5...6...7...8...9...10
Number of features: 272898
Seconds required: 0.662
```

```
L-BFGS optimization
c1: 0.000000
```

c2: 1.000000
num_memories: 6
max_iterations: 2147483647
epsilon: 0.000010
stop: 10
delta: 0.000010
linesearch: MoreThuente
linesearch.max_iterations: 20

***** Iteration #1 *****

Loss: 807954.639519
Feature norm: 1.000000
Error norm: 201980.900695
Active features: 270313
Line search trials: 1
Line search step: 0.000003
Seconds required for this iteration: 0.226
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (0, 0, 4361) (0.0000, 0.0000, 0.0000)
 -: (0, 0, 2166) (0.0000, 0.0000, 0.0000)
 |: (13461, 19989, 13461) (0.6734, 1.0000, 0.8048)
Macro-average precision, recall, F1: (0.168355, 0.250000, 0.201211)
Item accuracy: 13461 / 19989 (0.6734)
Instance accuracy: 55 / 999 (0.0551)

***** Iteration #2 *****

Loss: 674306.276580
Feature norm: 3.052376
Error norm: 250346.062864
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.111
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (0, 0, 4361) (0.0000, 0.0000, 0.0000)
 -: (0, 0, 2166) (0.0000, 0.0000, 0.0000)
 |: (13461, 19989, 13461) (0.6734, 1.0000, 0.8048)
Macro-average precision, recall, F1: (0.168355, 0.250000, 0.201211)
Item accuracy: 13461 / 19989 (0.6734)
Instance accuracy: 55 / 999 (0.0551)

***** Iteration #3 *****

Loss: 573621.847172
Feature norm: 3.439057
Error norm: 88998.972011
Active features: 272898

Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.112
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (848, 943, 4361) (0.8993, 0.1945, 0.3198)
 -: (0, 0, 2166) (0.0000, 0.0000, 0.0000)
 |: (13461, 19046, 13461) (0.7068, 1.0000, 0.8282)
 Macro-average precision, recall, F1: (0.401505, 0.298613, 0.286987)
 Item accuracy: 14309 / 19989 (0.7158)
 Instance accuracy: 84 / 999 (0.0841)

***** Iteration #4 *****
 Loss: 538665.549400
 Feature norm: 4.046224
 Error norm: 86766.406881
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.133
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (1606, 2014, 4361) (0.7974, 0.3683, 0.5038)
 -: (0, 0, 2166) (0.0000, 0.0000, 0.0000)
 |: (13175, 17975, 13461) (0.7330, 0.9788, 0.8382)
 Macro-average precision, recall, F1: (0.382595, 0.336754, 0.335514)
 Item accuracy: 14781 / 19989 (0.7395)
 Instance accuracy: 90 / 999 (0.0901)

***** Iteration #5 *****
 Loss: 488246.329969
 Feature norm: 5.097064
 Error norm: 109321.710836
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.132
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3651, 6766, 4361) (0.5396, 0.8372, 0.6562)
 -: (0, 0, 2166) (0.0000, 0.0000, 0.0000)
 |: (11151, 13223, 13461) (0.8433, 0.8284, 0.8358)
 Macro-average precision, recall, F1: (0.345728, 0.416397, 0.373006)
 Item accuracy: 14802 / 19989 (0.7405)
 Instance accuracy: 81 / 999 (0.0811)

***** Iteration #6 *****
 Loss: 431280.143056

Feature norm: 7.201213
 Error norm: 127163.558420
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.124
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3568, 5770, 4361) (0.6184, 0.8182, 0.7044)
 -: (810, 1751, 2166) (0.4626, 0.3740, 0.4136)
 |: (11001, 12468, 13461) (0.8823, 0.8172, 0.8485)
 Macro-average precision, recall, F1: (0.490826, 0.502343, 0.491626)
 Item accuracy: 15379 / 19989 (0.7694)
 Instance accuracy: 110 / 999 (0.1101)

***** Iteration #7 *****
 Loss: 387467.946512
 Feature norm: 8.453381
 Error norm: 81184.252196
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.119
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3078, 4393, 4361) (0.7007, 0.7058, 0.7032)
 -: (482, 977, 2166) (0.4933, 0.2225, 0.3067)
 |: (12122, 14619, 13461) (0.8292, 0.9005, 0.8634)
 Macro-average precision, recall, F1: (0.505801, 0.457215, 0.468331)
 Item accuracy: 15682 / 19989 (0.7845)
 Instance accuracy: 101 / 999 (0.1011)

***** Iteration #8 *****
 Loss: 336255.015964
 Feature norm: 11.675881
 Error norm: 26148.245092
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (2306, 2890, 4361) (0.7979, 0.5288, 0.6361)
 -: (788, 1413, 2166) (0.5577, 0.3638, 0.4403)
 |: (12551, 15686, 13461) (0.8001, 0.9324, 0.8612)
 Macro-average precision, recall, F1: (0.538936, 0.456245, 0.484404)
 Item accuracy: 15645 / 19989 (0.7827)
 Instance accuracy: 87 / 999 (0.0871)

```

***** Iteration #9 *****
Loss: 319636.408741
Feature norm: 13.758644
Error norm: 30025.412165
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (2457, 3075, 4361) (0.7990, 0.5634, 0.6608)
  -: (819, 1473, 2166) (0.5560, 0.3781, 0.4501)
  |: (12538, 15441, 13461) (0.8120, 0.9314, 0.8676)
Macro-average precision, recall, F1: (0.541757, 0.468238, 0.494646)
Item accuracy: 15814 / 19989 (0.7911)
Instance accuracy: 96 / 999 (0.0961)

```

```

***** Iteration #10 *****
Loss: 288994.994613
Feature norm: 18.262237
Error norm: 30469.256709
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (2875, 3698, 4361) (0.7774, 0.6593, 0.7135)
  -: (820, 1237, 2166) (0.6629, 0.3786, 0.4819)
  |: (12589, 15054, 13461) (0.8363, 0.9352, 0.8830)
Macro-average precision, recall, F1: (0.569149, 0.493263, 0.519597)
Item accuracy: 16284 / 19989 (0.8146)
Instance accuracy: 135 / 999 (0.1351)

```

```

***** Iteration #11 *****
Loss: 255861.948368
Feature norm: 25.310903
Error norm: 20690.241078
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (3221, 4123, 4361) (0.7812, 0.7386, 0.7593)
  -: (1119, 1875, 2166) (0.5968, 0.5166, 0.5538)
  |: (12299, 13991, 13461) (0.8791, 0.9137, 0.8960)

```

Macro-average precision, recall, F1: (0.564273, 0.542222, 0.552293)
Item accuracy: 16639 / 19989 (0.8324)
Instance accuracy: 203 / 999 (0.2032)

***** Iteration #12 *****

Loss: 250680.891134
Feature norm: 33.751359
Error norm: 74091.856371
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (2818, 3253, 4361) (0.8663, 0.6462, 0.7402)
 -: (1263, 1873, 2166) (0.6743, 0.5831, 0.6254)
 |: (12752, 14863, 13461) (0.8580, 0.9473, 0.9004)
Macro-average precision, recall, F1: (0.599642, 0.544153, 0.566514)
Item accuracy: 16833 / 19989 (0.8421)
Instance accuracy: 234 / 999 (0.2342)

***** Iteration #13 *****

Loss: 231192.177337
Feature norm: 35.958347
Error norm: 15901.350460
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3348, 4241, 4361) (0.7894, 0.7677, 0.7784)
 -: (1213, 1790, 2166) (0.6777, 0.5600, 0.6132)
 |: (12444, 13958, 13461) (0.8915, 0.9244, 0.9077)
Macro-average precision, recall, F1: (0.589655, 0.563045, 0.574840)
Item accuracy: 17005 / 19989 (0.8507)
Instance accuracy: 252 / 999 (0.2523)

***** Iteration #14 *****

Loss: 228883.643656
Feature norm: 36.730348
Error norm: 12019.610737
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)


```

    <: (3444, 4459, 4361) (0.7724, 0.7897, 0.7810)
    -: (1229, 1821, 2166) (0.6749, 0.5674, 0.6165)
    |: (12343, 13709, 13461) (0.9004, 0.9169, 0.9086)
Macro-average precision, recall, F1: (0.586908, 0.568519, 0.576508)
Item accuracy: 17016 / 19989 (0.8513)
Instance accuracy: 261 / 999 (0.2613)

***** Iteration #15 *****
Loss: 225229.360735
Feature norm: 38.095753
Error norm: 19480.574810
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (3532, 4632, 4361) (0.7625, 0.8099, 0.7855)
    -: (1300, 1937, 2166) (0.6711, 0.6002, 0.6337)
    |: (12212, 13420, 13461) (0.9100, 0.9072, 0.9086)
Macro-average precision, recall, F1: (0.585912, 0.579326, 0.581945)
Item accuracy: 17044 / 19989 (0.8527)
Instance accuracy: 263 / 999 (0.2633)

***** Iteration #16 *****
Loss: 219480.596835
Feature norm: 39.924495
Error norm: 27088.421981
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (3590, 4737, 4361) (0.7579, 0.8232, 0.7892)
    -: (1382, 2044, 2166) (0.6761, 0.6380, 0.6565)
    |: (12138, 13208, 13461) (0.9190, 0.9017, 0.9103)
Macro-average precision, recall, F1: (0.588244, 0.590741, 0.588997)
Item accuracy: 17110 / 19989 (0.8560)
Instance accuracy: 268 / 999 (0.2683)

***** Iteration #17 *****
Loss: 210655.711024
Feature norm: 42.326182
Error norm: 28789.599037
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```

Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3632, 4768, 4361) (0.7617, 0.8328, 0.7957)
 -: (1431, 2117, 2166) (0.6760, 0.6607, 0.6682)
 |: (12128, 13104, 13461) (0.9255, 0.9010, 0.9131)
Macro-average precision, recall, F1: (0.590805, 0.598619, 0.594253)
Item accuracy: 17191 / 19989 (0.8600)
Instance accuracy: 272 / 999 (0.2723)

***** Iteration #18 *****
Loss: 207832.669306
Feature norm: 42.408634
Error norm: 29122.677092
Active features: 272898
Line search trials: 2
Line search step: 0.139763
Seconds required for this iteration: 0.223
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3673, 4852, 4361) (0.7570, 0.8422, 0.7974)
 -: (1292, 1770, 2166) (0.7299, 0.5965, 0.6565)
 |: (12274, 13367, 13461) (0.9182, 0.9118, 0.9150)
Macro-average precision, recall, F1: (0.601296, 0.587637, 0.592217)
Item accuracy: 17239 / 19989 (0.8624)
Instance accuracy: 272 / 999 (0.2723)

***** Iteration #19 *****
Loss: 198118.364383
Feature norm: 43.296004
Error norm: 15896.657432
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3627, 4650, 4361) (0.7800, 0.8317, 0.8050)
 -: (1403, 1979, 2166) (0.7089, 0.6477, 0.6770)
 |: (12346, 13360, 13461) (0.9241, 0.9172, 0.9206)
Macro-average precision, recall, F1: (0.603261, 0.599149, 0.600650)
Item accuracy: 17376 / 19989 (0.8693)
Instance accuracy: 283 / 999 (0.2833)

***** Iteration #20 *****
Loss: 192277.539272
Feature norm: 43.097504
Error norm: 9946.438066

Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3630, 4558, 4361) (0.7964, 0.8324, 0.8140)
 -: (1435, 1940, 2166) (0.7397, 0.6625, 0.6990)
 |: (12460, 13491, 13461) (0.9236, 0.9256, 0.9246)
 Macro-average precision, recall, F1: (0.614918, 0.605132, 0.609394)
 Item accuracy: 17525 / 19989 (0.8767)
 Instance accuracy: 298 / 999 (0.2983)

***** Iteration #21 *****
 Loss: 187072.017965
 Feature norm: 42.883251
 Error norm: 14576.436281
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.118
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3609, 4392, 4361) (0.8217, 0.8276, 0.8246)
 -: (1492, 1963, 2166) (0.7601, 0.6888, 0.7227)
 |: (12589, 13634, 13461) (0.9234, 0.9352, 0.9292)
 Macro-average precision, recall, F1: (0.626284, 0.612903, 0.619143)
 Item accuracy: 17690 / 19989 (0.8850)
 Instance accuracy: 304 / 999 (0.3043)

***** Iteration #22 *****
 Loss: 181345.221306
 Feature norm: 43.374597
 Error norm: 14739.523175
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3708, 4608, 4361) (0.8047, 0.8503, 0.8268)
 -: (1514, 2000, 2166) (0.7570, 0.6990, 0.7268)
 |: (12469, 13381, 13461) (0.9318, 0.9263, 0.9291)
 Macro-average precision, recall, F1: (0.623383, 0.618888, 0.620688)
 Item accuracy: 17691 / 19989 (0.8850)
 Instance accuracy: 313 / 999 (0.3133)

***** Iteration #23 *****

Loss: 176901.901153
Feature norm: 43.564653
Error norm: 24512.413104
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3582, 4223, 4361) (0.8482, 0.8214, 0.8346)
 -: (1468, 1922, 2166) (0.7638, 0.6777, 0.7182)
 |: (12740, 13844, 13461) (0.9203, 0.9464, 0.9332)
Macro-average precision, recall, F1: (0.633064, 0.611389, 0.621484)
Item accuracy: 17790 / 19989 (0.8900)
Instance accuracy: 340 / 999 (0.3403)

***** Iteration #24 *****

Loss: 173704.264693
Feature norm: 43.909640
Error norm: 9614.808849
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3680, 4439, 4361) (0.8290, 0.8438, 0.8364)
 -: (1472, 1937, 2166) (0.7599, 0.6796, 0.7175)
 |: (12630, 13613, 13461) (0.9278, 0.9383, 0.9330)
Macro-average precision, recall, F1: (0.629186, 0.615426, 0.621721)
Item accuracy: 17782 / 19989 (0.8896)
Instance accuracy: 340 / 999 (0.3403)

***** Iteration #25 *****

Loss: 171123.083385
Feature norm: 44.743312
Error norm: 10406.688652
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3748, 4585, 4361) (0.8174, 0.8594, 0.8379)
 -: (1501, 1954, 2166) (0.7682, 0.6930, 0.7286)
 |: (12557, 13450, 13461) (0.9336, 0.9328, 0.9332)
Macro-average precision, recall, F1: (0.629806, 0.621315, 0.624945)
Item accuracy: 17806 / 19989 (0.8908)

Instance accuracy: 344 / 999 (0.3443)

***** Iteration #26 *****

Loss: 169105.620242

Feature norm: 45.661141

Error norm: 15887.587086

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.117

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (3766, 4638, 4361) (0.8120, 0.8636, 0.8370)

-: (1521, 1990, 2166) (0.7643, 0.7022, 0.7320)

|: (12519, 13361, 13461) (0.9370, 0.9300, 0.9335)

Macro-average precision, recall, F1: (0.628323, 0.623950, 0.625606)

Item accuracy: 17806 / 19989 (0.8908)

Instance accuracy: 345 / 999 (0.3453)

***** Iteration #27 *****

Loss: 164411.934288

Feature norm: 47.994988

Error norm: 21491.889190

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (3805, 4691, 4361) (0.8111, 0.8725, 0.8407)

-: (1563, 2007, 2166) (0.7788, 0.7216, 0.7491)

|: (12512, 13291, 13461) (0.9414, 0.9295, 0.9354)

Macro-average precision, recall, F1: (0.632823, 0.630903, 0.631302)

Item accuracy: 17880 / 19989 (0.8945)

Instance accuracy: 355 / 999 (0.3554)

***** Iteration #28 *****

Loss: 157168.476238

Feature norm: 51.875468

Error norm: 21617.378679

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.114

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (3833, 4680, 4361) (0.8190, 0.8789, 0.8479)

-: (1611, 2079, 2166) (0.7749, 0.7438, 0.7590)

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|: (12518, 13230, 13461) (0.9462, 0.9299, 0.9380)
Macro-average precision, recall, F1: (0.635023, 0.638160, 0.636230)
Item accuracy: 17962 / 19989 (0.8986)
Instance accuracy: 373 / 999 (0.3734)

***** Iteration #29 *****
Loss: 154945.844896
Feature norm: 53.896165
Error norm: 24744.712883
Active features: 272898
Line search trials: 2
Line search step: 0.205627
Seconds required for this iteration: 0.227
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (3868, 4787, 4361) (0.8080, 0.8870, 0.8456)
  -: (1520, 1867, 2166) (0.8141, 0.7018, 0.7538)
  |: (12562, 13335, 13461) (0.9420, 0.9332, 0.9376)
Macro-average precision, recall, F1: (0.641049, 0.630480, 0.634258)
Item accuracy: 17950 / 19989 (0.8980)
Instance accuracy: 371 / 999 (0.3714)

***** Iteration #30 *****
Loss: 149615.461314
Feature norm: 57.230446
Error norm: 13131.200335
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (3844, 4671, 4361) (0.8230, 0.8814, 0.8512)
  -: (1576, 1986, 2166) (0.7936, 0.7276, 0.7592)
  |: (12586, 13332, 13461) (0.9440, 0.9350, 0.9395)
Macro-average precision, recall, F1: (0.640137, 0.636014, 0.637462)
Item accuracy: 18006 / 19989 (0.9008)
Instance accuracy: 370 / 999 (0.3704)

***** Iteration #31 *****
Loss: 147080.793666
Feature norm: 59.072227
Error norm: 6165.783863
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):

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['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3794, 4505, 4361) (0.8422, 0.8700, 0.8559)
-: (1606, 2035, 2166) (0.7892, 0.7415, 0.7646)
|: (12671, 13449, 13461) (0.9422, 0.9413, 0.9417)
Macro-average precision, recall, F1: (0.643379, 0.638189, 0.640541)
Item accuracy: 18071 / 19989 (0.9040)
Instance accuracy: 386 / 999 (0.3864)

***** Iteration #32 *****
Loss: 145751.834951
Feature norm: 60.324682
Error norm: 8404.740880
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3775, 4466, 4361) (0.8453, 0.8656, 0.8553)
-: (1620, 2068, 2166) (0.7834, 0.7479, 0.7652)
|: (12673, 13455, 13461) (0.9419, 0.9415, 0.9417)
Macro-average precision, recall, F1: (0.642630, 0.638753, 0.640559)
Item accuracy: 18068 / 19989 (0.9039)
Instance accuracy: 389 / 999 (0.3894)

***** Iteration #33 *****
Loss: 143387.530871
Feature norm: 61.980885
Error norm: 8596.149006
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3797, 4469, 4361) (0.8496, 0.8707, 0.8600)
-: (1644, 2114, 2166) (0.7777, 0.7590, 0.7682)
|: (12661, 13406, 13461) (0.9444, 0.9406, 0.9425)
Macro-average precision, recall, F1: (0.642933, 0.642561, 0.642685)
Item accuracy: 18102 / 19989 (0.9056)
Instance accuracy: 395 / 999 (0.3954)

***** Iteration #34 *****
Loss: 142990.708797
Feature norm: 61.981760
Error norm: 36417.328174
Active features: 272898
Line search trials: 1

```

Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3715, 4215, 4361) (0.8814, 0.8519, 0.8664)
 -: (1585, 1911, 2166) (0.8294, 0.7318, 0.7775)
 |: (12929, 13863, 13461) (0.9326, 0.9605, 0.9463)
Macro-average precision, recall, F1: (0.660853, 0.636028, 0.647563)
Item accuracy: 18229 / 19989 (0.9120)
Instance accuracy: 416 / 999 (0.4164)

***** Iteration #35 *****

Loss: 137027.815479
Feature norm: 64.400482
Error norm: 11358.972520
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3828, 4463, 4361) (0.8577, 0.8778, 0.8676)
 -: (1647, 2076, 2166) (0.7934, 0.7604, 0.7765)
 |: (12728, 13450, 13461) (0.9463, 0.9455, 0.9459)
Macro-average precision, recall, F1: (0.649348, 0.645929, 0.647522)
Item accuracy: 18203 / 19989 (0.9107)
Instance accuracy: 405 / 999 (0.4054)

***** Iteration #36 *****

Loss: 135084.649876
Feature norm: 64.582757
Error norm: 5220.622002
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.127
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3879, 4548, 4361) (0.8529, 0.8895, 0.8708)
 -: (1663, 2099, 2166) (0.7923, 0.7678, 0.7798)
 |: (12684, 13342, 13461) (0.9507, 0.9423, 0.9465)
Macro-average precision, recall, F1: (0.648967, 0.649882, 0.649275)
Item accuracy: 18226 / 19989 (0.9118)
Instance accuracy: 411 / 999 (0.4114)

***** Iteration #37 *****

Loss: 133193.519316
Feature norm: 64.391811

Error norm: 8480.129345
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3909, 4603, 4361) (0.8492, 0.8964, 0.8722)
-: (1676, 2110, 2166) (0.7943, 0.7738, 0.7839)
|: (12661, 13276, 13461) (0.9537, 0.9406, 0.9471)
Macro-average precision, recall, F1: (0.649304, 0.652675, 0.650786)
Item accuracy: 18246 / 19989 (0.9128)
Instance accuracy: 416 / 999 (0.4164)

***** Iteration #38 *****
Loss: 131982.868836
Feature norm: 64.527621
Error norm: 9151.050792
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3902, 4580, 4361) (0.8520, 0.8947, 0.8728)
-: (1707, 2158, 2166) (0.7910, 0.7881, 0.7895)
|: (12655, 13251, 13461) (0.9550, 0.9401, 0.9475)
Macro-average precision, recall, F1: (0.649499, 0.655740, 0.652473)
Item accuracy: 18264 / 19989 (0.9137)
Instance accuracy: 417 / 999 (0.4174)

***** Iteration #39 *****
Loss: 130733.346339
Feature norm: 67.176121
Error norm: 19720.533676
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3943, 4686, 4361) (0.8414, 0.9042, 0.8717)
-: (1602, 1891, 2166) (0.8472, 0.7396, 0.7897)
|: (12744, 13412, 13461) (0.9502, 0.9467, 0.9485)
Macro-average precision, recall, F1: (0.659702, 0.647624, 0.652469)
Item accuracy: 18289 / 19989 (0.9150)
Instance accuracy: 418 / 999 (0.4184)

```

***** Iteration #40 *****
Loss: 127394.566397
Feature norm: 68.654514
Error norm: 6969.156125
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (3909, 4554, 4361) (0.8584, 0.8964, 0.8769)
  -: (1704, 2097, 2166) (0.8126, 0.7867, 0.7994)
  |: (12730, 13338, 13461) (0.9544, 0.9457, 0.9500)
Macro-average precision, recall, F1: (0.656343, 0.657188, 0.656605)
Item accuracy: 18343 / 19989 (0.9177)
Instance accuracy: 437 / 999 (0.4374)

```

```

***** Iteration #41 *****
Loss: 125920.563685
Feature norm: 70.688915
Error norm: 4159.377104
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (3889, 4497, 4361) (0.8648, 0.8918, 0.8781)
  -: (1738, 2155, 2166) (0.8065, 0.8024, 0.8044)
  |: (12738, 13337, 13461) (0.9551, 0.9463, 0.9507)
Macro-average precision, recall, F1: (0.656596, 0.660114, 0.658297)
Item accuracy: 18365 / 19989 (0.9188)
Instance accuracy: 444 / 999 (0.4444)

```

```

***** Iteration #42 *****
Loss: 124804.635370
Feature norm: 73.354892
Error norm: 4853.857126
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (3882, 4467, 4361) (0.8690, 0.8902, 0.8795)
  -: (1748, 2162, 2166) (0.8085, 0.8070, 0.8078)
  |: (12761, 13360, 13461) (0.9552, 0.9480, 0.9516)
Macro-average precision, recall, F1: (0.658179, 0.661295, 0.659701)

```

Item accuracy: 18391 / 19989 (0.9201)
Instance accuracy: 447 / 999 (0.4474)

***** Iteration #43 *****

Loss: 123175.858008
Feature norm: 76.310438
Error norm: 4250.425232
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3906, 4483, 4361) (0.8713, 0.8957, 0.8833)
 -: (1767, 2193, 2166) (0.8057, 0.8158, 0.8107)
 |: (12752, 13313, 13461) (0.9579, 0.9473, 0.9526)
Macro-average precision, recall, F1: (0.658724, 0.664696, 0.661653)
Item accuracy: 18425 / 19989 (0.9218)
Instance accuracy: 451 / 999 (0.4515)

***** Iteration #44 *****

Loss: 122167.852788
Feature norm: 78.481947
Error norm: 10024.454151
Active features: 272898
Line search trials: 2
Line search step: 0.223603
Seconds required for this iteration: 0.221
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3891, 4419, 4361) (0.8805, 0.8922, 0.8863)
 -: (1734, 2095, 2166) (0.8277, 0.8006, 0.8139)
 |: (12849, 13475, 13461) (0.9535, 0.9545, 0.9540)
Macro-average precision, recall, F1: (0.665436, 0.661829, 0.663566)
Item accuracy: 18474 / 19989 (0.9242)
Instance accuracy: 456 / 999 (0.4565)

***** Iteration #45 *****

Loss: 120004.457281
Feature norm: 82.057666
Error norm: 4478.246175
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3924, 4474, 4361) (0.8771, 0.8998, 0.8883)

```

-: (1757, 2137, 2166) (0.8222, 0.8112, 0.8166)
|: (12809, 13378, 13461) (0.9575, 0.9516, 0.9545)
Macro-average precision, recall, F1: (0.664179, 0.665633, 0.664858)
Item accuracy: 18490 / 19989 (0.9250)
Instance accuracy: 461 / 999 (0.4615)

***** Iteration #46 *****
Loss: 118585.066497
Feature norm: 84.004939
Error norm: 3760.670348
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.121
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3941, 4512, 4361) (0.8734, 0.9037, 0.8883)
-: (1762, 2147, 2166) (0.8207, 0.8135, 0.8171)
|: (12782, 13330, 13461) (0.9589, 0.9496, 0.9542)
Macro-average precision, recall, F1: (0.663255, 0.666683, 0.664895)
Item accuracy: 18485 / 19989 (0.9248)
Instance accuracy: 455 / 999 (0.4555)

***** Iteration #47 *****
Loss: 117238.914900
Feature norm: 85.430172
Error norm: 3671.055276
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3949, 4502, 4361) (0.8772, 0.9055, 0.8911)
-: (1791, 2198, 2166) (0.8148, 0.8269, 0.8208)
|: (12772, 13289, 13461) (0.9611, 0.9488, 0.9549)
Macro-average precision, recall, F1: (0.663273, 0.670303, 0.666711)
Item accuracy: 18512 / 19989 (0.9261)
Instance accuracy: 454 / 999 (0.4545)

***** Iteration #48 *****
Loss: 116560.030483
Feature norm: 85.625631
Error norm: 9665.437058
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

```

Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3984, 4592, 4361) (0.8676, 0.9136, 0.8900)
-: (1732, 2070, 2166) (0.8367, 0.7996, 0.8178)
|: (12796, 13327, 13461) (0.9602, 0.9506, 0.9554)
Macro-average precision, recall, F1: (0.666117, 0.665945, 0.665772)
Item accuracy: 18512 / 19989 (0.9261)
Instance accuracy: 455 / 999 (0.4555)

***** Iteration #49 *****

Loss: 115342.057162
Feature norm: 85.558156
Error norm: 3886.603964
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3967, 4523, 4361) (0.8771, 0.9097, 0.8931)
-: (1776, 2142, 2166) (0.8291, 0.8199, 0.8245)
|: (12811, 13324, 13461) (0.9615, 0.9517, 0.9566)
Macro-average precision, recall, F1: (0.666926, 0.670328, 0.668540)
Item accuracy: 18554 / 19989 (0.9282)
Instance accuracy: 461 / 999 (0.4615)

***** Iteration #50 *****

Loss: 114197.935212
Feature norm: 84.631484
Error norm: 4521.853491
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (3943, 4448, 4361) (0.8865, 0.9042, 0.8952)
-: (1804, 2181, 2166) (0.8271, 0.8329, 0.8300)
|: (12842, 13360, 13461) (0.9612, 0.9540, 0.9576)
Macro-average precision, recall, F1: (0.668709, 0.672759, 0.670707)
Item accuracy: 18589 / 19989 (0.9300)
Instance accuracy: 470 / 999 (0.4705)

***** Iteration #51 *****

Loss: 113296.389665
Feature norm: 84.218659
Error norm: 6227.349057
Active features: 272898

Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3932, 4418, 4361) (0.8900, 0.9016, 0.8958)
 -: (1813, 2187, 2166) (0.8290, 0.8370, 0.8330)
 |: (12856, 13384, 13461) (0.9605, 0.9551, 0.9578)
Macro-average precision, recall, F1: (0.669884, 0.673428, 0.671639)
Item accuracy: 18601 / 19989 (0.9306)
Instance accuracy: 476 / 999 (0.4765)

***** Iteration #52 *****

Loss: 111972.945950
Feature norm: 83.566505
Error norm: 6977.700676
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3943, 4437, 4361) (0.8887, 0.9042, 0.8963)
 -: (1831, 2229, 2166) (0.8214, 0.8453, 0.8332)
 |: (12828, 13323, 13461) (0.9628, 0.9530, 0.9579)
Macro-average precision, recall, F1: (0.668239, 0.675616, 0.671861)
Item accuracy: 18602 / 19989 (0.9306)
Instance accuracy: 483 / 999 (0.4835)

***** Iteration #53 *****

Loss: 110545.562117
Feature norm: 83.838307
Error norm: 6660.067498
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3953, 4439, 4361) (0.8905, 0.9064, 0.8984)
 -: (1784, 2107, 2166) (0.8467, 0.8236, 0.8350)
 |: (12906, 13443, 13461) (0.9601, 0.9588, 0.9594)
Macro-average precision, recall, F1: (0.674318, 0.672213, 0.673208)
Item accuracy: 18643 / 19989 (0.9327)
Instance accuracy: 490 / 999 (0.4905)

***** Iteration #54 *****

Loss: 108983.849238

Feature norm: 84.333879
 Error norm: 4736.940710
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3979, 4491, 4361) (0.8860, 0.9124, 0.8990)
 -: (1786, 2104, 2166) (0.8489, 0.8246, 0.8365)
 |: (12883, 13394, 13461) (0.9618, 0.9571, 0.9594)
 Macro-average precision, recall, F1: (0.674176, 0.673507, 0.673747)
 Item accuracy: 18648 / 19989 (0.9329)
 Instance accuracy: 488 / 999 (0.4885)

***** Iteration #55 *****
 Loss: 106868.935318
 Feature norm: 85.557969
 Error norm: 5251.413045
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4015, 4541, 4361) (0.8842, 0.9207, 0.9020)
 -: (1789, 2095, 2166) (0.8539, 0.8259, 0.8397)
 |: (12878, 13353, 13461) (0.9644, 0.9567, 0.9605)
 Macro-average precision, recall, F1: (0.675633, 0.675824, 0.675574)
 Item accuracy: 18682 / 19989 (0.9346)
 Instance accuracy: 490 / 999 (0.4905)

***** Iteration #56 *****
 Loss: 105916.231624
 Feature norm: 86.804251
 Error norm: 4792.726938
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (3990, 4449, 4361) (0.8968, 0.9149, 0.9058)
 -: (1814, 2170, 2166) (0.8359, 0.8375, 0.8367)
 |: (12892, 13370, 13461) (0.9642, 0.9577, 0.9610)
 Macro-average precision, recall, F1: (0.674256, 0.677536, 0.675871)
 Item accuracy: 18696 / 19989 (0.9353)
 Instance accuracy: 502 / 999 (0.5025)

```

***** Iteration #57 *****
Loss: 105329.727758
Feature norm: 87.473240
Error norm: 3731.768447
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (3999, 4465, 4361) (0.8956, 0.9170, 0.9062)
  -: (1811, 2149, 2166) (0.8427, 0.8361, 0.8394)
  |: (12897, 13375, 13461) (0.9643, 0.9581, 0.9612)
Macro-average precision, recall, F1: (0.675653, 0.677799, 0.676689)
Item accuracy: 18707 / 19989 (0.9359)
Instance accuracy: 500 / 999 (0.5005)

```

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***** Iteration #58 *****
Loss: 103691.210920
Feature norm: 89.265085
Error norm: 2653.794004
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4011, 4472, 4361) (0.8969, 0.9197, 0.9082)
  -: (1817, 2139, 2166) (0.8495, 0.8389, 0.8441)
  |: (12910, 13378, 13461) (0.9650, 0.9591, 0.9620)
Macro-average precision, recall, F1: (0.677848, 0.679421, 0.678588)
Item accuracy: 18738 / 19989 (0.9374)
Instance accuracy: 504 / 999 (0.5045)

```

```

***** Iteration #59 *****
Loss: 102185.072972
Feature norm: 91.008212
Error norm: 3200.760622
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4015, 4459, 4361) (0.9004, 0.9207, 0.9104)
  -: (1809, 2106, 2166) (0.8590, 0.8352, 0.8469)
  |: (12944, 13424, 13461) (0.9642, 0.9616, 0.9629)

```


Macro-average precision, recall, F1: (0.680911, 0.679358, 0.680064)
Item accuracy: 18768 / 19989 (0.9389)
Instance accuracy: 517 / 999 (0.5175)

***** Iteration #60 *****

Loss: 101535.117703
Feature norm: 92.297764
Error norm: 6890.959999
Active features: 272898
Line search trials: 2
Line search step: 0.374907
Seconds required for this iteration: 0.220
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4042, 4501, 4361) (0.8980, 0.9269, 0.9122)
 -: (1843, 2177, 2166) (0.8466, 0.8509, 0.8487)
 |: (12897, 13311, 13461) (0.9689, 0.9581, 0.9635)
Macro-average precision, recall, F1: (0.678375, 0.683958, 0.681100)
Item accuracy: 18782 / 19989 (0.9396)
Instance accuracy: 516 / 999 (0.5165)

***** Iteration #61 *****

Loss: 100338.159716
Feature norm: 93.546240
Error norm: 3946.829093
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4029, 4472, 4361) (0.9009, 0.9239, 0.9123)
 -: (1831, 2157, 2166) (0.8489, 0.8453, 0.8471)
 |: (12923, 13360, 13461) (0.9673, 0.9600, 0.9636)
Macro-average precision, recall, F1: (0.679273, 0.682310, 0.680751)
Item accuracy: 18783 / 19989 (0.9397)
Instance accuracy: 518 / 999 (0.5185)

***** Iteration #62 *****

Loss: 99014.886459
Feature norm: 94.983082
Error norm: 2983.784274
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

```

    <: (4027, 4451, 4361) (0.9047, 0.9234, 0.9140)
    -: (1828, 2145, 2166) (0.8522, 0.8440, 0.8481)
    |: (12947, 13393, 13461) (0.9667, 0.9618, 0.9643)
Macro-average precision, recall, F1: (0.680913, 0.682295, 0.681574)
Item accuracy: 18802 / 19989 (0.9406)
Instance accuracy: 530 / 999 (0.5305)

***** Iteration #63 *****
Loss: 97384.639518
Feature norm: 96.946004
Error norm: 3437.439691
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.130
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4025, 4435, 4361) (0.9076, 0.9230, 0.9152)
    -: (1836, 2134, 2166) (0.8604, 0.8476, 0.8540)
    |: (12971, 13420, 13461) (0.9665, 0.9636, 0.9651)
Macro-average precision, recall, F1: (0.683613, 0.683549, 0.683553)
Item accuracy: 18832 / 19989 (0.9421)
Instance accuracy: 538 / 999 (0.5385)

***** Iteration #64 *****
Loss: 96993.773811
Feature norm: 97.424415
Error norm: 5135.963562
Active features: 272898
Line search trials: 2
Line search step: 0.349783
Seconds required for this iteration: 0.232
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4057, 4490, 4361) (0.9036, 0.9303, 0.9167)
    -: (1817, 2086, 2166) (0.8710, 0.8389, 0.8547)
    |: (12976, 13413, 13461) (0.9674, 0.9640, 0.9657)
Macro-average precision, recall, F1: (0.685507, 0.683284, 0.684270)
Item accuracy: 18850 / 19989 (0.9430)
Instance accuracy: 541 / 999 (0.5415)

***** Iteration #65 *****
Loss: 96332.410080
Feature norm: 98.052008
Error norm: 2464.673341
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```

Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4053, 4473, 4361) (0.9061, 0.9294, 0.9176)
 -: (1832, 2122, 2166) (0.8633, 0.8458, 0.8545)
 |: (12969, 13394, 13461) (0.9683, 0.9634, 0.9659)
Macro-average precision, recall, F1: (0.684427, 0.684656, 0.684481)
Item accuracy: 18854 / 19989 (0.9432)
Instance accuracy: 536 / 999 (0.5365)

***** Iteration #66 *****
Loss: 95925.141502
Feature norm: 98.182630
Error norm: 1978.754959
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4056, 4472, 4361) (0.9070, 0.9301, 0.9184)
 -: (1835, 2129, 2166) (0.8619, 0.8472, 0.8545)
 |: (12967, 13388, 13461) (0.9686, 0.9633, 0.9659)
Macro-average precision, recall, F1: (0.684359, 0.685137, 0.684694)
Item accuracy: 18858 / 19989 (0.9434)
Instance accuracy: 535 / 999 (0.5355)

***** Iteration #67 *****
Loss: 95010.128717
Feature norm: 98.664730
Error norm: 2431.211710
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4057, 4468, 4361) (0.9080, 0.9303, 0.9190)
 -: (1850, 2155, 2166) (0.8585, 0.8541, 0.8563)
 |: (12957, 13366, 13461) (0.9694, 0.9626, 0.9660)
Macro-average precision, recall, F1: (0.683970, 0.686740, 0.685317)
Item accuracy: 18864 / 19989 (0.9437)
Instance accuracy: 539 / 999 (0.5395)

***** Iteration #68 *****
Loss: 94057.323730
Feature norm: 100.373775
Error norm: 7071.314554

Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4071, 4506, 4361) (0.9035, 0.9335, 0.9182)
 -: (1867, 2196, 2166) (0.8502, 0.8620, 0.8560)
 |: (12917, 13287, 13461) (0.9722, 0.9596, 0.9658)
 Macro-average precision, recall, F1: (0.681449, 0.688761, 0.685024)
 Item accuracy: 18855 / 19989 (0.9433)
 Instance accuracy: 538 / 999 (0.5385)

***** Iteration #69 *****
 Loss: 92750.007978
 Feature norm: 102.901024
 Error norm: 3824.900184
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4073, 4503, 4361) (0.9045, 0.9340, 0.9190)
 -: (1840, 2127, 2166) (0.8651, 0.8495, 0.8572)
 |: (12962, 13359, 13461) (0.9703, 0.9629, 0.9666)
 Macro-average precision, recall, F1: (0.684965, 0.686596, 0.685700)
 Item accuracy: 18875 / 19989 (0.9443)
 Instance accuracy: 539 / 999 (0.5395)

***** Iteration #70 *****
 Loss: 91764.964226
 Feature norm: 104.785537
 Error norm: 2540.328142
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4068, 4504, 4361) (0.9032, 0.9328, 0.9178)
 -: (1838, 2133, 2166) (0.8617, 0.8486, 0.8551)
 |: (12953, 13352, 13461) (0.9701, 0.9623, 0.9662)
 Macro-average precision, recall, F1: (0.683753, 0.685911, 0.684756)
 Item accuracy: 18859 / 19989 (0.9435)
 Instance accuracy: 535 / 999 (0.5355)

***** Iteration #71 *****

Loss: 90928.593084
 Feature norm: 106.607837
 Error norm: 2451.407805
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4065, 4484, 4361) (0.9066, 0.9321, 0.9192)
 -: (1831, 2111, 2166) (0.8674, 0.8453, 0.8562)
 |: (12984, 13394, 13461) (0.9694, 0.9646, 0.9670)
 Macro-average precision, recall, F1: (0.685827, 0.685507, 0.685585)
 Item accuracy: 18880 / 19989 (0.9445)
 Instance accuracy: 542 / 999 (0.5425)

***** Iteration #72 *****

Loss: 90397.219544
 Feature norm: 107.902109
 Error norm: 3846.552774
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4070, 4512, 4361) (0.9020, 0.9333, 0.9174)
 -: (1844, 2122, 2166) (0.8690, 0.8513, 0.8601)
 |: (12960, 13355, 13461) (0.9704, 0.9628, 0.9666)
 Macro-average precision, recall, F1: (0.685363, 0.686848, 0.686013)
 Item accuracy: 18874 / 19989 (0.9442)
 Instance accuracy: 537 / 999 (0.5375)

***** Iteration #73 *****

Loss: 89912.773813
 Feature norm: 108.897595
 Error norm: 2860.300966
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.121
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4048, 4459, 4361) (0.9078, 0.9282, 0.9179)
 -: (1863, 2178, 2166) (0.8554, 0.8601, 0.8577)
 |: (12967, 13352, 13461) (0.9712, 0.9633, 0.9672)
 Macro-average precision, recall, F1: (0.683591, 0.687910, 0.685717)
 Item accuracy: 18878 / 19989 (0.9444)

Instance accuracy: 545 / 999 (0.5455)

***** Iteration #74 *****

Loss: 89429.226957

Feature norm: 109.371609

Error norm: 2449.749358

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.115

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4054, 4462, 4361) (0.9086, 0.9296, 0.9190)

-: (1864, 2163, 2166) (0.8618, 0.8606, 0.8612)

|: (12980, 13364, 13461) (0.9713, 0.9643, 0.9678)

Macro-average precision, recall, F1: (0.685398, 0.688611, 0.686971)

Item accuracy: 18898 / 19989 (0.9454)

Instance accuracy: 552 / 999 (0.5526)

***** Iteration #75 *****

Loss: 88257.232800

Feature norm: 111.336406

Error norm: 2258.050518

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4070, 4464, 4361) (0.9117, 0.9333, 0.9224)

-: (1879, 2171, 2166) (0.8655, 0.8675, 0.8665)

|: (12990, 13354, 13461) (0.9727, 0.9650, 0.9689)

Macro-average precision, recall, F1: (0.687495, 0.691445, 0.689434)

Item accuracy: 18939 / 19989 (0.9475)

Instance accuracy: 566 / 999 (0.5666)

***** Iteration #76 *****

Loss: 87805.507112

Feature norm: 112.311833

Error norm: 4756.926299

Active features: 272898

Line search trials: 2

Line search step: 0.278980

Seconds required for this iteration: 0.221

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4067, 4436, 4361) (0.9168, 0.9326, 0.9246)

-: (1863, 2131, 2166) (0.8742, 0.8601, 0.8671)

```

|: (13032, 13422, 13461) (0.9709, 0.9681, 0.9695)
Macro-average precision, recall, F1: (0.690499, 0.690206, 0.690321)
Item accuracy: 18962 / 19989 (0.9486)
Instance accuracy: 569 / 999 (0.5696)

***** Iteration #77 *****
Loss: 87098.452094
Feature norm: 114.149920
Error norm: 1868.631726
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4086, 4471, 4361) (0.9139, 0.9369, 0.9253)
  -: (1867, 2141, 2166) (0.8720, 0.8620, 0.8670)
  |: (13010, 13377, 13461) (0.9726, 0.9665, 0.9695)
Macro-average precision, recall, F1: (0.689619, 0.691349, 0.690438)
Item accuracy: 18963 / 19989 (0.9487)
Instance accuracy: 572 / 999 (0.5726)

***** Iteration #78 *****
Loss: 86853.903797
Feature norm: 114.700484
Error norm: 1618.714042
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4088, 4476, 4361) (0.9133, 0.9374, 0.9252)
  -: (1865, 2141, 2166) (0.8711, 0.8610, 0.8660)
  |: (13004, 13372, 13461) (0.9725, 0.9661, 0.9693)
Macro-average precision, recall, F1: (0.689221, 0.691121, 0.690122)
Item accuracy: 18957 / 19989 (0.9484)
Instance accuracy: 569 / 999 (0.5696)

***** Iteration #79 *****
Loss: 86292.806043
Feature norm: 115.840523
Error norm: 2011.630817
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):

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    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4095, 4487, 4361) (0.9126, 0.9390, 0.9256)
    -: (1875, 2152, 2166) (0.8713, 0.8657, 0.8685)
    |: (12997, 13350, 13461) (0.9736, 0.9655, 0.9695)
Macro-average precision, recall, F1: (0.689369, 0.692546, 0.690904)
Item accuracy: 18967 / 19989 (0.9489)
Instance accuracy: 567 / 999 (0.5676)

***** Iteration #80 *****
Loss: 86012.044530
Feature norm: 116.742264
Error norm: 3541.052181
Active features: 272898
Line search trials: 2
Line search step: 0.504240
Seconds required for this iteration: 0.225
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4084, 4451, 4361) (0.9175, 0.9365, 0.9269)
    -: (1892, 2186, 2166) (0.8655, 0.8735, 0.8695)
    |: (13004, 13352, 13461) (0.9739, 0.9661, 0.9700)
Macro-average precision, recall, F1: (0.689248, 0.694008, 0.691595)
Item accuracy: 18980 / 19989 (0.9495)
Instance accuracy: 576 / 999 (0.5766)

***** Iteration #81 *****
Loss: 85558.386508
Feature norm: 117.355194
Error norm: 1923.410027
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4095, 4476, 4361) (0.9149, 0.9390, 0.9268)
    -: (1882, 2167, 2166) (0.8685, 0.8689, 0.8687)
    |: (13002, 13346, 13461) (0.9742, 0.9659, 0.9700)
Macro-average precision, recall, F1: (0.689396, 0.693447, 0.691378)
Item accuracy: 18979 / 19989 (0.9495)
Instance accuracy: 574 / 999 (0.5746)

***** Iteration #82 *****
Loss: 85179.778319
Feature norm: 117.584084
Error norm: 2001.226724
Active features: 272898
Line search trials: 1

```


Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4097, 4477, 4361) (0.9151, 0.9395, 0.9271)
 -: (1883, 2162, 2166) (0.8710, 0.8693, 0.8701)
 |: (13004, 13350, 13461) (0.9741, 0.9661, 0.9700)
Macro-average precision, recall, F1: (0.690039, 0.693714, 0.691833)
Item accuracy: 18984 / 19989 (0.9497)
Instance accuracy: 575 / 999 (0.5756)

***** Iteration #83 *****
Loss: 84444.134565
Feature norm: 117.853890
Error norm: 2635.391507
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4101, 4488, 4361) (0.9138, 0.9404, 0.9269)
 -: (1890, 2166, 2166) (0.8726, 0.8726, 0.8726)
 |: (12998, 13335, 13461) (0.9747, 0.9656, 0.9701)
Macro-average precision, recall, F1: (0.690269, 0.694640, 0.692401)
Item accuracy: 18989 / 19989 (0.9500)
Instance accuracy: 573 / 999 (0.5736)

***** Iteration #84 *****
Loss: 84208.767289
Feature norm: 118.197856
Error norm: 3457.022225
Active features: 272898
Line search trials: 2
Line search step: 0.254324
Seconds required for this iteration: 0.222
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4103, 4464, 4361) (0.9191, 0.9408, 0.9299)
 -: (1873, 2114, 2166) (0.8860, 0.8647, 0.8752)
 |: (13048, 13411, 13461) (0.9729, 0.9693, 0.9711)
Macro-average precision, recall, F1: (0.694515, 0.693721, 0.694054)
Item accuracy: 19024 / 19989 (0.9517)
Instance accuracy: 586 / 999 (0.5866)

***** Iteration #85 *****
Loss: 83901.489461
Feature norm: 118.384217

Error norm: 2227.013812
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4103, 4468, 4361) (0.9183, 0.9408, 0.9294)
 -: (1878, 2121, 2166) (0.8854, 0.8670, 0.8761)
 |: (13045, 13400, 13461) (0.9735, 0.9691, 0.9713)
Macro-average precision, recall, F1: (0.694312, 0.694243, 0.694218)
Item accuracy: 19026 / 19989 (0.9518)
Instance accuracy: 585 / 999 (0.5856)

***** Iteration #86 *****
Loss: 83463.526997
Feature norm: 119.082626
Error norm: 1495.192395
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4106, 4470, 4361) (0.9186, 0.9415, 0.9299)
 -: (1888, 2149, 2166) (0.8785, 0.8717, 0.8751)
 |: (13026, 13370, 13461) (0.9743, 0.9677, 0.9710)
Macro-average precision, recall, F1: (0.692847, 0.695216, 0.693990)
Item accuracy: 19020 / 19989 (0.9515)
Instance accuracy: 579 / 999 (0.5796)

***** Iteration #87 *****
Loss: 83116.815759
Feature norm: 119.811631
Error norm: 1715.236621
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4110, 4470, 4361) (0.9195, 0.9424, 0.9308)
 -: (1899, 2160, 2166) (0.8792, 0.8767, 0.8779)
 |: (13026, 13359, 13461) (0.9751, 0.9677, 0.9714)
Macro-average precision, recall, F1: (0.693426, 0.696715, 0.695031)
Item accuracy: 19035 / 19989 (0.9523)
Instance accuracy: 581 / 999 (0.5816)

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***** Iteration #88 *****
Loss: 82512.965575
Feature norm: 121.116803
Error norm: 3057.452539
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4111, 4451, 4361) (0.9236, 0.9427, 0.9330)
  -: (1909, 2184, 2166) (0.8741, 0.8813, 0.8777)
  |: (13032, 13354, 13461) (0.9759, 0.9681, 0.9720)
Macro-average precision, recall, F1: (0.693396, 0.698038, 0.695685)
Item accuracy: 19052 / 19989 (0.9531)
Instance accuracy: 589 / 999 (0.5896)

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***** Iteration #89 *****
Loss: 82174.499694
Feature norm: 122.139620
Error norm: 3412.156959
Active features: 272898
Line search trials: 2
Line search step: 0.432026
Seconds required for this iteration: 0.222
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4130, 4486, 4361) (0.9206, 0.9470, 0.9336)
  -: (1904, 2154, 2166) (0.8839, 0.8790, 0.8815)
  |: (13033, 13349, 13461) (0.9763, 0.9682, 0.9722)
Macro-average precision, recall, F1: (0.695227, 0.698569, 0.696845)
Item accuracy: 19067 / 19989 (0.9539)
Instance accuracy: 587 / 999 (0.5876)

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***** Iteration #90 *****
Loss: 81768.757612
Feature norm: 122.674336
Error norm: 1720.964293
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4126, 4477, 4361) (0.9216, 0.9461, 0.9337)
  -: (1896, 2148, 2166) (0.8827, 0.8753, 0.8790)
  |: (13037, 13364, 13461) (0.9755, 0.9685, 0.9720)
Macro-average precision, recall, F1: (0.694953, 0.697490, 0.696174)

```

Item accuracy: 19059 / 19989 (0.9535)
Instance accuracy: 584 / 999 (0.5846)

***** Iteration #91 *****

Loss: 81516.072350
Feature norm: 122.804627
Error norm: 1359.505578
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4122, 4466, 4361) (0.9230, 0.9452, 0.9340)
 -: (1898, 2142, 2166) (0.8861, 0.8763, 0.8812)
 |: (13047, 13381, 13461) (0.9750, 0.9692, 0.9721)
Macro-average precision, recall, F1: (0.696025, 0.697678, 0.696809)
Item accuracy: 19067 / 19989 (0.9539)
Instance accuracy: 589 / 999 (0.5896)

***** Iteration #92 *****

Loss: 81304.036556
Feature norm: 122.857110
Error norm: 1565.048067
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4120, 4463, 4361) (0.9231, 0.9447, 0.9338)
 -: (1903, 2153, 2166) (0.8839, 0.8786, 0.8812)
 |: (13043, 13373, 13461) (0.9753, 0.9689, 0.9721)
Macro-average precision, recall, F1: (0.695588, 0.698066, 0.696791)
Item accuracy: 19066 / 19989 (0.9538)
Instance accuracy: 589 / 999 (0.5896)

***** Iteration #93 *****

Loss: 81070.985115
Feature norm: 123.142380
Error norm: 3879.466639
Active features: 272898
Line search trials: 2
Line search step: 0.457220
Seconds required for this iteration: 0.224
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4123, 4452, 4361) (0.9261, 0.9454, 0.9357)

```

-: (1892, 2113, 2166) (0.8954, 0.8735, 0.8843)
|: (13077, 13424, 13461) (0.9742, 0.9715, 0.9728)
Macro-average precision, recall, F1: (0.698915, 0.697600, 0.698198)
Item accuracy: 19092 / 19989 (0.9551)
Instance accuracy: 600 / 999 (0.6006)

***** Iteration #94 *****
Loss: 80699.745143
Feature norm: 123.484059
Error norm: 2504.343030
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4120, 4450, 4361) (0.9258, 0.9447, 0.9352)
-: (1899, 2135, 2166) (0.8895, 0.8767, 0.8831)
|: (13067, 13404, 13461) (0.9749, 0.9707, 0.9728)
Macro-average precision, recall, F1: (0.697541, 0.698050, 0.697759)
Item accuracy: 19086 / 19989 (0.9548)
Instance accuracy: 596 / 999 (0.5966)

***** Iteration #95 *****
Loss: 80159.278579
Feature norm: 124.484316
Error norm: 1544.265684
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4110, 4453, 4361) (0.9230, 0.9424, 0.9326)
-: (1906, 2152, 2166) (0.8857, 0.8800, 0.8828)
|: (13053, 13384, 13461) (0.9753, 0.9697, 0.9725)
Macro-average precision, recall, F1: (0.695982, 0.698024, 0.696974)
Item accuracy: 19069 / 19989 (0.9540)
Instance accuracy: 592 / 999 (0.5926)

***** Iteration #96 *****
Loss: 79906.508029
Feature norm: 125.126094
Error norm: 1542.621961
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

```

Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4119, 4461, 4361) (0.9233, 0.9445, 0.9338)
 -: (1905, 2145, 2166) (0.8881, 0.8795, 0.8838)
 |: (13055, 13383, 13461) (0.9755, 0.9698, 0.9727)
 Macro-average precision, recall, F1: (0.696735, 0.698462, 0.697561)
 Item accuracy: 19079 / 19989 (0.9545)
 Instance accuracy: 597 / 999 (0.5976)

***** Iteration #97 *****

Loss: 79655.484169
 Feature norm: 126.256488
 Error norm: 4873.872110
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4103, 4427, 4361) (0.9268, 0.9408, 0.9338)
 -: (1923, 2194, 2166) (0.8765, 0.8878, 0.8821)
 |: (13046, 13368, 13461) (0.9759, 0.9692, 0.9725)
 Macro-average precision, recall, F1: (0.694802, 0.699455, 0.697103)
 Item accuracy: 19072 / 19989 (0.9541)
 Instance accuracy: 595 / 999 (0.5956)

***** Iteration #98 *****

Loss: 79304.255735
 Feature norm: 126.818111
 Error norm: 1229.027637
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4127, 4468, 4361) (0.9237, 0.9463, 0.9349)
 -: (1910, 2144, 2166) (0.8909, 0.8818, 0.8863)
 |: (13056, 13377, 13461) (0.9760, 0.9699, 0.9729)
 Macro-average precision, recall, F1: (0.697635, 0.699516, 0.698533)
 Item accuracy: 19093 / 19989 (0.9552)
 Instance accuracy: 603 / 999 (0.6036)

***** Iteration #99 *****

Loss: 79184.054358
 Feature norm: 126.691368
 Error norm: 1120.534500
 Active features: 272898

Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4129, 4466, 4361) (0.9245, 0.9468, 0.9355)
 -: (1908, 2140, 2166) (0.8916, 0.8809, 0.8862)
 |: (13061, 13383, 13461) (0.9759, 0.9703, 0.9731)
Macro-average precision, recall, F1: (0.698017, 0.699493, 0.698712)
Item accuracy: 19098 / 19989 (0.9554)
Instance accuracy: 602 / 999 (0.6026)

***** Iteration #100 *****

Loss: 78888.967198
Feature norm: 126.705127
Error norm: 1341.250945
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4133, 4463, 4361) (0.9261, 0.9477, 0.9368)
 -: (1912, 2145, 2166) (0.8914, 0.8827, 0.8870)
 |: (13063, 13381, 13461) (0.9762, 0.9704, 0.9733)
Macro-average precision, recall, F1: (0.698417, 0.700221, 0.699280)
Item accuracy: 19108 / 19989 (0.9559)
Instance accuracy: 606 / 999 (0.6066)

***** Iteration #101 *****

Loss: 78751.097789
Feature norm: 126.937322
Error norm: 3770.144528
Active features: 272898
Line search trials: 2
Line search step: 0.286679
Seconds required for this iteration: 0.233
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4131, 4455, 4361) (0.9273, 0.9473, 0.9372)
 -: (1900, 2108, 2166) (0.9013, 0.8772, 0.8891)
 |: (13092, 13426, 13461) (0.9751, 0.9726, 0.9739)
Macro-average precision, recall, F1: (0.700931, 0.699260, 0.700028)
Item accuracy: 19123 / 19989 (0.9567)
Instance accuracy: 617 / 999 (0.6176)

***** Iteration #102 *****

Loss: 78393.267664

Feature norm: 127.318113
Error norm: 2774.631040
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4137, 4459, 4361) (0.9278, 0.9486, 0.9381)
 -: (1909, 2130, 2166) (0.8962, 0.8813, 0.8887)
 |: (13081, 13400, 13461) (0.9762, 0.9718, 0.9740)
Macro-average precision, recall, F1: (0.700056, 0.700439, 0.700202)
Item accuracy: 19127 / 19989 (0.9569)
Instance accuracy: 617 / 999 (0.6176)

***** Iteration #103 *****
Loss: 77880.802929
Feature norm: 128.259512
Error norm: 1505.922057
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4135, 4464, 4361) (0.9263, 0.9482, 0.9371)
 -: (1921, 2151, 2166) (0.8931, 0.8869, 0.8900)
 |: (13068, 13374, 13461) (0.9771, 0.9708, 0.9740)
Macro-average precision, recall, F1: (0.699123, 0.701467, 0.700258)
Item accuracy: 19124 / 19989 (0.9567)
Instance accuracy: 613 / 999 (0.6136)

***** Iteration #104 *****
Loss: 77633.072097
Feature norm: 129.045734
Error norm: 1144.025529
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4138, 4460, 4361) (0.9278, 0.9489, 0.9382)
 -: (1931, 2159, 2166) (0.8944, 0.8915, 0.8929)
 |: (13070, 13370, 13461) (0.9776, 0.9710, 0.9742)
Macro-average precision, recall, F1: (0.699940, 0.702831, 0.701352)
Item accuracy: 19139 / 19989 (0.9575)
Instance accuracy: 618 / 999 (0.6186)


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***** Iteration #105 *****
Loss: 77467.675620
Feature norm: 129.769725
Error norm: 2815.826812
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4146, 4484, 4361) (0.9246, 0.9507, 0.9375)
  -: (1928, 2146, 2166) (0.8984, 0.8901, 0.8942)
  |: (13063, 13359, 13461) (0.9778, 0.9704, 0.9741)
Macro-average precision, recall, F1: (0.700220, 0.702813, 0.701463)
Item accuracy: 19137 / 19989 (0.9574)
Instance accuracy: 611 / 999 (0.6116)

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***** Iteration #106 *****
Loss: 77246.605502
Feature norm: 130.297923
Error norm: 1339.659435
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4139, 4460, 4361) (0.9280, 0.9491, 0.9384)
  -: (1937, 2163, 2166) (0.8955, 0.8943, 0.8949)
  |: (13069, 13366, 13461) (0.9778, 0.9709, 0.9743)
Macro-average precision, recall, F1: (0.700330, 0.703562, 0.701914)
Item accuracy: 19145 / 19989 (0.9578)
Instance accuracy: 620 / 999 (0.6206)

```

```

***** Iteration #107 *****
Loss: 77081.790709
Feature norm: 130.331063
Error norm: 1388.719510
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4141, 4460, 4361) (0.9285, 0.9496, 0.9389)
  -: (1934, 2158, 2166) (0.8962, 0.8929, 0.8945)
  |: (13071, 13371, 13461) (0.9776, 0.9710, 0.9743)

```

Macro-average precision, recall, F1: (0.700560, 0.703368, 0.701931)
Item accuracy: 19146 / 19989 (0.9578)
Instance accuracy: 621 / 999 (0.6216)

***** Iteration #108 *****

Loss: 76661.061283
Feature norm: 130.815668
Error norm: 2528.243526
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4140, 4451, 4361) (0.9301, 0.9493, 0.9396)
 -: (1924, 2140, 2166) (0.8991, 0.8883, 0.8936)
 |: (13090, 13398, 13461) (0.9770, 0.9724, 0.9747)
Macro-average precision, recall, F1: (0.701551, 0.702509, 0.701996)
Item accuracy: 19154 / 19989 (0.9582)
Instance accuracy: 622 / 999 (0.6226)

***** Iteration #109 *****

Loss: 76618.165495
Feature norm: 131.051800
Error norm: 3713.082397
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4145, 4471, 4361) (0.9271, 0.9505, 0.9386)
 -: (1947, 2190, 2166) (0.8890, 0.8989, 0.8939)
 |: (13052, 13328, 13461) (0.9793, 0.9696, 0.9744)
Macro-average precision, recall, F1: (0.698855, 0.704744, 0.701750)
Item accuracy: 19144 / 19989 (0.9577)
Instance accuracy: 613 / 999 (0.6136)

***** Iteration #110 *****

Loss: 76320.321279
Feature norm: 131.234501
Error norm: 1277.939503
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

```

    <: (4151, 4471, 4361) (0.9284, 0.9518, 0.9400)
    -: (1929, 2144, 2166) (0.8997, 0.8906, 0.8951)
    |: (13080, 13374, 13461) (0.9780, 0.9717, 0.9748)
Macro-average precision, recall, F1: (0.701541, 0.703531, 0.702491)
Item accuracy: 19160 / 19989 (0.9585)
Instance accuracy: 622 / 999 (0.6226)

***** Iteration #111 *****
Loss: 76213.438389
Feature norm: 131.463152
Error norm: 1376.023574
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4152, 4466, 4361) (0.9297, 0.9521, 0.9407)
    -: (1932, 2144, 2166) (0.9011, 0.8920, 0.8965)
    |: (13087, 13379, 13461) (0.9782, 0.9722, 0.9752)
Macro-average precision, recall, F1: (0.702246, 0.704065, 0.703114)
Item accuracy: 19171 / 19989 (0.9591)
Instance accuracy: 625 / 999 (0.6256)

***** Iteration #112 *****
Loss: 75974.665460
Feature norm: 132.009770
Error norm: 1382.650371
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4151, 4463, 4361) (0.9301, 0.9518, 0.9408)
    -: (1933, 2151, 2166) (0.8987, 0.8924, 0.8955)
    |: (13084, 13375, 13461) (0.9782, 0.9720, 0.9751)
Macro-average precision, recall, F1: (0.701747, 0.704067, 0.702870)
Item accuracy: 19168 / 19989 (0.9589)
Instance accuracy: 624 / 999 (0.6246)

***** Iteration #113 *****
Loss: 75877.109271
Feature norm: 132.972987
Error norm: 5783.634353
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```

Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4176, 4519, 4361) (0.9241, 0.9576, 0.9405)
 -: (1918, 2114, 2166) (0.9073, 0.8855, 0.8963)
 |: (13068, 13356, 13461) (0.9784, 0.9708, 0.9746)
Macro-average precision, recall, F1: (0.702455, 0.703472, 0.702852)
Item accuracy: 19162 / 19989 (0.9586)
Instance accuracy: 617 / 999 (0.6176)

***** Iteration #114 *****
Loss: 75465.930329
Feature norm: 133.201568
Error norm: 1635.201319
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4161, 4474, 4361) (0.9300, 0.9541, 0.9419)
 -: (1933, 2141, 2166) (0.9028, 0.8924, 0.8976)
 |: (13085, 13374, 13461) (0.9784, 0.9721, 0.9752)
Macro-average precision, recall, F1: (0.702820, 0.704659, 0.703691)
Item accuracy: 19179 / 19989 (0.9595)
Instance accuracy: 624 / 999 (0.6246)

***** Iteration #115 *****
Loss: 75316.626670
Feature norm: 133.245079
Error norm: 1161.773767
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4157, 4463, 4361) (0.9314, 0.9532, 0.9422)
 -: (1937, 2150, 2166) (0.9009, 0.8943, 0.8976)
 |: (13088, 13376, 13461) (0.9785, 0.9723, 0.9754)
Macro-average precision, recall, F1: (0.702709, 0.704947, 0.703791)
Item accuracy: 19182 / 19989 (0.9596)
Instance accuracy: 625 / 999 (0.6256)

***** Iteration #116 *****
Loss: 75179.986323
Feature norm: 133.422986
Error norm: 1282.414878

Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4157, 4462, 4361) (0.9316, 0.9532, 0.9423)
 -: (1941, 2156, 2166) (0.9003, 0.8961, 0.8982)
 |: (13086, 13371, 13461) (0.9787, 0.9721, 0.9754)
 Macro-average precision, recall, F1: (0.702652, 0.705371, 0.703977)
 Item accuracy: 19184 / 19989 (0.9597)
 Instance accuracy: 624 / 999 (0.6246)

***** Iteration #117 *****
 Loss: 74922.716133
 Feature norm: 133.776563
 Error norm: 2252.202068
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4166, 4474, 4361) (0.9312, 0.9553, 0.9431)
 -: (1955, 2182, 2166) (0.8960, 0.9026, 0.8993)
 |: (13070, 13333, 13461) (0.9803, 0.9710, 0.9756)
 Macro-average precision, recall, F1: (0.701850, 0.707206, 0.704481)
 Item accuracy: 19191 / 19989 (0.9601)
 Instance accuracy: 624 / 999 (0.6246)

***** Iteration #118 *****
 Loss: 74785.536481
 Feature norm: 134.821339
 Error norm: 3541.597848
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4160, 4435, 4361) (0.9380, 0.9539, 0.9459)
 -: (1943, 2142, 2166) (0.9071, 0.8970, 0.9020)
 |: (13123, 13412, 13461) (0.9785, 0.9749, 0.9767)
 Macro-average precision, recall, F1: (0.705885, 0.706461, 0.706149)
 Item accuracy: 19226 / 19989 (0.9618)
 Instance accuracy: 645 / 999 (0.6456)

***** Iteration #119 *****

Loss: 74577.964790
 Feature norm: 134.690171
 Error norm: 913.389287
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4165, 4462, 4361) (0.9334, 0.9551, 0.9441)
 -: (1950, 2171, 2166) (0.8982, 0.9003, 0.8992)
 |: (13086, 13356, 13461) (0.9798, 0.9721, 0.9759)
 Macro-average precision, recall, F1: (0.702856, 0.706869, 0.704828)
 Item accuracy: 19201 / 19989 (0.9606)
 Instance accuracy: 636 / 999 (0.6366)

***** Iteration #120 *****

Loss: 74538.961306
 Feature norm: 134.693933
 Error norm: 1018.627196
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4165, 4464, 4361) (0.9330, 0.9551, 0.9439)
 -: (1951, 2175, 2166) (0.8970, 0.9007, 0.8989)
 |: (13081, 13350, 13461) (0.9799, 0.9718, 0.9758)
 Macro-average precision, recall, F1: (0.702470, 0.706891, 0.704644)
 Item accuracy: 19197 / 19989 (0.9604)
 Instance accuracy: 633 / 999 (0.6336)

***** Iteration #121 *****

Loss: 74422.134443
 Feature norm: 134.869691
 Error norm: 1105.515091
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4166, 4467, 4361) (0.9326, 0.9553, 0.9438)
 -: (1951, 2173, 2166) (0.8978, 0.9007, 0.8993)
 |: (13080, 13349, 13461) (0.9798, 0.9717, 0.9758)
 Macro-average precision, recall, F1: (0.702576, 0.706930, 0.704714)
 Item accuracy: 19197 / 19989 (0.9604)

Instance accuracy: 633 / 999 (0.6336)

***** Iteration #122 *****

Loss: 74350.217620

Feature norm: 135.177782

Error norm: 2749.285982

Active features: 272898

Line search trials: 2

Line search step: 0.217210

Seconds required for this iteration: 0.221

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4174, 4493, 4361) (0.9290, 0.9571, 0.9429)

-: (1944, 2145, 2166) (0.9063, 0.8975, 0.9019)

|: (13081, 13351, 13461) (0.9798, 0.9718, 0.9758)

Macro-average precision, recall, F1: (0.703768, 0.706599, 0.705122)

Item accuracy: 19199 / 19989 (0.9605)

Instance accuracy: 633 / 999 (0.6336)

***** Iteration #123 *****

Loss: 74149.441427

Feature norm: 135.671025

Error norm: 1733.484961

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.121

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4168, 4479, 4361) (0.9306, 0.9557, 0.9430)

-: (1940, 2149, 2166) (0.9027, 0.8957, 0.8992)

|: (13084, 13361, 13461) (0.9793, 0.9720, 0.9756)

Macro-average precision, recall, F1: (0.703145, 0.705849, 0.704448)

Item accuracy: 19192 / 19989 (0.9601)

Instance accuracy: 629 / 999 (0.6296)

***** Iteration #124 *****

Loss: 73976.637198

Feature norm: 136.166543

Error norm: 981.672860

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4165, 4462, 4361) (0.9334, 0.9551, 0.9441)

-: (1946, 2155, 2166) (0.9030, 0.8984, 0.9007)

```

|: (13095, 13372, 13461) (0.9793, 0.9728, 0.9760)
Macro-average precision, recall, F1: (0.703935, 0.706574, 0.705219)
Item accuracy: 19206 / 19989 (0.9608)
Instance accuracy: 635 / 999 (0.6356)

***** Iteration #125 *****
Loss: 73807.926025
Feature norm: 136.725128
Error norm: 1062.802012
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4165, 4454, 4361) (0.9351, 0.9551, 0.9450)
  -: (1956, 2160, 2166) (0.9056, 0.9030, 0.9043)
  |: (13104, 13375, 13461) (0.9797, 0.9735, 0.9766)
Macro-average precision, recall, F1: (0.705102, 0.707896, 0.706470)
Item accuracy: 19225 / 19989 (0.9618)
Instance accuracy: 640 / 999 (0.6406)

***** Iteration #126 *****
Loss: 73637.706280
Feature norm: 137.237897
Error norm: 1449.679992
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4163, 4450, 4361) (0.9355, 0.9546, 0.9450)
  -: (1963, 2167, 2166) (0.9059, 0.9063, 0.9061)
  |: (13104, 13372, 13461) (0.9800, 0.9735, 0.9767)
Macro-average precision, recall, F1: (0.705331, 0.708589, 0.706933)
Item accuracy: 19230 / 19989 (0.9620)
Instance accuracy: 638 / 999 (0.6386)

***** Iteration #127 *****
Loss: 73547.139449
Feature norm: 137.675937
Error norm: 2162.561913
Active features: 272898
Line search trials: 2
Line search step: 0.414991
Seconds required for this iteration: 0.223
Performance by label (#match, #model, #ref) (precision, recall, F1):

```



```

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4170, 4460, 4361) (0.9350, 0.9562, 0.9455)
-: (1976, 2190, 2166) (0.9023, 0.9123, 0.9073)
|: (13088, 13339, 13461) (0.9812, 0.9723, 0.9767)
Macro-average precision, recall, F1: (0.704611, 0.710193, 0.707360)
Item accuracy: 19234 / 19989 (0.9622)
Instance accuracy: 637 / 999 (0.6376)

***** Iteration #128 *****
Loss: 73379.908610
Feature norm: 137.999856
Error norm: 1104.144636
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4171, 4455, 4361) (0.9363, 0.9564, 0.9462)
-: (1973, 2177, 2166) (0.9063, 0.9109, 0.9086)
|: (13106, 13357, 13461) (0.9812, 0.9736, 0.9774)
Macro-average precision, recall, F1: (0.705938, 0.710239, 0.708056)
Item accuracy: 19250 / 19989 (0.9630)
Instance accuracy: 642 / 999 (0.6426)

***** Iteration #129 *****
Loss: 73290.236912
Feature norm: 138.153010
Error norm: 894.708881
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4173, 4464, 4361) (0.9348, 0.9569, 0.9457)
-: (1967, 2163, 2166) (0.9094, 0.9081, 0.9088)
|: (13107, 13362, 13461) (0.9809, 0.9737, 0.9773)
Macro-average precision, recall, F1: (0.706278, 0.709680, 0.707943)
Item accuracy: 19247 / 19989 (0.9629)
Instance accuracy: 641 / 999 (0.6416)

***** Iteration #130 *****
Loss: 73193.886409
Feature norm: 138.157833
Error norm: 982.277889
Active features: 272898
Line search trials: 1

```

Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4176, 4462, 4361) (0.9359, 0.9576, 0.9466)
 -: (1970, 2173, 2166) (0.9066, 0.9095, 0.9080)
 |: (13104, 13354, 13461) (0.9813, 0.9735, 0.9774)
Macro-average precision, recall, F1: (0.705941, 0.710142, 0.708006)
Item accuracy: 19250 / 19989 (0.9630)
Instance accuracy: 641 / 999 (0.6416)

***** Iteration #131 *****
Loss: 73095.467847
Feature norm: 138.446662
Error norm: 3275.001372
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.124
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4184, 4486, 4361) (0.9327, 0.9594, 0.9459)
 -: (1962, 2151, 2166) (0.9121, 0.9058, 0.9090)
 |: (13102, 13352, 13461) (0.9813, 0.9733, 0.9773)
Macro-average precision, recall, F1: (0.706522, 0.709640, 0.708027)
Item accuracy: 19248 / 19989 (0.9629)
Instance accuracy: 644 / 999 (0.6446)

***** Iteration #132 *****
Loss: 72947.675734
Feature norm: 138.528199
Error norm: 1871.383217
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4179, 4471, 4361) (0.9347, 0.9583, 0.9463)
 -: (1968, 2164, 2166) (0.9094, 0.9086, 0.9090)
 |: (13104, 13354, 13461) (0.9813, 0.9735, 0.9774)
Macro-average precision, recall, F1: (0.706349, 0.710083, 0.708175)
Item accuracy: 19251 / 19989 (0.9631)
Instance accuracy: 647 / 999 (0.6476)

***** Iteration #133 *****
Loss: 72759.810059
Feature norm: 138.907541

Error norm: 982.169747
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4170, 4454, 4361) (0.9362, 0.9562, 0.9461)
 -: (1971, 2176, 2166) (0.9058, 0.9100, 0.9079)
 |: (13104, 13359, 13461) (0.9809, 0.9735, 0.9772)
Macro-average precision, recall, F1: (0.705735, 0.709913, 0.707793)
Item accuracy: 19245 / 19989 (0.9628)
Instance accuracy: 645 / 999 (0.6456)

***** Iteration #134 *****
Loss: 72703.862140
Feature norm: 139.113805
Error norm: 1115.169702
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4169, 4454, 4361) (0.9360, 0.9560, 0.9459)
 -: (1973, 2177, 2166) (0.9063, 0.9109, 0.9086)
 |: (13104, 13358, 13461) (0.9810, 0.9735, 0.9772)
Macro-average precision, recall, F1: (0.705823, 0.710087, 0.707923)
Item accuracy: 19246 / 19989 (0.9628)
Instance accuracy: 645 / 999 (0.6456)

***** Iteration #135 *****
Loss: 72614.565236
Feature norm: 139.852096
Error norm: 2259.858177
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4176, 4451, 4361) (0.9382, 0.9576, 0.9478)
 -: (1963, 2146, 2166) (0.9147, 0.9063, 0.9105)
 |: (13134, 13392, 13461) (0.9807, 0.9757, 0.9782)
Macro-average precision, recall, F1: (0.708419, 0.709891, 0.709124)
Item accuracy: 19273 / 19989 (0.9642)
Instance accuracy: 661 / 999 (0.6617)

```

***** Iteration #136 *****
Loss: 72501.570214
Feature norm: 139.950081
Error norm: 1007.017894
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4179, 4464, 4361) (0.9362, 0.9583, 0.9471)
  -: (1967, 2161, 2166) (0.9102, 0.9081, 0.9092)
  |: (13115, 13364, 13461) (0.9814, 0.9743, 0.9778)
Macro-average precision, recall, F1: (0.706938, 0.710172, 0.708519)
Item accuracy: 19261 / 19989 (0.9636)
Instance accuracy: 653 / 999 (0.6537)

```

```

***** Iteration #137 *****
Loss: 72441.566461
Feature norm: 139.899894
Error norm: 1043.864795
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4180, 4469, 4361) (0.9353, 0.9585, 0.9468)
  -: (1966, 2161, 2166) (0.9098, 0.9077, 0.9087)
  |: (13110, 13359, 13461) (0.9814, 0.9739, 0.9776)
Macro-average precision, recall, F1: (0.706614, 0.710021, 0.708278)
Item accuracy: 19256 / 19989 (0.9633)
Instance accuracy: 651 / 999 (0.6517)

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```

***** Iteration #138 *****
Loss: 72289.077506
Feature norm: 140.020565
Error norm: 1185.189058
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4182, 4470, 4361) (0.9356, 0.9590, 0.9471)
  -: (1966, 2162, 2166) (0.9093, 0.9077, 0.9085)
  |: (13109, 13357, 13461) (0.9814, 0.9739, 0.9776)
Macro-average precision, recall, F1: (0.706587, 0.710117, 0.708312)

```

Item accuracy: 19257 / 19989 (0.9634)
Instance accuracy: 653 / 999 (0.6537)

***** Iteration #139 *****

Loss: 72056.297697

Feature norm: 140.402037

Error norm: 1077.889721

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4175, 4464, 4361) (0.9353, 0.9573, 0.9462)

-: (1963, 2161, 2166) (0.9084, 0.9063, 0.9073)

|: (13112, 13364, 13461) (0.9811, 0.9741, 0.9776)

Macro-average precision, recall, F1: (0.706195, 0.709425, 0.707774)

Item accuracy: 19250 / 19989 (0.9630)

Instance accuracy: 648 / 999 (0.6486)

***** Iteration #140 *****

Loss: 71977.946705

Feature norm: 140.578403

Error norm: 2812.449277

Active features: 272898

Line search trials: 2

Line search step: 0.347117

Seconds required for this iteration: 0.224

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4189, 4490, 4361) (0.9330, 0.9606, 0.9466)

-: (1955, 2136, 2166) (0.9153, 0.9026, 0.9089)

|: (13113, 13363, 13461) (0.9813, 0.9741, 0.9777)

Macro-average precision, recall, F1: (0.707379, 0.709323, 0.708286)

Item accuracy: 19257 / 19989 (0.9634)

Instance accuracy: 648 / 999 (0.6486)

***** Iteration #141 *****

Loss: 71790.568613

Feature norm: 140.934402

Error norm: 1329.320759

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.117

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4182, 4466, 4361) (0.9364, 0.9590, 0.9475)

-: (1961, 2152, 2166) (0.9112, 0.9054, 0.9083)
|: (13119, 13371, 13461) (0.9812, 0.9746, 0.9779)
Macro-average precision, recall, F1: (0.707202, 0.709726, 0.708425)
Item accuracy: 19262 / 19989 (0.9636)
Instance accuracy: 655 / 999 (0.6557)

***** Iteration #142 *****

Loss: 71690.736622
Feature norm: 141.081831
Error norm: 693.814695
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4180, 4460, 4361) (0.9372, 0.9585, 0.9477)
-: (1963, 2152, 2166) (0.9122, 0.9063, 0.9092)
|: (13122, 13377, 13461) (0.9809, 0.9748, 0.9779)
Macro-average precision, recall, F1: (0.707583, 0.709898, 0.708706)
Item accuracy: 19265 / 19989 (0.9638)
Instance accuracy: 653 / 999 (0.6537)

***** Iteration #143 *****

Loss: 71596.857563
Feature norm: 141.121842
Error norm: 782.820864
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4177, 4454, 4361) (0.9378, 0.9578, 0.9477)
-: (1962, 2152, 2166) (0.9117, 0.9058, 0.9088)
|: (13123, 13383, 13461) (0.9806, 0.9749, 0.9777)
Macro-average precision, recall, F1: (0.707523, 0.709629, 0.708545)
Item accuracy: 19262 / 19989 (0.9636)
Instance accuracy: 652 / 999 (0.6527)

***** Iteration #144 *****

Loss: 71511.391602
Feature norm: 141.178298
Error norm: 1520.340756
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4187, 4465, 4361) (0.9377, 0.9601, 0.9488)
-: (1967, 2166, 2166) (0.9081, 0.9081, 0.9081)
|: (13113, 13358, 13461) (0.9817, 0.9741, 0.9779)
Macro-average precision, recall, F1: (0.706881, 0.710594, 0.708701)
Item accuracy: 19267 / 19989 (0.9639)
Instance accuracy: 655 / 999 (0.6557)

***** Iteration #145 *****

Loss: 71398.649708
Feature norm: 141.151202
Error norm: 1166.552038
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4184, 4455, 4361) (0.9392, 0.9594, 0.9492)
-: (1961, 2145, 2166) (0.9142, 0.9054, 0.9098)
|: (13130, 13389, 13461) (0.9807, 0.9754, 0.9780)
Macro-average precision, recall, F1: (0.708511, 0.710045, 0.709244)
Item accuracy: 19275 / 19989 (0.9643)
Instance accuracy: 657 / 999 (0.6577)

***** Iteration #146 *****

Loss: 71302.894513
Feature norm: 141.156821
Error norm: 768.817814
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4189, 4464, 4361) (0.9384, 0.9606, 0.9493)
-: (1964, 2143, 2166) (0.9165, 0.9067, 0.9116)
|: (13129, 13382, 13461) (0.9811, 0.9753, 0.9782)
Macro-average precision, recall, F1: (0.708991, 0.710659, 0.709784)
Item accuracy: 19282 / 19989 (0.9646)
Instance accuracy: 659 / 999 (0.6597)

***** Iteration #147 *****

Loss: 71180.569547
Feature norm: 141.271261
Error norm: 790.004652
Active features: 272898

Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4191, 4464, 4361) (0.9388, 0.9610, 0.9498)
 -: (1967, 2146, 2166) (0.9166, 0.9081, 0.9123)
 |: (13131, 13379, 13461) (0.9815, 0.9755, 0.9785)
Macro-average precision, recall, F1: (0.709224, 0.711157, 0.710151)
Item accuracy: 19289 / 19989 (0.9650)
Instance accuracy: 660 / 999 (0.6607)

***** Iteration #148 *****
Loss: 71138.203488
Feature norm: 141.281484
Error norm: 1210.185918
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4185, 4448, 4361) (0.9409, 0.9596, 0.9502)
 -: (1973, 2167, 2166) (0.9105, 0.9109, 0.9107)
 |: (13128, 13374, 13461) (0.9816, 0.9753, 0.9784)
Macro-average precision, recall, F1: (0.708238, 0.711450, 0.709818)
Item accuracy: 19286 / 19989 (0.9648)
Instance accuracy: 660 / 999 (0.6607)

***** Iteration #149 *****
Loss: 71079.413345
Feature norm: 141.319508
Error norm: 839.462755
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4186, 4449, 4361) (0.9409, 0.9599, 0.9503)
 -: (1972, 2165, 2166) (0.9109, 0.9104, 0.9106)
 |: (13129, 13375, 13461) (0.9816, 0.9753, 0.9785)
Macro-average precision, recall, F1: (0.708337, 0.711410, 0.709847)
Item accuracy: 19287 / 19989 (0.9649)
Instance accuracy: 660 / 999 (0.6607)

***** Iteration #150 *****
Loss: 70900.420390

Feature norm: 141.445231
Error norm: 987.756192
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4194, 4459, 4361) (0.9406, 0.9617, 0.9510)
 -: (1967, 2148, 2166) (0.9157, 0.9081, 0.9119)
 |: (13135, 13382, 13461) (0.9815, 0.9758, 0.9787)
Macro-average precision, recall, F1: (0.709462, 0.711403, 0.710397)
Item accuracy: 19296 / 19989 (0.9653)
Instance accuracy: 665 / 999 (0.6657)

***** Iteration #151 *****
Loss: 70795.831748
Feature norm: 141.533661
Error norm: 1237.906302
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4200, 4463, 4361) (0.9411, 0.9631, 0.9519)
 -: (1980, 2151, 2166) (0.9205, 0.9141, 0.9173)
 |: (13140, 13375, 13461) (0.9824, 0.9762, 0.9793)
Macro-average precision, recall, F1: (0.711001, 0.713341, 0.712134)
Item accuracy: 19320 / 19989 (0.9665)
Instance accuracy: 675 / 999 (0.6757)

***** Iteration #152 *****
Loss: 70755.033130
Feature norm: 141.562988
Error norm: 1285.131687
Active features: 272898
Line search trials: 2
Line search step: 0.362763
Seconds required for this iteration: 0.222
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4203, 4456, 4361) (0.9432, 0.9638, 0.9534)
 -: (1970, 2132, 2166) (0.9240, 0.9095, 0.9167)
 |: (13156, 13401, 13461) (0.9817, 0.9773, 0.9795)
Macro-average precision, recall, F1: (0.712239, 0.712656, 0.712404)
Item accuracy: 19329 / 19989 (0.9670)
Instance accuracy: 680 / 999 (0.6807)

```

***** Iteration #153 *****
Loss: 70711.374238
Feature norm: 141.574412
Error norm: 724.565348
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4201, 4458, 4361) (0.9424, 0.9633, 0.9527)
  -: (1974, 2142, 2166) (0.9216, 0.9114, 0.9164)
  |: (13149, 13389, 13461) (0.9821, 0.9768, 0.9794)
Macro-average precision, recall, F1: (0.711499, 0.712873, 0.712148)
Item accuracy: 19324 / 19989 (0.9667)
Instance accuracy: 678 / 999 (0.6787)

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***** Iteration #154 *****
Loss: 70669.947483
Feature norm: 141.569986
Error norm: 685.145303
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4199, 4456, 4361) (0.9423, 0.9629, 0.9525)
  -: (1975, 2146, 2166) (0.9203, 0.9118, 0.9160)
  |: (13146, 13387, 13461) (0.9820, 0.9766, 0.9793)
Macro-average precision, recall, F1: (0.711160, 0.712818, 0.711954)
Item accuracy: 19320 / 19989 (0.9665)
Instance accuracy: 678 / 999 (0.6787)

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```

***** Iteration #155 *****
Loss: 70588.453893
Feature norm: 141.579218
Error norm: 986.941546
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4197, 4454, 4361) (0.9423, 0.9624, 0.9522)
  -: (1976, 2154, 2166) (0.9174, 0.9123, 0.9148)
  |: (13143, 13381, 13461) (0.9822, 0.9764, 0.9793)

```

Macro-average precision, recall, F1: (0.710469, 0.712763, 0.711585)
Item accuracy: 19316 / 19989 (0.9663)
Instance accuracy: 679 / 999 (0.6797)

***** Iteration #156 *****

Loss: 70495.121105
Feature norm: 141.610867
Error norm: 1085.434116
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4199, 4471, 4361) (0.9392, 0.9629, 0.9509)
 -: (1967, 2138, 2166) (0.9200, 0.9081, 0.9140)
 |: (13138, 13380, 13461) (0.9819, 0.9760, 0.9790)
Macro-average precision, recall, F1: (0.710274, 0.711746, 0.710961)
Item accuracy: 19304 / 19989 (0.9657)
Instance accuracy: 670 / 999 (0.6707)

***** Iteration #157 *****

Loss: 70459.531768
Feature norm: 141.652895
Error norm: 2376.376521
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4182, 4435, 4361) (0.9430, 0.9590, 0.9509)
 -: (1968, 2152, 2166) (0.9145, 0.9086, 0.9115)
 |: (13150, 13402, 13461) (0.9812, 0.9769, 0.9790)
Macro-average precision, recall, F1: (0.709662, 0.711109, 0.710365)
Item accuracy: 19300 / 19989 (0.9655)
Instance accuracy: 672 / 999 (0.6727)

***** Iteration #158 *****

Loss: 70362.455755
Feature norm: 141.661650
Error norm: 746.717459
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

```

    <: (4193, 4459, 4361) (0.9403, 0.9615, 0.9508)
    -: (1969, 2139, 2166) (0.9205, 0.9090, 0.9148)
    |: (13145, 13391, 13461) (0.9816, 0.9765, 0.9791)
Macro-average precision, recall, F1: (0.710625, 0.711763, 0.711154)
Item accuracy: 19307 / 19989 (0.9659)
Instance accuracy: 673 / 999 (0.6737)

***** Iteration #159 *****
Loss: 70317.996890
Feature norm: 141.676279
Error norm: 785.115932
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4197, 4466, 4361) (0.9398, 0.9624, 0.9509)
    -: (1968, 2131, 2166) (0.9235, 0.9086, 0.9160)
    |: (13148, 13392, 13461) (0.9818, 0.9767, 0.9793)
Macro-average precision, recall, F1: (0.711264, 0.711932, 0.711548)
Item accuracy: 19313 / 19989 (0.9662)
Instance accuracy: 675 / 999 (0.6757)

***** Iteration #160 *****
Loss: 70273.430869
Feature norm: 141.696250
Error norm: 1009.490597
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4197, 4468, 4361) (0.9393, 0.9624, 0.9507)
    -: (1973, 2135, 2166) (0.9241, 0.9109, 0.9175)
    |: (13145, 13386, 13461) (0.9820, 0.9765, 0.9793)
Macro-average precision, recall, F1: (0.711366, 0.712454, 0.711861)
Item accuracy: 19315 / 19989 (0.9663)
Instance accuracy: 673 / 999 (0.6737)

***** Iteration #161 *****
Loss: 70194.960496
Feature norm: 141.745102
Error norm: 1322.632250
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```

Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4194, 4463, 4361) (0.9397, 0.9617, 0.9506)
 -: (1972, 2128, 2166) (0.9267, 0.9104, 0.9185)
 |: (13152, 13398, 13461) (0.9816, 0.9770, 0.9793)
Macro-average precision, recall, F1: (0.712014, 0.712296, 0.712104)
Item accuracy: 19318 / 19989 (0.9664)
Instance accuracy: 675 / 999 (0.6757)

***** Iteration #162 *****
Loss: 70153.497713
Feature norm: 141.782072
Error norm: 1090.880836
Active features: 272898
Line search trials: 2
Line search step: 0.431657
Seconds required for this iteration: 0.221
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4199, 4468, 4361) (0.9398, 0.9629, 0.9512)
 -: (1979, 2146, 2166) (0.9222, 0.9137, 0.9179)
 |: (13139, 13375, 13461) (0.9824, 0.9761, 0.9792)
Macro-average precision, recall, F1: (0.711083, 0.713149, 0.712074)
Item accuracy: 19317 / 19989 (0.9664)
Instance accuracy: 673 / 999 (0.6737)

***** Iteration #163 *****
Loss: 70122.027971
Feature norm: 141.808708
Error norm: 545.583571
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4195, 4461, 4361) (0.9404, 0.9619, 0.9510)
 -: (1981, 2150, 2166) (0.9214, 0.9146, 0.9180)
 |: (13140, 13378, 13461) (0.9822, 0.9762, 0.9792)
Macro-average precision, recall, F1: (0.710994, 0.713169, 0.712046)
Item accuracy: 19316 / 19989 (0.9663)
Instance accuracy: 674 / 999 (0.6747)

***** Iteration #164 *****
Loss: 70101.798709
Feature norm: 141.827755
Error norm: 528.877001

Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4192, 4455, 4361) (0.9410, 0.9612, 0.9510)
 -: (1981, 2154, 2166) (0.9197, 0.9146, 0.9171)
 |: (13140, 13380, 13461) (0.9821, 0.9762, 0.9791)
Macro-average precision, recall, F1: (0.710678, 0.712997, 0.711807)
Item accuracy: 19313 / 19989 (0.9662)
Instance accuracy: 675 / 999 (0.6757)

***** Iteration #165 *****
Loss: 70072.974596
Feature norm: 141.848270
Error norm: 479.049139
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4195, 4457, 4361) (0.9412, 0.9619, 0.9515)
 -: (1982, 2152, 2166) (0.9210, 0.9151, 0.9180)
 |: (13144, 13380, 13461) (0.9824, 0.9765, 0.9794)
Macro-average precision, recall, F1: (0.711145, 0.713359, 0.712219)
Item accuracy: 19321 / 19989 (0.9666)
Instance accuracy: 678 / 999 (0.6787)

***** Iteration #166 *****
Loss: 70041.829688
Feature norm: 141.856716
Error norm: 1559.833735
Active features: 272898
Line search trials: 2
Line search step: 0.269188
Seconds required for this iteration: 0.224
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4188, 4438, 4361) (0.9437, 0.9603, 0.9519)
 -: (1987, 2167, 2166) (0.9169, 0.9174, 0.9171)
 |: (13148, 13384, 13461) (0.9824, 0.9767, 0.9795)
Macro-average precision, recall, F1: (0.710743, 0.713609, 0.712156)
Item accuracy: 19323 / 19989 (0.9667)
Instance accuracy: 680 / 999 (0.6807)

***** Iteration #167 *****

Loss: 69963.852417
Feature norm: 141.907314
Error norm: 916.470007
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4192, 4445, 4361) (0.9431, 0.9612, 0.9521)
 -: (1985, 2161, 2166) (0.9186, 0.9164, 0.9175)
 |: (13150, 13383, 13461) (0.9826, 0.9769, 0.9797)
Macro-average precision, recall, F1: (0.711057, 0.713645, 0.712327)
Item accuracy: 19327 / 19989 (0.9669)
Instance accuracy: 680 / 999 (0.6807)

***** Iteration #168 *****

Loss: 69874.782991
Feature norm: 141.944296
Error norm: 586.763451
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4194, 4450, 4361) (0.9425, 0.9617, 0.9520)
 -: (1986, 2158, 2166) (0.9203, 0.9169, 0.9186)
 |: (13150, 13381, 13461) (0.9827, 0.9769, 0.9798)
Macro-average precision, recall, F1: (0.711376, 0.713875, 0.712598)
Item accuracy: 19330 / 19989 (0.9670)
Instance accuracy: 681 / 999 (0.6817)

***** Iteration #169 *****

Loss: 69799.878314
Feature norm: 141.946951
Error norm: 769.191871
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4194, 4454, 4361) (0.9416, 0.9617, 0.9516)
 -: (1983, 2148, 2166) (0.9232, 0.9155, 0.9193)
 |: (13153, 13387, 13461) (0.9825, 0.9771, 0.9798)
Macro-average precision, recall, F1: (0.711833, 0.713584, 0.712676)
Item accuracy: 19330 / 19989 (0.9670)

Instance accuracy: 679 / 999 (0.6797)

***** Iteration #170 *****

Loss: 69717.310231

Feature norm: 141.957653

Error norm: 702.822004

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4194, 4456, 4361) (0.9412, 0.9617, 0.9513)

-: (1981, 2145, 2166) (0.9235, 0.9146, 0.9190)

|: (13153, 13388, 13461) (0.9824, 0.9771, 0.9798)

Macro-average precision, recall, F1: (0.711798, 0.713354, 0.712541)

Item accuracy: 19328 / 19989 (0.9669)

Instance accuracy: 677 / 999 (0.6777)

***** Iteration #171 *****

Loss: 69698.358270

Feature norm: 141.946073

Error norm: 919.523951

Active features: 272898

Line search trials: 2

Line search step: 0.224848

Seconds required for this iteration: 0.224

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4195, 4452, 4361) (0.9423, 0.9619, 0.9520)

-: (1981, 2140, 2166) (0.9257, 0.9146, 0.9201)

|: (13162, 13397, 13461) (0.9825, 0.9778, 0.9801)

Macro-average precision, recall, F1: (0.712608, 0.713578, 0.713058)

Item accuracy: 19338 / 19989 (0.9674)

Instance accuracy: 683 / 999 (0.6837)

***** Iteration #172 *****

Loss: 69665.247289

Feature norm: 141.926279

Error norm: 639.493280

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4193, 4448, 4361) (0.9427, 0.9615, 0.9520)

-: (1986, 2147, 2166) (0.9250, 0.9169, 0.9209)


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|: (13160, 13394, 13461) (0.9825, 0.9776, 0.9801)
Macro-average precision, recall, F1: (0.712553, 0.714003, 0.713249)
Item accuracy: 19339 / 19989 (0.9675)
Instance accuracy: 680 / 999 (0.6807)

***** Iteration #173 *****
Loss: 69632.676296
Feature norm: 141.878381
Error norm: 565.856710
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4193, 4452, 4361) (0.9418, 0.9615, 0.9515)
  -: (1983, 2145, 2166) (0.9245, 0.9155, 0.9200)
  |: (13158, 13392, 13461) (0.9825, 0.9775, 0.9800)
Macro-average precision, recall, F1: (0.712207, 0.713620, 0.712881)
Item accuracy: 19334 / 19989 (0.9672)
Instance accuracy: 681 / 999 (0.6817)

***** Iteration #174 *****
Loss: 69590.987454
Feature norm: 141.816683
Error norm: 1299.858011
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4186, 4439, 4361) (0.9430, 0.9599, 0.9514)
  -: (1988, 2165, 2166) (0.9182, 0.9178, 0.9180)
  |: (13152, 13385, 13461) (0.9826, 0.9770, 0.9798)
Macro-average precision, recall, F1: (0.710961, 0.713684, 0.712302)
Item accuracy: 19326 / 19989 (0.9668)
Instance accuracy: 679 / 999 (0.6797)

***** Iteration #175 *****
Loss: 69541.953208
Feature norm: 141.789700
Error norm: 701.405415
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):

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```

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4192, 4451, 4361) (0.9418, 0.9612, 0.9514)
-: (1985, 2153, 2166) (0.9220, 0.9164, 0.9192)
|: (13152, 13385, 13461) (0.9826, 0.9770, 0.9798)
Macro-average precision, recall, F1: (0.711593, 0.713682, 0.712609)
Item accuracy: 19329 / 19989 (0.9670)
Instance accuracy: 679 / 999 (0.6797)

***** Iteration #176 *****
Loss: 69479.818799
Feature norm: 141.768916
Error norm: 487.052624
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4192, 4458, 4361) (0.9403, 0.9612, 0.9507)
-: (1978, 2144, 2166) (0.9226, 0.9132, 0.9179)
|: (13150, 13387, 13461) (0.9823, 0.9769, 0.9796)
Macro-average precision, recall, F1: (0.711301, 0.712837, 0.712032)
Item accuracy: 19320 / 19989 (0.9665)
Instance accuracy: 675 / 999 (0.6757)

***** Iteration #177 *****
Loss: 69447.483079
Feature norm: 141.767205
Error norm: 524.519143
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4193, 4459, 4361) (0.9403, 0.9615, 0.9508)
-: (1978, 2144, 2166) (0.9226, 0.9132, 0.9179)
|: (13150, 13386, 13461) (0.9824, 0.9769, 0.9796)
Macro-average precision, recall, F1: (0.711322, 0.712894, 0.712071)
Item accuracy: 19321 / 19989 (0.9666)
Instance accuracy: 677 / 999 (0.6777)

***** Iteration #178 *****
Loss: 69371.313560
Feature norm: 141.757897
Error norm: 692.612583
Active features: 272898
Line search trials: 1

```

Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4198, 4458, 4361) (0.9417, 0.9626, 0.9520)
 -: (1978, 2137, 2166) (0.9256, 0.9132, 0.9194)
 |: (13160, 13394, 13461) (0.9825, 0.9776, 0.9801)
Macro-average precision, recall, F1: (0.712451, 0.713367, 0.712868)
Item accuracy: 19336 / 19989 (0.9673)
Instance accuracy: 683 / 999 (0.6837)

***** Iteration #179 *****
Loss: 69349.090000
Feature norm: 141.773362
Error norm: 1045.877272
Active features: 272898
Line search trials: 2
Line search step: 0.234804
Seconds required for this iteration: 0.227
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4197, 4462, 4361) (0.9406, 0.9624, 0.9514)
 -: (1985, 2149, 2166) (0.9237, 0.9164, 0.9200)
 |: (13151, 13378, 13461) (0.9830, 0.9770, 0.9800)
Macro-average precision, recall, F1: (0.711832, 0.713950, 0.712854)
Item accuracy: 19333 / 19989 (0.9672)
Instance accuracy: 681 / 999 (0.6817)

***** Iteration #180 *****
Loss: 69297.357704
Feature norm: 141.785678
Error norm: 496.187757
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4197, 4456, 4361) (0.9419, 0.9624, 0.9520)
 -: (1985, 2151, 2166) (0.9228, 0.9164, 0.9196)
 |: (13154, 13382, 13461) (0.9830, 0.9772, 0.9801)
Macro-average precision, recall, F1: (0.711916, 0.714006, 0.712928)
Item accuracy: 19336 / 19989 (0.9673)
Instance accuracy: 684 / 999 (0.6847)

***** Iteration #181 *****
Loss: 69265.600777
Feature norm: 141.791082

Error norm: 473.735939
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4196, 4454, 4361) (0.9421, 0.9622, 0.9520)
-: (1982, 2148, 2166) (0.9227, 0.9151, 0.9189)
|: (13154, 13387, 13461) (0.9826, 0.9772, 0.9799)
Macro-average precision, recall, F1: (0.711847, 0.713602, 0.712692)
Item accuracy: 19332 / 19989 (0.9671)
Instance accuracy: 682 / 999 (0.6827)

***** Iteration #182 *****
Loss: 69214.020290
Feature norm: 141.803188
Error norm: 616.530706
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4193, 4446, 4361) (0.9431, 0.9615, 0.9522)
-: (1988, 2160, 2166) (0.9204, 0.9178, 0.9191)
|: (13153, 13383, 13461) (0.9828, 0.9771, 0.9800)
Macro-average precision, recall, F1: (0.711570, 0.714104, 0.712812)
Item accuracy: 19334 / 19989 (0.9672)
Instance accuracy: 682 / 999 (0.6827)

***** Iteration #183 *****
Loss: 69195.658546
Feature norm: 141.809618
Error norm: 797.270963
Active features: 272898
Line search trials: 2
Line search step: 0.336597
Seconds required for this iteration: 0.223
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4197, 4461, 4361) (0.9408, 0.9624, 0.9515)
-: (1981, 2147, 2166) (0.9227, 0.9146, 0.9186)
|: (13150, 13381, 13461) (0.9827, 0.9769, 0.9798)
Macro-average precision, recall, F1: (0.711560, 0.713470, 0.712478)
Item accuracy: 19328 / 19989 (0.9669)
Instance accuracy: 679 / 999 (0.6797)

```

***** Iteration #184 *****
Loss: 69161.373624
Feature norm: 141.816097
Error norm: 516.252165
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4190, 4449, 4361) (0.9418, 0.9608, 0.9512)
  -: (1987, 2159, 2166) (0.9203, 0.9174, 0.9188)
  |: (13151, 13381, 13461) (0.9828, 0.9770, 0.9799)
Macro-average precision, recall, F1: (0.711232, 0.713780, 0.712480)
Item accuracy: 19328 / 19989 (0.9669)
Instance accuracy: 678 / 999 (0.6787)

```

```

***** Iteration #185 *****
Loss: 69125.036666
Feature norm: 141.820263
Error norm: 466.681981
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.129
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4192, 4450, 4361) (0.9420, 0.9612, 0.9515)
  -: (1990, 2161, 2166) (0.9209, 0.9187, 0.9198)
  |: (13152, 13378, 13461) (0.9831, 0.9770, 0.9801)
Macro-average precision, recall, F1: (0.711500, 0.714259, 0.712853)
Item accuracy: 19334 / 19989 (0.9672)
Instance accuracy: 676 / 999 (0.6767)

```

```

***** Iteration #186 *****
Loss: 69115.820255
Feature norm: 141.859766
Error norm: 2509.222400
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4195, 4454, 4361) (0.9419, 0.9619, 0.9518)
  -: (1998, 2176, 2166) (0.9182, 0.9224, 0.9203)
  |: (13144, 13359, 13461) (0.9839, 0.9765, 0.9802)
Macro-average precision, recall, F1: (0.710989, 0.715206, 0.713066)

```

Item accuracy: 19337 / 19989 (0.9674)
Instance accuracy: 682 / 999 (0.6827)

***** Iteration #187 *****

Loss: 69040.707890
Feature norm: 141.852795
Error norm: 718.783274
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4195, 4443, 4361) (0.9442, 0.9619, 0.9530)
 -: (1989, 2152, 2166) (0.9243, 0.9183, 0.9213)
 |: (13166, 13394, 13461) (0.9830, 0.9781, 0.9805)
Macro-average precision, recall, F1: (0.712854, 0.714576, 0.713690)
Item accuracy: 19350 / 19989 (0.9680)
Instance accuracy: 688 / 999 (0.6887)

***** Iteration #188 *****

Loss: 69028.166553
Feature norm: 141.843940
Error norm: 438.683561
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4199, 4450, 4361) (0.9436, 0.9629, 0.9531)
 -: (1992, 2155, 2166) (0.9244, 0.9197, 0.9220)
 |: (13163, 13384, 13461) (0.9835, 0.9779, 0.9807)
Macro-average precision, recall, F1: (0.712861, 0.715096, 0.713951)
Item accuracy: 19354 / 19989 (0.9682)
Instance accuracy: 689 / 999 (0.6897)

***** Iteration #189 *****

Loss: 69010.036399
Feature norm: 141.845599
Error norm: 389.000378
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4199, 4452, 4361) (0.9432, 0.9629, 0.9529)

-: (1994, 2157, 2166) (0.9244, 0.9206, 0.9225)
|: (13161, 13380, 13461) (0.9836, 0.9777, 0.9807)
Macro-average precision, recall, F1: (0.712809, 0.715289, 0.714020)
Item accuracy: 19354 / 19989 (0.9682)
Instance accuracy: 689 / 999 (0.6897)

***** Iteration #190 *****

Loss: 68981.758091
Feature norm: 141.854798
Error norm: 695.023225
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4200, 4456, 4361) (0.9425, 0.9631, 0.9527)
-: (1993, 2154, 2166) (0.9253, 0.9201, 0.9227)
|: (13159, 13379, 13461) (0.9836, 0.9776, 0.9806)
Macro-average precision, recall, F1: (0.712840, 0.715194, 0.713985)
Item accuracy: 19352 / 19989 (0.9681)
Instance accuracy: 688 / 999 (0.6887)

***** Iteration #191 *****

Loss: 68953.011894
Feature norm: 141.865018
Error norm: 656.441958
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4198, 4447, 4361) (0.9440, 0.9626, 0.9532)
-: (1994, 2159, 2166) (0.9236, 0.9206, 0.9221)
|: (13161, 13383, 13461) (0.9834, 0.9777, 0.9806)
Macro-average precision, recall, F1: (0.712749, 0.715232, 0.713965)
Item accuracy: 19353 / 19989 (0.9682)
Instance accuracy: 688 / 999 (0.6887)

***** Iteration #192 *****

Loss: 68927.667194
Feature norm: 141.890621
Error norm: 784.082452
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4200, 4451, 4361) (0.9436, 0.9631, 0.9532)
-: (1994, 2156, 2166) (0.9249, 0.9206, 0.9227)
|: (13161, 13382, 13461) (0.9835, 0.9777, 0.9806)
Macro-average precision, recall, F1: (0.712989, 0.715347, 0.714139)
Item accuracy: 19355 / 19989 (0.9683)
Instance accuracy: 687 / 999 (0.6877)

***** Iteration #193 *****

Loss: 68910.632698
Feature norm: 141.888653
Error norm: 494.000623
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.123
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4199, 4449, 4361) (0.9438, 0.9629, 0.9532)
-: (1994, 2156, 2166) (0.9249, 0.9206, 0.9227)
|: (13162, 13384, 13461) (0.9834, 0.9778, 0.9806)
Macro-average precision, recall, F1: (0.713020, 0.715308, 0.714137)
Item accuracy: 19355 / 19989 (0.9683)
Instance accuracy: 687 / 999 (0.6877)

***** Iteration #194 *****

Loss: 68887.424176
Feature norm: 141.881298
Error norm: 422.320623
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4198, 4447, 4361) (0.9440, 0.9626, 0.9532)
-: (1998, 2164, 2166) (0.9233, 0.9224, 0.9229)
|: (13158, 13378, 13461) (0.9836, 0.9775, 0.9805)
Macro-average precision, recall, F1: (0.712713, 0.715638, 0.714150)
Item accuracy: 19354 / 19989 (0.9682)
Instance accuracy: 685 / 999 (0.6857)

***** Iteration #195 *****

Loss: 68875.401557
Feature norm: 141.873661
Error norm: 780.290764
Active features: 272898

Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.117
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4196, 4440, 4361) (0.9450, 0.9622, 0.9535)
 -: (1995, 2163, 2166) (0.9223, 0.9211, 0.9217)
 |: (13162, 13386, 13461) (0.9833, 0.9778, 0.9805)
 Macro-average precision, recall, F1: (0.712660, 0.715251, 0.713935)
 Item accuracy: 19353 / 19989 (0.9682)
 Instance accuracy: 688 / 999 (0.6887)

***** Iteration #196 *****
 Loss: 68854.526640
 Feature norm: 141.863536
 Error norm: 771.469022
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4198, 4441, 4361) (0.9453, 0.9626, 0.9539)
 -: (1997, 2164, 2166) (0.9228, 0.9220, 0.9224)
 |: (13162, 13384, 13461) (0.9834, 0.9778, 0.9806)
 Macro-average precision, recall, F1: (0.712881, 0.715597, 0.714217)
 Item accuracy: 19357 / 19989 (0.9684)
 Instance accuracy: 689 / 999 (0.6897)

***** Iteration #197 *****
 Loss: 68811.503542
 Feature norm: 141.836990
 Error norm: 691.298627
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.117
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4197, 4442, 4361) (0.9448, 0.9624, 0.9535)
 -: (1992, 2153, 2166) (0.9252, 0.9197, 0.9224)
 |: (13164, 13394, 13461) (0.9828, 0.9779, 0.9804)
 Macro-average precision, recall, F1: (0.713223, 0.714999, 0.714088)
 Item accuracy: 19353 / 19989 (0.9682)
 Instance accuracy: 686 / 999 (0.6867)

***** Iteration #198 *****
 Loss: 68777.081936

Feature norm: 141.819034
Error norm: 547.488060
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4200, 4452, 4361) (0.9434, 0.9631, 0.9531)
 -: (1994, 2155, 2166) (0.9253, 0.9206, 0.9229)
 |: (13159, 13382, 13461) (0.9833, 0.9776, 0.9804)
Macro-average precision, recall, F1: (0.713006, 0.715309, 0.714128)
Item accuracy: 19353 / 19989 (0.9682)
Instance accuracy: 686 / 999 (0.6867)

***** Iteration #199 *****
Loss: 68755.257196
Feature norm: 141.808514
Error norm: 550.051425
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4205, 4454, 4361) (0.9441, 0.9642, 0.9541)
 -: (1991, 2150, 2166) (0.9260, 0.9192, 0.9226)
 |: (13162, 13385, 13461) (0.9833, 0.9778, 0.9806)
Macro-average precision, recall, F1: (0.713370, 0.715305, 0.714306)
Item accuracy: 19358 / 19989 (0.9684)
Instance accuracy: 689 / 999 (0.6897)

***** Iteration #200 *****
Loss: 68726.608938
Feature norm: 141.791749
Error norm: 957.754521
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4461, 4361) (0.9440, 0.9656, 0.9547)
 -: (1986, 2132, 2166) (0.9315, 0.9169, 0.9242)
 |: (13169, 13396, 13461) (0.9831, 0.9783, 0.9807)
Macro-average precision, recall, F1: (0.714633, 0.715202, 0.714871)
Item accuracy: 19366 / 19989 (0.9688)
Instance accuracy: 695 / 999 (0.6957)

```

***** Iteration #201 *****
Loss: 68703.058011
Feature norm: 141.770932
Error norm: 591.628863
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4204, 4449, 4361) (0.9449, 0.9640, 0.9544)
  -: (1985, 2137, 2166) (0.9289, 0.9164, 0.9226)
  |: (13172, 13403, 13461) (0.9828, 0.9785, 0.9806)
Macro-average precision, recall, F1: (0.714142, 0.714741, 0.714406)
Item accuracy: 19361 / 19989 (0.9686)
Instance accuracy: 691 / 999 (0.6917)

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```

***** Iteration #202 *****
Loss: 68684.661033
Feature norm: 141.771621
Error norm: 362.931611
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4203, 4447, 4361) (0.9451, 0.9638, 0.9544)
  -: (1987, 2139, 2166) (0.9289, 0.9174, 0.9231)
  |: (13172, 13403, 13461) (0.9828, 0.9785, 0.9806)
Macro-average precision, recall, F1: (0.714209, 0.714915, 0.714529)
Item accuracy: 19362 / 19989 (0.9686)
Instance accuracy: 690 / 999 (0.6907)

```

```

***** Iteration #203 *****
Loss: 68666.547704
Feature norm: 141.766030
Error norm: 318.645532
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4202, 4444, 4361) (0.9455, 0.9635, 0.9545)
  -: (1987, 2145, 2166) (0.9263, 0.9174, 0.9218)
  |: (13169, 13400, 13461) (0.9828, 0.9783, 0.9805)

```

Macro-average precision, recall, F1: (0.713662, 0.714802, 0.714204)
Item accuracy: 19358 / 19989 (0.9684)
Instance accuracy: 689 / 999 (0.6897)

***** Iteration #204 *****

Loss: 68644.105824
Feature norm: 141.752824
Error norm: 344.653890
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4203, 4447, 4361) (0.9451, 0.9638, 0.9544)
 -: (1985, 2144, 2166) (0.9258, 0.9164, 0.9211)
 |: (13168, 13398, 13461) (0.9828, 0.9782, 0.9805)
Macro-average precision, recall, F1: (0.713451, 0.714610, 0.714000)
Item accuracy: 19356 / 19989 (0.9683)
Instance accuracy: 690 / 999 (0.6907)

***** Iteration #205 *****

Loss: 68622.161518
Feature norm: 141.733777
Error norm: 1179.414885
Active features: 272898
Line search trials: 2
Line search step: 0.467461
Seconds required for this iteration: 0.239
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4201, 4445, 4361) (0.9451, 0.9633, 0.9541)
 -: (1983, 2138, 2166) (0.9275, 0.9155, 0.9215)
 |: (13173, 13406, 13461) (0.9826, 0.9786, 0.9806)
Macro-average precision, recall, F1: (0.713807, 0.714357, 0.714050)
Item accuracy: 19357 / 19989 (0.9684)
Instance accuracy: 691 / 999 (0.6917)

***** Iteration #206 *****

Loss: 68586.834751
Feature norm: 141.704300
Error norm: 356.397706
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.121
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

```

    <: (4205, 4448, 4361) (0.9454, 0.9642, 0.9547)
    -: (1988, 2143, 2166) (0.9277, 0.9178, 0.9227)
    |: (13171, 13398, 13461) (0.9831, 0.9785, 0.9808)
Macro-average precision, recall, F1: (0.714024, 0.715126, 0.714544)
Item accuracy: 19364 / 19989 (0.9687)
Instance accuracy: 694 / 999 (0.6947)

***** Iteration #207 *****
Loss: 68576.773840
Feature norm: 141.699004
Error norm: 293.291471
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4205, 4448, 4361) (0.9454, 0.9642, 0.9547)
    -: (1988, 2143, 2166) (0.9277, 0.9178, 0.9227)
    |: (13171, 13398, 13461) (0.9831, 0.9785, 0.9808)
Macro-average precision, recall, F1: (0.714024, 0.715126, 0.714544)
Item accuracy: 19364 / 19989 (0.9687)
Instance accuracy: 694 / 999 (0.6947)

***** Iteration #208 *****
Loss: 68555.560953
Feature norm: 141.684674
Error norm: 467.704713
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4205, 4455, 4361) (0.9439, 0.9642, 0.9539)
    -: (1983, 2133, 2166) (0.9297, 0.9155, 0.9225)
    |: (13169, 13401, 13461) (0.9827, 0.9783, 0.9805)
Macro-average precision, recall, F1: (0.714062, 0.714512, 0.714245)
Item accuracy: 19357 / 19989 (0.9684)
Instance accuracy: 692 / 999 (0.6927)

***** Iteration #209 *****
Loss: 68548.231118
Feature norm: 141.667613
Error norm: 1069.085120
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```

Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4201, 4445, 4361) (0.9451, 0.9633, 0.9541)
 -: (1994, 2149, 2166) (0.9279, 0.9206, 0.9242)
 |: (13168, 13395, 13461) (0.9831, 0.9782, 0.9806)
Macro-average precision, recall, F1: (0.714008, 0.715534, 0.714744)
Item accuracy: 19363 / 19989 (0.9687)
Instance accuracy: 691 / 999 (0.6917)

***** Iteration #210 *****
Loss: 68531.495427
Feature norm: 141.666133
Error norm: 336.519262
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4202, 4449, 4361) (0.9445, 0.9635, 0.9539)
 -: (1990, 2140, 2166) (0.9299, 0.9187, 0.9243)
 |: (13171, 13400, 13461) (0.9829, 0.9785, 0.9807)
Macro-average precision, recall, F1: (0.714325, 0.715185, 0.714721)
Item accuracy: 19363 / 19989 (0.9687)
Instance accuracy: 690 / 999 (0.6907)

***** Iteration #211 *****
Loss: 68525.509771
Feature norm: 141.663408
Error norm: 329.277995
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4202, 4450, 4361) (0.9443, 0.9635, 0.9538)
 -: (1991, 2140, 2166) (0.9304, 0.9192, 0.9248)
 |: (13171, 13399, 13461) (0.9830, 0.9785, 0.9807)
Macro-average precision, recall, F1: (0.714407, 0.715301, 0.714820)
Item accuracy: 19364 / 19989 (0.9687)
Instance accuracy: 691 / 999 (0.6917)

***** Iteration #212 *****
Loss: 68514.457741
Feature norm: 141.654556
Error norm: 373.536559

Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4204, 4452, 4361) (0.9443, 0.9640, 0.9540)
 -: (1991, 2138, 2166) (0.9312, 0.9192, 0.9252)
 |: (13173, 13399, 13461) (0.9831, 0.9786, 0.9809)
 Macro-average precision, recall, F1: (0.714668, 0.715452, 0.715024)
 Item accuracy: 19368 / 19989 (0.9689)
 Instance accuracy: 691 / 999 (0.6917)

***** Iteration #213 *****
 Loss: 68503.565380
 Feature norm: 141.638962
 Error norm: 912.842932
 Active features: 272898
 Line search trials: 2
 Line search step: 0.516084
 Seconds required for this iteration: 0.224
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4207, 4453, 4361) (0.9448, 0.9647, 0.9546)
 -: (1995, 2148, 2166) (0.9288, 0.9211, 0.9249)
 |: (13168, 13388, 13461) (0.9836, 0.9782, 0.9809)
 Macro-average precision, recall, F1: (0.714274, 0.715993, 0.715102)
 Item accuracy: 19370 / 19989 (0.9690)
 Instance accuracy: 694 / 999 (0.6947)

***** Iteration #214 *****
 Loss: 68485.450102
 Feature norm: 141.620873
 Error norm: 389.942279
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4208, 4456, 4361) (0.9443, 0.9649, 0.9545)
 -: (1990, 2138, 2166) (0.9308, 0.9187, 0.9247)
 |: (13172, 13395, 13461) (0.9834, 0.9785, 0.9809)
 Macro-average precision, recall, F1: (0.714618, 0.715548, 0.715044)
 Item accuracy: 19370 / 19989 (0.9690)
 Instance accuracy: 695 / 999 (0.6957)

***** Iteration #215 *****

Loss: 68475.986694
 Feature norm: 141.611815
 Error norm: 240.709884
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4455, 4361) (0.9448, 0.9651, 0.9549)
 -: (1990, 2138, 2166) (0.9308, 0.9187, 0.9247)
 |: (13174, 13396, 13461) (0.9834, 0.9787, 0.9810)
 Macro-average precision, recall, F1: (0.714746, 0.715642, 0.715156)
 Item accuracy: 19373 / 19989 (0.9692)
 Instance accuracy: 695 / 999 (0.6957)

***** Iteration #216 *****

Loss: 68468.022158
 Feature norm: 141.604871
 Error norm: 280.638804
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.114
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4455, 4361) (0.9448, 0.9651, 0.9549)
 -: (1988, 2135, 2166) (0.9311, 0.9178, 0.9244)
 |: (13176, 13399, 13461) (0.9834, 0.9788, 0.9811)
 Macro-average precision, recall, F1: (0.714821, 0.715449, 0.715095)
 Item accuracy: 19373 / 19989 (0.9692)
 Instance accuracy: 695 / 999 (0.6957)

***** Iteration #217 *****

Loss: 68454.655515
 Feature norm: 141.595201
 Error norm: 354.173949
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4457, 4361) (0.9446, 0.9654, 0.9549)
 -: (1986, 2133, 2166) (0.9311, 0.9169, 0.9239)
 |: (13176, 13399, 13461) (0.9834, 0.9788, 0.9811)
 Macro-average precision, recall, F1: (0.714755, 0.715275, 0.714972)
 Item accuracy: 19372 / 19989 (0.9691)

Instance accuracy: 696 / 999 (0.6967)

***** Iteration #218 *****

Loss: 68443.037409

Feature norm: 141.580787

Error norm: 547.217489

Active features: 272898

Line search trials: 2

Line search step: 0.465573

Seconds required for this iteration: 0.230

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4210, 4453, 4361) (0.9454, 0.9654, 0.9553)

-: (1992, 2137, 2166) (0.9321, 0.9197, 0.9259)

|: (13178, 13399, 13461) (0.9835, 0.9790, 0.9812)

Macro-average precision, recall, F1: (0.715271, 0.716005, 0.715600)

Item accuracy: 19380 / 19989 (0.9695)

Instance accuracy: 696 / 999 (0.6967)

***** Iteration #219 *****

Loss: 68427.353723

Feature norm: 141.572163

Error norm: 329.498672

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.117

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4207, 4451, 4361) (0.9452, 0.9647, 0.9548)

-: (1993, 2142, 2166) (0.9304, 0.9201, 0.9253)

|: (13174, 13396, 13461) (0.9834, 0.9787, 0.9810)

Macro-average precision, recall, F1: (0.714762, 0.715874, 0.715284)

Item accuracy: 19374 / 19989 (0.9692)

Instance accuracy: 694 / 999 (0.6947)

***** Iteration #220 *****

Loss: 68420.007927

Feature norm: 141.568296

Error norm: 299.658004

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4209, 4455, 4361) (0.9448, 0.9651, 0.9549)

-: (1992, 2141, 2166) (0.9304, 0.9197, 0.9250)

```

|: (13173, 13393, 13461) (0.9836, 0.9786, 0.9811)
Macro-average precision, recall, F1: (0.714690, 0.715855, 0.715236)
Item accuracy: 19374 / 19989 (0.9692)
Instance accuracy: 694 / 999 (0.6947)

***** Iteration #221 *****
Loss: 68415.056990
Feature norm: 141.566570
Error norm: 384.844876
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4206, 4449, 4361) (0.9454, 0.9645, 0.9548)
  -: (1992, 2141, 2166) (0.9304, 0.9197, 0.9250)
  |: (13176, 13399, 13461) (0.9834, 0.9788, 0.9811)
Macro-average precision, recall, F1: (0.714786, 0.715738, 0.715229)
Item accuracy: 19374 / 19989 (0.9692)
Instance accuracy: 693 / 999 (0.6937)

***** Iteration #222 *****
Loss: 68409.840576
Feature norm: 141.565336
Error norm: 296.167094
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4205, 4450, 4361) (0.9449, 0.9642, 0.9545)
  -: (1993, 2144, 2166) (0.9296, 0.9201, 0.9248)
  |: (13172, 13395, 13461) (0.9834, 0.9785, 0.9809)
Macro-average precision, recall, F1: (0.714467, 0.715722, 0.715063)
Item accuracy: 19370 / 19989 (0.9690)
Instance accuracy: 691 / 999 (0.6917)

***** Iteration #223 *****
Loss: 68402.331474
Feature norm: 141.562724
Error norm: 390.995524
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):

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```

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4207, 4454, 4361) (0.9445, 0.9647, 0.9545)
-: (1994, 2146, 2166) (0.9292, 0.9206, 0.9249)
|: (13170, 13389, 13461) (0.9836, 0.9784, 0.9810)
Macro-average precision, recall, F1: (0.714340, 0.715915, 0.715094)
Item accuracy: 19371 / 19989 (0.9691)
Instance accuracy: 694 / 999 (0.6947)

***** Iteration #224 *****
Loss: 68386.816714
Feature norm: 141.554425
Error norm: 450.327364
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4207, 4454, 4361) (0.9445, 0.9647, 0.9545)
-: (1994, 2148, 2166) (0.9283, 0.9206, 0.9244)
|: (13168, 13387, 13461) (0.9836, 0.9782, 0.9809)
Macro-average precision, recall, F1: (0.714123, 0.715878, 0.714968)
Item accuracy: 19369 / 19989 (0.9690)
Instance accuracy: 693 / 999 (0.6937)

***** Iteration #225 *****
Loss: 68369.548688
Feature norm: 141.544300
Error norm: 677.437660
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4212, 4459, 4361) (0.9446, 0.9658, 0.9551)
-: (1994, 2138, 2166) (0.9326, 0.9206, 0.9266)
|: (13175, 13392, 13461) (0.9838, 0.9788, 0.9813)
Macro-average precision, recall, F1: (0.715263, 0.716294, 0.715738)
Item accuracy: 19381 / 19989 (0.9696)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #226 *****
Loss: 68360.474916
Feature norm: 141.538761
Error norm: 477.289348
Active features: 272898
Line search trials: 2

```

Line search step: 0.489101
Seconds required for this iteration: 0.246
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4457, 4361) (0.9448, 0.9656, 0.9551)
 -: (1994, 2140, 2166) (0.9318, 0.9206, 0.9261)
 |: (13175, 13392, 13461) (0.9838, 0.9788, 0.9813)
Macro-average precision, recall, F1: (0.715094, 0.716237, 0.715627)
Item accuracy: 19380 / 19989 (0.9695)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #227 *****
Loss: 68354.221765
Feature norm: 141.537631
Error norm: 234.842171
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4456, 4361) (0.9446, 0.9651, 0.9547)
 -: (1991, 2135, 2166) (0.9326, 0.9192, 0.9258)
 |: (13176, 13398, 13461) (0.9834, 0.9788, 0.9811)
Macro-average precision, recall, F1: (0.715138, 0.715795, 0.715425)
Item accuracy: 19376 / 19989 (0.9693)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #228 *****
Loss: 68349.363878
Feature norm: 141.537205
Error norm: 256.000751
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4208, 4454, 4361) (0.9448, 0.9649, 0.9547)
 -: (1991, 2133, 2166) (0.9334, 0.9192, 0.9263)
 |: (13179, 13402, 13461) (0.9834, 0.9791, 0.9812)
Macro-average precision, recall, F1: (0.715389, 0.715793, 0.715550)
Item accuracy: 19378 / 19989 (0.9694)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #229 *****
Loss: 68344.371355
Feature norm: 141.536850

Error norm: 252.654133
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.122
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4455, 4361) (0.9450, 0.9654, 0.9551)
-: (1992, 2134, 2166) (0.9335, 0.9197, 0.9265)
|: (13179, 13400, 13461) (0.9835, 0.9791, 0.9813)
Macro-average precision, recall, F1: (0.715493, 0.716023, 0.715717)
Item accuracy: 19381 / 19989 (0.9696)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #230 *****
Loss: 68340.333552
Feature norm: 141.537313
Error norm: 733.967383
Active features: 272898
Line search trials: 2
Line search step: 0.281614
Seconds required for this iteration: 0.262
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4207, 4448, 4361) (0.9458, 0.9647, 0.9552)
-: (1994, 2139, 2166) (0.9322, 0.9206, 0.9264)
|: (13181, 13402, 13461) (0.9835, 0.9792, 0.9813)
Macro-average precision, recall, F1: (0.715385, 0.716119, 0.715719)
Item accuracy: 19382 / 19989 (0.9696)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #231 *****
Loss: 68331.037477
Feature norm: 141.537130
Error norm: 508.991563
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.134
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4209, 4451, 4361) (0.9456, 0.9651, 0.9553)
-: (1995, 2139, 2166) (0.9327, 0.9211, 0.9268)
|: (13180, 13399, 13461) (0.9837, 0.9791, 0.9814)
Macro-average precision, recall, F1: (0.715491, 0.716331, 0.715876)
Item accuracy: 19384 / 19989 (0.9697)
Instance accuracy: 698 / 999 (0.6987)

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***** Iteration #232 *****
Loss: 68318.772490
Feature norm: 141.538224
Error norm: 261.529675
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.130
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4209, 4451, 4361) (0.9456, 0.9651, 0.9553)
  -: (1997, 2142, 2166) (0.9323, 0.9220, 0.9271)
  |: (13179, 13396, 13461) (0.9838, 0.9791, 0.9814)
Macro-average precision, recall, F1: (0.715434, 0.716543, 0.715955)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 697 / 999 (0.6977)

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***** Iteration #233 *****
Loss: 68311.165061
Feature norm: 141.539818
Error norm: 255.306335
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.126
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4210, 4452, 4361) (0.9456, 0.9654, 0.9554)
  -: (1998, 2145, 2166) (0.9315, 0.9224, 0.9269)
  |: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715264, 0.716679, 0.715939)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 700 / 999 (0.7007)

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***** Iteration #234 *****
Loss: 68304.955171
Feature norm: 141.542426
Error norm: 379.534251
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4455, 4361) (0.9452, 0.9656, 0.9553)
  -: (1997, 2140, 2166) (0.9332, 0.9220, 0.9275)
  |: (13179, 13394, 13461) (0.9839, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715589, 0.716658, 0.716086)

```

Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #235 *****

Loss: 68302.270434
Feature norm: 141.546036
Error norm: 529.485349
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4453, 4361) (0.9454, 0.9654, 0.9553)
 -: (1998, 2143, 2166) (0.9323, 0.9224, 0.9274)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715429, 0.716697, 0.716029)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #236 *****

Loss: 68297.587324
Feature norm: 141.545234
Error norm: 186.624759
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4452, 4361) (0.9456, 0.9654, 0.9554)
 -: (1998, 2142, 2166) (0.9328, 0.9224, 0.9276)
 |: (13180, 13395, 13461) (0.9839, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715591, 0.716734, 0.716129)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #237 *****

Loss: 68295.238488
Feature norm: 141.545606
Error norm: 178.277582
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4453, 4361) (0.9457, 0.9656, 0.9555)

```

-: (1998, 2139, 2166) (0.9341, 0.9224, 0.9282)
|: (13182, 13397, 13461) (0.9840, 0.9793, 0.9816)
Macro-average precision, recall, F1: (0.715922, 0.716829, 0.716339)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #238 *****
Loss: 68291.688637
Feature norm: 141.547270
Error norm: 255.992470
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4455, 4361) (0.9450, 0.9654, 0.9551)
-: (1996, 2136, 2166) (0.9345, 0.9215, 0.9279)
|: (13181, 13398, 13461) (0.9838, 0.9792, 0.9815)
Macro-average precision, recall, F1: (0.715817, 0.716522, 0.716130)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #239 *****
Loss: 68285.451103
Feature norm: 141.552745
Error norm: 603.278252
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4453, 4361) (0.9454, 0.9654, 0.9553)
-: (1996, 2135, 2166) (0.9349, 0.9215, 0.9282)
|: (13183, 13401, 13461) (0.9837, 0.9793, 0.9815)
Macro-average precision, recall, F1: (0.716014, 0.716559, 0.716247)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #240 *****
Loss: 68275.856691
Feature norm: 141.561525
Error norm: 433.181477
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.125

```


Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4454, 4361) (0.9452, 0.9654, 0.9552)
-: (1996, 2135, 2166) (0.9349, 0.9215, 0.9282)
|: (13182, 13400, 13461) (0.9837, 0.9793, 0.9815)
Macro-average precision, recall, F1: (0.715961, 0.716541, 0.716211)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #241 *****

Loss: 68267.302418
Feature norm: 141.571465
Error norm: 239.305860
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4208, 4451, 4361) (0.9454, 0.9649, 0.9551)
-: (1999, 2141, 2166) (0.9337, 0.9229, 0.9283)
|: (13180, 13397, 13461) (0.9838, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715721, 0.716735, 0.716194)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #242 *****

Loss: 68261.737663
Feature norm: 141.579145
Error norm: 160.236356
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
-: (2002, 2146, 2166) (0.9329, 0.9243, 0.9286)
|: (13179, 13392, 13461) (0.9841, 0.9791, 0.9816)
Macro-average precision, recall, F1: (0.715768, 0.717235, 0.716470)
Item accuracy: 19392 / 19989 (0.9701)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #243 *****

Loss: 68257.976660
Feature norm: 141.584400
Error norm: 224.041215
Active features: 272898

Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
 -: (2002, 2148, 2166) (0.9320, 0.9243, 0.9281)
 |: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)
 Macro-average precision, recall, F1: (0.715548, 0.717159, 0.716323)
 Item accuracy: 19390 / 19989 (0.9700)
 Instance accuracy: 699 / 999 (0.6997)

***** Iteration #244 *****
 Loss: 68252.318782
 Feature norm: 141.596134
 Error norm: 183.226073
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
 -: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
 |: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)
 Macro-average precision, recall, F1: (0.715482, 0.716985, 0.716201)
 Item accuracy: 19389 / 19989 (0.9700)
 Instance accuracy: 699 / 999 (0.6997)

***** Iteration #245 *****
 Loss: 68249.905847
 Feature norm: 141.607952
 Error norm: 736.378651
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4449, 4361) (0.9465, 0.9656, 0.9560)
 -: (2004, 2153, 2166) (0.9308, 0.9252, 0.9280)
 |: (13177, 13387, 13461) (0.9843, 0.9789, 0.9816)
 Macro-average precision, recall, F1: (0.715403, 0.717428, 0.716388)
 Item accuracy: 19392 / 19989 (0.9701)
 Instance accuracy: 701 / 999 (0.7017)

***** Iteration #246 *****
 Loss: 68242.412054

Feature norm: 141.611061
Error norm: 290.609568
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4449, 4361) (0.9465, 0.9656, 0.9560)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13179, 13391, 13461) (0.9842, 0.9791, 0.9816)
Macro-average precision, recall, F1: (0.715567, 0.717235, 0.716372)
Item accuracy: 19392 / 19989 (0.9701)
Instance accuracy: 701 / 999 (0.7017)

***** Iteration #247 *****
Loss: 68238.267612
Feature norm: 141.615576
Error norm: 170.229604
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4450, 4361) (0.9465, 0.9658, 0.9561)
 -: (2003, 2150, 2166) (0.9316, 0.9247, 0.9282)
 |: (13178, 13389, 13461) (0.9842, 0.9790, 0.9816)
Macro-average precision, recall, F1: (0.715596, 0.717389, 0.716463)
Item accuracy: 19393 / 19989 (0.9702)
Instance accuracy: 702 / 999 (0.7027)

***** Iteration #248 *****
Loss: 68236.120517
Feature norm: 141.620058
Error norm: 220.721567
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4213, 4450, 4361) (0.9467, 0.9661, 0.9563)
 -: (2003, 2148, 2166) (0.9325, 0.9247, 0.9286)
 |: (13180, 13391, 13461) (0.9842, 0.9791, 0.9817)
Macro-average precision, recall, F1: (0.715870, 0.717483, 0.716647)
Item accuracy: 19396 / 19989 (0.9703)
Instance accuracy: 702 / 999 (0.7027)

```

***** Iteration #249 *****
Loss: 68235.178102
Feature norm: 141.635144
Error norm: 657.571665
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4209, 4451, 4361) (0.9456, 0.9651, 0.9553)
  -: (2004, 2153, 2166) (0.9308, 0.9252, 0.9280)
  |: (13173, 13385, 13461) (0.9842, 0.9786, 0.9814)
Macro-average precision, recall, F1: (0.715146, 0.717240, 0.716164)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

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***** Iteration #250 *****
Loss: 68230.372432
Feature norm: 141.636121
Error norm: 215.846255
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4209, 4450, 4361) (0.9458, 0.9651, 0.9554)
  -: (2004, 2153, 2166) (0.9308, 0.9252, 0.9280)
  |: (13174, 13386, 13461) (0.9842, 0.9787, 0.9814)
Macro-average precision, recall, F1: (0.715200, 0.717258, 0.716201)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

```

```

***** Iteration #251 *****
Loss: 68227.252119
Feature norm: 141.639776
Error norm: 195.531563
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
  -: (2004, 2152, 2166) (0.9312, 0.9252, 0.9282)
  |: (13178, 13390, 13461) (0.9842, 0.9790, 0.9816)

```

Macro-average precision, recall, F1: (0.715469, 0.717332, 0.716374)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #252 *****

Loss: 68223.407803
Feature norm: 141.649122
Error norm: 336.981945
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
 -: (2003, 2151, 2166) (0.9312, 0.9247, 0.9280)
 |: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715442, 0.717217, 0.716302)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #253 *****

Loss: 68218.619918
Feature norm: 141.663811
Error norm: 323.150164
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715416, 0.717102, 0.716231)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #254 *****

Loss: 68214.038445
Feature norm: 141.689297
Error norm: 358.961129
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

```

    <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
    -: (2003, 2153, 2166) (0.9303, 0.9247, 0.9275)
    |: (13176, 13389, 13461) (0.9841, 0.9788, 0.9815)
Macro-average precision, recall, F1: (0.715225, 0.717180, 0.716176)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #255 *****
Loss: 68208.902530
Feature norm: 141.698741
Error norm: 218.922838
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
    -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
    |: (13179, 13393, 13461) (0.9840, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715525, 0.717120, 0.716294)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #256 *****
Loss: 68202.749805
Feature norm: 141.711842
Error norm: 262.201281
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4211, 4450, 4361) (0.9463, 0.9656, 0.9559)
    -: (2001, 2149, 2166) (0.9311, 0.9238, 0.9275)
    |: (13177, 13390, 13461) (0.9841, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715379, 0.717082, 0.716201)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #257 *****
Loss: 68199.328868
Feature norm: 141.727698
Error norm: 219.607245
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```

Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4212, 4450, 4361) (0.9465, 0.9658, 0.9561)
-: (2002, 2146, 2166) (0.9329, 0.9243, 0.9286)
|: (13181, 13393, 13461) (0.9842, 0.9792, 0.9817)
Macro-average precision, recall, F1: (0.715897, 0.717329, 0.716582)
Item accuracy: 19395 / 19989 (0.9703)
Instance accuracy: 702 / 999 (0.7027)

***** Iteration #258 *****
Loss: 68196.279058
Feature norm: 141.734894
Error norm: 172.033177
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4212, 4450, 4361) (0.9465, 0.9658, 0.9561)
-: (2002, 2146, 2166) (0.9329, 0.9243, 0.9286)
|: (13181, 13393, 13461) (0.9842, 0.9792, 0.9817)
Macro-average precision, recall, F1: (0.715897, 0.717329, 0.716582)
Item accuracy: 19395 / 19989 (0.9703)
Instance accuracy: 702 / 999 (0.7027)

***** Iteration #259 *****
Loss: 68191.196856
Feature norm: 141.751742
Error norm: 255.680049
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4449, 4361) (0.9465, 0.9656, 0.9560)
-: (2002, 2146, 2166) (0.9329, 0.9243, 0.9286)
|: (13181, 13394, 13461) (0.9841, 0.9792, 0.9816)
Macro-average precision, recall, F1: (0.715875, 0.717272, 0.716543)
Item accuracy: 19394 / 19989 (0.9702)
Instance accuracy: 701 / 999 (0.7017)

***** Iteration #260 *****
Loss: 68183.640187
Feature norm: 141.781974
Error norm: 198.452703

Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4452, 4361) (0.9461, 0.9658, 0.9559)
 -: (2001, 2146, 2166) (0.9324, 0.9238, 0.9281)
 |: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715654, 0.717158, 0.716374)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #261 *****
Loss: 68178.307703
Feature norm: 141.809436
Error norm: 294.798038
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2004, 2152, 2166) (0.9312, 0.9252, 0.9282)
 |: (13176, 13388, 13461) (0.9842, 0.9788, 0.9815)
Macro-average precision, recall, F1: (0.715418, 0.717353, 0.716357)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #262 *****
Loss: 68174.855407
Feature norm: 141.845139
Error norm: 393.502290
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4213, 4452, 4361) (0.9463, 0.9661, 0.9561)
 -: (1999, 2144, 2166) (0.9324, 0.9229, 0.9276)
 |: (13179, 13393, 13461) (0.9840, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715677, 0.717003, 0.716307)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #263 *****

Loss: 68171.837879
 Feature norm: 141.838641
 Error norm: 151.272210
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
 -: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
 |: (13176, 13390, 13461) (0.9840, 0.9788, 0.9814)
 Macro-average precision, recall, F1: (0.715354, 0.716948, 0.716120)
 Item accuracy: 19387 / 19989 (0.9699)
 Instance accuracy: 698 / 999 (0.6987)

***** Iteration #264 *****

Loss: 68170.380415
 Feature norm: 141.836894
 Error norm: 139.657501
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4452, 4361) (0.9461, 0.9658, 0.9559)
 -: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
 |: (13176, 13389, 13461) (0.9841, 0.9788, 0.9815)
 Macro-average precision, recall, F1: (0.715320, 0.717006, 0.716132)
 Item accuracy: 19388 / 19989 (0.9699)
 Instance accuracy: 698 / 999 (0.6987)

***** Iteration #265 *****

Loss: 68166.892247
 Feature norm: 141.854064
 Error norm: 325.313196
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4213, 4456, 4361) (0.9455, 0.9661, 0.9557)
 -: (1999, 2146, 2166) (0.9315, 0.9229, 0.9272)
 |: (13174, 13387, 13461) (0.9841, 0.9787, 0.9814)
 Macro-average precision, recall, F1: (0.715264, 0.716910, 0.716053)
 Item accuracy: 19386 / 19989 (0.9698)

Instance accuracy: 698 / 999 (0.6987)

***** Iteration #266 *****

Loss: 68163.589972

Feature norm: 141.882805

Error norm: 313.856414

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.117

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4214, 4456, 4361) (0.9457, 0.9663, 0.9559)

-: (1997, 2144, 2166) (0.9314, 0.9220, 0.9267)

|: (13176, 13389, 13461) (0.9841, 0.9788, 0.9815)

Macro-average precision, recall, F1: (0.715305, 0.716774, 0.716004)

Item accuracy: 19387 / 19989 (0.9699)

Instance accuracy: 701 / 999 (0.7017)

***** Iteration #267 *****

Loss: 68161.088450

Feature norm: 141.894360

Error norm: 178.587952

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4213, 4456, 4361) (0.9455, 0.9661, 0.9557)

-: (1997, 2144, 2166) (0.9314, 0.9220, 0.9267)

|: (13175, 13389, 13461) (0.9840, 0.9788, 0.9814)

Macro-average precision, recall, F1: (0.715230, 0.716698, 0.715929)

Item accuracy: 19385 / 19989 (0.9698)

Instance accuracy: 699 / 999 (0.6997)

***** Iteration #268 *****

Loss: 68158.324443

Feature norm: 141.914311

Error norm: 147.911342

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.133

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4212, 4454, 4361) (0.9457, 0.9658, 0.9556)

-: (1999, 2144, 2166) (0.9324, 0.9229, 0.9276)

```

|: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715514, 0.716909, 0.716177)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #269 *****
Loss: 68155.780803
Feature norm: 141.929003
Error norm: 196.794852
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.135
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4212, 4454, 4361) (0.9457, 0.9658, 0.9556)
  -: (1999, 2141, 2166) (0.9337, 0.9229, 0.9283)
  |: (13179, 13394, 13461) (0.9839, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715823, 0.716946, 0.716348)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #270 *****
Loss: 68154.514330
Feature norm: 141.963202
Error norm: 365.626264
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.135
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4209, 4452, 4361) (0.9454, 0.9651, 0.9552)
  -: (1999, 2148, 2166) (0.9306, 0.9229, 0.9268)
  |: (13174, 13389, 13461) (0.9839, 0.9787, 0.9813)
Macro-average precision, recall, F1: (0.714998, 0.716681, 0.715808)
Item accuracy: 19382 / 19989 (0.9696)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #271 *****
Loss: 68151.883533
Feature norm: 141.958825
Error norm: 134.804853
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.132
Performance by label (#match, #model, #ref) (precision, recall, F1):

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['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4452, 4361) (0.9456, 0.9654, 0.9554)
-: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
|: (13175, 13390, 13461) (0.9839, 0.9788, 0.9813)
Macro-average precision, recall, F1: (0.715280, 0.716872, 0.716044)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #272 *****
Loss: 68150.967911
Feature norm: 141.954872
Error norm: 136.813524
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.122
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
-: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
|: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715444, 0.716948, 0.716164)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #273 *****
Loss: 68147.214712
Feature norm: 141.964045
Error norm: 248.187513
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.134
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
-: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
|: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715278, 0.716891, 0.716054)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #274 *****
Loss: 68144.986805
Feature norm: 141.985634
Error norm: 367.249143
Active features: 272898
Line search trials: 1

```

Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4213, 4455, 4361) (0.9457, 0.9661, 0.9558)
 -: (2000, 2144, 2166) (0.9328, 0.9234, 0.9281)
 |: (13177, 13390, 13461) (0.9841, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715652, 0.717081, 0.716332)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 701 / 999 (0.7017)

***** Iteration #275 *****
Loss: 68141.376045
Feature norm: 142.008265
Error norm: 157.377740
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
 |: (13174, 13387, 13461) (0.9841, 0.9787, 0.9814)
Macro-average precision, recall, F1: (0.715225, 0.717142, 0.716154)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #276 *****
Loss: 68140.305855
Feature norm: 142.017202
Error norm: 130.629022
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
 -: (2003, 2149, 2166) (0.9321, 0.9247, 0.9284)
 |: (13175, 13388, 13461) (0.9841, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715505, 0.717276, 0.716359)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #277 *****
Loss: 68137.923725
Feature norm: 142.037754

Error norm: 94.853403
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
-: (2003, 2149, 2166) (0.9321, 0.9247, 0.9284)
|: (13175, 13388, 13461) (0.9841, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715505, 0.717276, 0.716359)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #278 *****
Loss: 68137.421075
Feature norm: 142.063830
Error norm: 548.334762
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4451, 4361) (0.9459, 0.9654, 0.9555)
-: (2003, 2152, 2166) (0.9308, 0.9247, 0.9277)
|: (13173, 13386, 13461) (0.9841, 0.9786, 0.9813)
Macro-average precision, recall, F1: (0.715176, 0.717181, 0.716150)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #279 *****
Loss: 68134.053788
Feature norm: 142.065580
Error norm: 166.390787
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4452, 4361) (0.9456, 0.9654, 0.9554)
-: (2001, 2149, 2166) (0.9311, 0.9238, 0.9275)
|: (13174, 13388, 13461) (0.9840, 0.9787, 0.9813)
Macro-average precision, recall, F1: (0.715197, 0.716969, 0.716052)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 699 / 999 (0.6997)

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***** Iteration #280 *****
Loss: 68132.828232
Feature norm: 142.063642
Error norm: 111.556670
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
  -: (2001, 2149, 2166) (0.9311, 0.9238, 0.9275)
  |: (13175, 13388, 13461) (0.9841, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715272, 0.717045, 0.716128)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

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***** Iteration #281 *****
Loss: 68131.575956
Feature norm: 142.065278
Error norm: 122.838956
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
  -: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
  |: (13175, 13389, 13461) (0.9840, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715246, 0.716930, 0.716056)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 699 / 999 (0.6997)

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***** Iteration #282 *****
Loss: 68130.603854
Feature norm: 142.068029
Error norm: 246.055218
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
  -: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
  |: (13176, 13390, 13461) (0.9840, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715354, 0.716948, 0.716120)

```

Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #283 *****

Loss: 68129.082426
Feature norm: 142.080919
Error norm: 100.987922
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.121
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
 -: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
 |: (13175, 13389, 13461) (0.9840, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715246, 0.716930, 0.716056)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #284 *****

Loss: 68127.712600
Feature norm: 142.093016
Error norm: 120.599112
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.120
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4452, 4361) (0.9459, 0.9656, 0.9556)
 -: (2000, 2151, 2166) (0.9298, 0.9234, 0.9266)
 |: (13172, 13386, 13461) (0.9840, 0.9785, 0.9813)
Macro-average precision, recall, F1: (0.714920, 0.716874, 0.715867)
Item accuracy: 19383 / 19989 (0.9697)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #285 *****

Loss: 68126.050815
Feature norm: 142.110049
Error norm: 155.831507
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4452, 4361) (0.9456, 0.9654, 0.9554)

-: (2000, 2150, 2166) (0.9302, 0.9234, 0.9268)
|: (13172, 13387, 13461) (0.9839, 0.9785, 0.9812)
Macro-average precision, recall, F1: (0.714954, 0.716817, 0.715855)
Item accuracy: 19382 / 19989 (0.9696)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #286 *****

Loss: 68122.358881
Feature norm: 142.146974
Error norm: 202.639684
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
-: (2000, 2150, 2166) (0.9302, 0.9234, 0.9268)
|: (13174, 13388, 13461) (0.9840, 0.9787, 0.9813)
Macro-average precision, recall, F1: (0.715082, 0.716911, 0.715967)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #287 *****

Loss: 68121.428995
Feature norm: 142.159502
Error norm: 286.553022
Active features: 272898
Line search trials: 2
Line search step: 0.195042
Seconds required for this iteration: 0.221
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4450, 4361) (0.9463, 0.9656, 0.9559)
-: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
|: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715443, 0.716967, 0.716174)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #288 *****

Loss: 68118.649067
Feature norm: 142.194422
Error norm: 139.264494
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.114

Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4450, 4361) (0.9463, 0.9656, 0.9559)
-: (1998, 2146, 2166) (0.9310, 0.9224, 0.9267)
|: (13177, 13393, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715300, 0.716736, 0.715987)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #289 *****

Loss: 68117.255076
Feature norm: 142.202946
Error norm: 89.335116
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4212, 4451, 4361) (0.9463, 0.9658, 0.9560)
-: (1998, 2143, 2166) (0.9323, 0.9224, 0.9274)
|: (13179, 13395, 13461) (0.9839, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715629, 0.716830, 0.716197)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #290 *****

Loss: 68115.179078
Feature norm: 142.212979
Error norm: 102.241501
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4213, 4453, 4361) (0.9461, 0.9661, 0.9560)
-: (1998, 2143, 2166) (0.9323, 0.9224, 0.9274)
|: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715597, 0.716869, 0.716199)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 701 / 999 (0.7017)

***** Iteration #291 *****

Loss: 68114.501955
Feature norm: 142.216069
Error norm: 255.000259
Active features: 272898

Line search trials: 2
 Line search step: 0.271573
 Seconds required for this iteration: 0.233
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4213, 4453, 4361) (0.9461, 0.9661, 0.9560)
 -: (1998, 2143, 2166) (0.9323, 0.9224, 0.9274)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
 Macro-average precision, recall, F1: (0.715597, 0.716869, 0.716199)
 Item accuracy: 19389 / 19989 (0.9700)
 Instance accuracy: 701 / 999 (0.7017)

***** Iteration #292 *****
 Loss: 68113.191656
 Feature norm: 142.221574
 Error norm: 197.782043
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.117
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4213, 4453, 4361) (0.9461, 0.9661, 0.9560)
 -: (1998, 2143, 2166) (0.9323, 0.9224, 0.9274)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
 Macro-average precision, recall, F1: (0.715597, 0.716869, 0.716199)
 Item accuracy: 19389 / 19989 (0.9700)
 Instance accuracy: 701 / 999 (0.7017)

***** Iteration #293 *****
 Loss: 68110.543666
 Feature norm: 142.242770
 Error norm: 87.631979
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4449, 4361) (0.9467, 0.9658, 0.9562)
 -: (2002, 2147, 2166) (0.9325, 0.9243, 0.9284)
 |: (13179, 13393, 13461) (0.9840, 0.9791, 0.9815)
 Macro-average precision, recall, F1: (0.715804, 0.717292, 0.716518)
 Item accuracy: 19393 / 19989 (0.9702)
 Instance accuracy: 701 / 999 (0.7017)

***** Iteration #294 *****
 Loss: 68109.361493

Feature norm: 142.259120
Error norm: 108.145104
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4449, 4361) (0.9467, 0.9658, 0.9562)
 -: (2002, 2147, 2166) (0.9325, 0.9243, 0.9284)
 |: (13179, 13393, 13461) (0.9840, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715804, 0.717292, 0.716518)
Item accuracy: 19393 / 19989 (0.9702)
Instance accuracy: 701 / 999 (0.7017)

***** Iteration #295 *****
Loss: 68108.738683
Feature norm: 142.280168
Error norm: 260.417850
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4451, 4361) (0.9463, 0.9658, 0.9560)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13176, 13389, 13461) (0.9841, 0.9788, 0.9815)
Macro-average precision, recall, F1: (0.715498, 0.717236, 0.716337)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #296 *****
Loss: 68107.459602
Feature norm: 142.285942
Error norm: 118.344139
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4449, 4361) (0.9467, 0.9658, 0.9562)
 -: (2002, 2148, 2166) (0.9320, 0.9243, 0.9281)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715695, 0.717274, 0.716455)
Item accuracy: 19392 / 19989 (0.9701)
Instance accuracy: 701 / 999 (0.7017)

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***** Iteration #297 *****
Loss: 68106.556226
Feature norm: 142.288436
Error norm: 130.848005
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4212, 4448, 4361) (0.9469, 0.9658, 0.9563)
  -: (2002, 2147, 2166) (0.9325, 0.9243, 0.9284)
  |: (13180, 13394, 13461) (0.9840, 0.9791, 0.9816)
Macro-average precision, recall, F1: (0.715857, 0.717311, 0.716555)
Item accuracy: 19394 / 19989 (0.9702)
Instance accuracy: 701 / 999 (0.7017)

```

```

***** Iteration #298 *****
Loss: 68105.199366
Feature norm: 142.298660
Error norm: 162.597129
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4212, 4449, 4361) (0.9467, 0.9658, 0.9562)
  -: (2002, 2147, 2166) (0.9325, 0.9243, 0.9284)
  |: (13179, 13393, 13461) (0.9840, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715804, 0.717292, 0.716518)
Item accuracy: 19393 / 19989 (0.9702)
Instance accuracy: 700 / 999 (0.7007)

```

```

***** Iteration #299 *****
Loss: 68104.118412
Feature norm: 142.313385
Error norm: 213.771279
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4212, 4449, 4361) (0.9467, 0.9658, 0.9562)
  -: (2002, 2148, 2166) (0.9320, 0.9243, 0.9281)
  |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)

```

Macro-average precision, recall, F1: (0.715695, 0.717274, 0.716455)
Item accuracy: 19392 / 19989 (0.9701)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #300 *****

Loss: 68102.982684
Feature norm: 142.333159
Error norm: 132.651080
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4213, 4450, 4361) (0.9467, 0.9661, 0.9563)
 -: (2002, 2148, 2166) (0.9320, 0.9243, 0.9281)
 |: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715716, 0.717331, 0.716493)
Item accuracy: 19393 / 19989 (0.9702)
Instance accuracy: 701 / 999 (0.7017)

***** Iteration #301 *****

Loss: 68102.379500
Feature norm: 142.335668
Error norm: 119.279215
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4448, 4361) (0.9467, 0.9656, 0.9561)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715475, 0.717198, 0.716308)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #302 *****

Loss: 68101.329071
Feature norm: 142.347834
Error norm: 92.390765
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

```

    <: (4211, 4448, 4361) (0.9467, 0.9656, 0.9561)
    -: (2004, 2153, 2166) (0.9308, 0.9252, 0.9280)
    |: (13176, 13388, 13461) (0.9842, 0.9788, 0.9815)
Macro-average precision, recall, F1: (0.715419, 0.717410, 0.716387)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #303 *****
Loss: 68100.506553
Feature norm: 142.361404
Error norm: 72.846543
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4211, 4447, 4361) (0.9469, 0.9656, 0.9562)
    -: (2004, 2151, 2166) (0.9317, 0.9252, 0.9284)
    |: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715671, 0.717447, 0.716532)
Item accuracy: 19393 / 19989 (0.9702)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #304 *****
Loss: 68098.872669
Feature norm: 142.394334
Error norm: 157.894246
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4210, 4447, 4361) (0.9467, 0.9654, 0.9559)
    -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
    |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715472, 0.717159, 0.716288)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #305 *****
Loss: 68097.706789
Feature norm: 142.426539
Error norm: 234.018075
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```

Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4447, 4361) (0.9467, 0.9654, 0.9559)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715472, 0.717159, 0.716288)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #306 *****
Loss: 68096.625212
Feature norm: 142.427772
Error norm: 137.868397
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4447, 4361) (0.9467, 0.9654, 0.9559)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715472, 0.717159, 0.716288)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #307 *****
Loss: 68095.512913
Feature norm: 142.429756
Error norm: 87.000860
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4448, 4361) (0.9465, 0.9654, 0.9558)
 -: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715385, 0.716928, 0.716127)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #308 *****
Loss: 68094.977643
Feature norm: 142.432459
Error norm: 79.885317

Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4448, 4361) (0.9465, 0.9654, 0.9558)
-: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
|: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715276, 0.716909, 0.716064)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #309 *****
Loss: 68094.343480
Feature norm: 142.465717
Error norm: 304.959386
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4212, 4450, 4361) (0.9465, 0.9658, 0.9561)
-: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
|: (13177, 13390, 13461) (0.9841, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715319, 0.717024, 0.716142)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #310 *****
Loss: 68093.225486
Feature norm: 142.463074
Error norm: 74.564349
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4448, 4361) (0.9467, 0.9656, 0.9561)
-: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
|: (13177, 13390, 13461) (0.9841, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715385, 0.717198, 0.716264)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #311 *****

Loss: 68092.937807
 Feature norm: 142.467174
 Error norm: 59.983555
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4448, 4361) (0.9467, 0.9656, 0.9561)
 -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
 |: (13177, 13390, 13461) (0.9841, 0.9789, 0.9815)
 Macro-average precision, recall, F1: (0.715385, 0.717198, 0.716264)
 Item accuracy: 19390 / 19989 (0.9700)
 Instance accuracy: 700 / 999 (0.7007)

***** Iteration #312 *****

Loss: 68092.676734
 Feature norm: 142.473621
 Error norm: 68.764567
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4448, 4361) (0.9467, 0.9656, 0.9561)
 -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
 |: (13177, 13390, 13461) (0.9841, 0.9789, 0.9815)
 Macro-average precision, recall, F1: (0.715385, 0.717198, 0.716264)
 Item accuracy: 19390 / 19989 (0.9700)
 Instance accuracy: 700 / 999 (0.7007)

***** Iteration #313 *****

Loss: 68091.594605
 Feature norm: 142.493606
 Error norm: 101.003782
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.116
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4447, 4361) (0.9467, 0.9654, 0.9559)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
 Macro-average precision, recall, F1: (0.715472, 0.717159, 0.716288)
 Item accuracy: 19390 / 19989 (0.9700)

Instance accuracy: 699 / 999 (0.6997)

***** Iteration #314 *****

Loss: 68090.928205

Feature norm: 142.530766

Error norm: 279.924424

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.116

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4210, 4444, 4361) (0.9473, 0.9654, 0.9563)

-: (2005, 2154, 2166) (0.9308, 0.9257, 0.9282)

|: (13179, 13391, 13461) (0.9842, 0.9791, 0.9816)

Macro-average precision, recall, F1: (0.715585, 0.717524, 0.716530)

Item accuracy: 19394 / 19989 (0.9702)

Instance accuracy: 701 / 999 (0.7017)

***** Iteration #315 *****

Loss: 68089.208981

Feature norm: 142.555510

Error norm: 98.057337

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.127

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4210, 4445, 4361) (0.9471, 0.9654, 0.9562)

-: (2004, 2153, 2166) (0.9308, 0.9252, 0.9280)

|: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)

Macro-average precision, recall, F1: (0.715505, 0.717390, 0.716422)

Item accuracy: 19392 / 19989 (0.9701)

Instance accuracy: 700 / 999 (0.7007)

***** Iteration #316 *****

Loss: 68088.677786

Feature norm: 142.549113

Error norm: 66.534084

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.125

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4210, 4445, 4361) (0.9471, 0.9654, 0.9562)

-: (2004, 2153, 2166) (0.9308, 0.9252, 0.9280)

```

|: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715505, 0.717390, 0.716422)
Item accuracy: 19392 / 19989 (0.9701)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #317 *****
Loss: 68087.967884
Feature norm: 142.550358
Error norm: 56.285948
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4447, 4361) (0.9469, 0.9656, 0.9562)
  -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
  |: (13178, 13391, 13461) (0.9841, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715439, 0.717216, 0.716300)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #318 *****
Loss: 68087.353883
Feature norm: 142.552021
Error norm: 144.684048
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4448, 4361) (0.9467, 0.9656, 0.9561)
  -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
  |: (13177, 13390, 13461) (0.9841, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715385, 0.717198, 0.716264)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #319 *****
Loss: 68086.462257
Feature norm: 142.555582
Error norm: 103.317686
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):

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```

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4448, 4361) (0.9467, 0.9656, 0.9561)
-: (2002, 2152, 2166) (0.9303, 0.9243, 0.9273)
|: (13176, 13389, 13461) (0.9841, 0.9788, 0.9815)
Macro-average precision, recall, F1: (0.715277, 0.717179, 0.716200)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #320 *****
Loss: 68085.551156
Feature norm: 142.562835
Error norm: 111.216908
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
-: (2002, 2153, 2166) (0.9299, 0.9243, 0.9271)
|: (13174, 13387, 13461) (0.9841, 0.9787, 0.9814)
Macro-average precision, recall, F1: (0.715059, 0.717085, 0.716044)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #321 *****
Loss: 68083.974059
Feature norm: 142.573118
Error norm: 78.184542
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
-: (2001, 2152, 2166) (0.9298, 0.9238, 0.9268)
|: (13175, 13388, 13461) (0.9841, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715051, 0.716988, 0.715991)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #322 *****
Loss: 68082.837450
Feature norm: 142.579998
Error norm: 67.671415
Active features: 272898
Line search trials: 1

```

Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
 -: (2001, 2152, 2166) (0.9298, 0.9238, 0.9268)
 |: (13175, 13389, 13461) (0.9840, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715029, 0.716930, 0.715952)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #323 *****
Loss: 68082.584394
Feature norm: 142.581817
Error norm: 134.896542
Active features: 272898
Line search trials: 2
Line search step: 0.225136
Seconds required for this iteration: 0.224
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4449, 4361) (0.9461, 0.9651, 0.9555)
 -: (2000, 2151, 2166) (0.9298, 0.9234, 0.9266)
 |: (13174, 13389, 13461) (0.9839, 0.9787, 0.9813)
Macro-average precision, recall, F1: (0.714949, 0.716796, 0.715845)
Item accuracy: 19383 / 19989 (0.9697)
Instance accuracy: 696 / 999 (0.6967)

***** Iteration #324 *****
Loss: 68081.990458
Feature norm: 142.583533
Error norm: 83.213364
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4449, 4361) (0.9461, 0.9651, 0.9555)
 -: (2000, 2151, 2166) (0.9298, 0.9234, 0.9266)
 |: (13174, 13389, 13461) (0.9839, 0.9787, 0.9813)
Macro-average precision, recall, F1: (0.714949, 0.716796, 0.715845)
Item accuracy: 19383 / 19989 (0.9697)
Instance accuracy: 696 / 999 (0.6967)

***** Iteration #325 *****
Loss: 68081.396587
Feature norm: 142.584754

Error norm: 60.992673
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
 -: (2002, 2153, 2166) (0.9299, 0.9243, 0.9271)
 |: (13174, 13388, 13461) (0.9840, 0.9787, 0.9813)
Macro-average precision, recall, F1: (0.715037, 0.717027, 0.716005)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 696 / 999 (0.6967)

***** Iteration #326 *****
Loss: 68080.710854
Feature norm: 142.587096
Error norm: 132.450236
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
 -: (2002, 2153, 2166) (0.9299, 0.9243, 0.9271)
 |: (13174, 13388, 13461) (0.9840, 0.9787, 0.9813)
Macro-average precision, recall, F1: (0.715037, 0.717027, 0.716005)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 696 / 999 (0.6967)

***** Iteration #327 *****
Loss: 68080.275338
Feature norm: 142.590267
Error norm: 144.502443
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
 |: (13175, 13389, 13461) (0.9840, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715257, 0.717103, 0.716152)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 696 / 999 (0.6967)

```

***** Iteration #328 *****
Loss: 68079.934173
Feature norm: 142.590499
Error norm: 77.678997
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
  -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
  |: (13175, 13389, 13461) (0.9840, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715257, 0.717103, 0.716152)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 696 / 999 (0.6967)

```

```

***** Iteration #329 *****
Loss: 68079.579671
Feature norm: 142.592035
Error norm: 57.707746
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
  -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
  |: (13175, 13390, 13461) (0.9839, 0.9788, 0.9813)
Macro-average precision, recall, F1: (0.715169, 0.716872, 0.715991)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 696 / 999 (0.6967)

```

```

***** Iteration #330 *****
Loss: 68079.340158
Feature norm: 142.593866
Error norm: 74.986423
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4209, 4449, 4361) (0.9461, 0.9651, 0.9555)
  -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
  |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715166, 0.716834, 0.715971)

```


Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #331 *****

Loss: 68078.945168
Feature norm: 142.598094
Error norm: 84.566282
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4449, 4361) (0.9461, 0.9651, 0.9555)
 -: (2000, 2150, 2166) (0.9302, 0.9234, 0.9268)
 |: (13175, 13390, 13461) (0.9839, 0.9788, 0.9813)
Macro-average precision, recall, F1: (0.715058, 0.716815, 0.715908)
Item accuracy: 19384 / 19989 (0.9697)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #332 *****

Loss: 68078.588977
Feature norm: 142.601675
Error norm: 101.799613
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.121
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4449, 4361) (0.9461, 0.9651, 0.9555)
 -: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
 |: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715275, 0.716852, 0.716034)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #333 *****

Loss: 68078.232112
Feature norm: 142.601732
Error norm: 67.979459
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4449, 4361) (0.9461, 0.9651, 0.9555)

-: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
|: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715166, 0.716834, 0.715971)
Item accuracy: 19385 / 19989 (0.9698)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #334 *****

Loss: 68077.276999
Feature norm: 142.602538
Error norm: 99.795583
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
-: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
|: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #335 *****

Loss: 68076.709858
Feature norm: 142.604507
Error norm: 97.904149
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
-: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
|: (13176, 13390, 13461) (0.9840, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715254, 0.717064, 0.716131)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #336 *****

Loss: 68076.100532
Feature norm: 142.607346
Error norm: 137.124982
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117

Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
 -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
 |: (13176, 13390, 13461) (0.9840, 0.9788, 0.9814)
 Macro-average precision, recall, F1: (0.715254, 0.717064, 0.716131)
 Item accuracy: 19387 / 19989 (0.9699)
 Instance accuracy: 697 / 999 (0.6977)

***** Iteration #337 *****

Loss: 68075.499280
 Feature norm: 142.609922
 Error norm: 70.148164
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
 -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
 |: (13176, 13390, 13461) (0.9840, 0.9788, 0.9814)
 Macro-average precision, recall, F1: (0.715254, 0.717064, 0.716131)
 Item accuracy: 19387 / 19989 (0.9699)
 Instance accuracy: 697 / 999 (0.6977)

***** Iteration #338 *****

Loss: 68075.148586
 Feature norm: 142.611527
 Error norm: 68.938427
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.115
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
 -: (2002, 2151, 2166) (0.9307, 0.9243, 0.9275)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
 Macro-average precision, recall, F1: (0.715307, 0.717083, 0.716168)
 Item accuracy: 19388 / 19989 (0.9699)
 Instance accuracy: 698 / 999 (0.6987)

***** Iteration #339 *****

Loss: 68074.763276
 Feature norm: 142.613163
 Error norm: 64.132495
 Active features: 272898

Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715416, 0.717102, 0.716231)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #340 *****
Loss: 68074.399069
Feature norm: 142.615131
Error norm: 166.336866
Active features: 272898
Line search trials: 2
Line search step: 0.288264
Seconds required for this iteration: 0.236
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715362, 0.717083, 0.716195)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #341 *****
Loss: 68073.635300
Feature norm: 142.618611
Error norm: 100.213586
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715362, 0.717083, 0.716195)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #342 *****
Loss: 68072.958851

Feature norm: 142.622300
Error norm: 49.772027
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715416, 0.717102, 0.716231)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #343 *****
Loss: 68072.584523
Feature norm: 142.624404
Error norm: 57.053226
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4448, 4361) (0.9463, 0.9651, 0.9556)
 -: (2002, 2150, 2166) (0.9312, 0.9243, 0.9277)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715362, 0.717083, 0.716195)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #344 *****
Loss: 68072.390821
Feature norm: 142.626112
Error norm: 106.809368
Active features: 272898
Line search trials: 2
Line search step: 0.501428
Seconds required for this iteration: 0.223
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4209, 4446, 4361) (0.9467, 0.9651, 0.9558)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13180, 13394, 13461) (0.9840, 0.9791, 0.9816)
Macro-average precision, recall, F1: (0.715578, 0.717139, 0.716331)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 699 / 999 (0.6997)

```

***** Iteration #345 *****
Loss: 68072.084785
Feature norm: 142.627140
Error norm: 67.329365
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4209, 4447, 4361) (0.9465, 0.9651, 0.9557)
  -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
  |: (13179, 13393, 13461) (0.9840, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715525, 0.717120, 0.716294)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

```

```

***** Iteration #346 *****
Loss: 68071.670043
Feature norm: 142.628504
Error norm: 54.186887
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
  -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
  |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

```

```

***** Iteration #347 *****
Loss: 68071.163148
Feature norm: 142.631174
Error norm: 67.768541
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
  -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
  |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)

```

Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #348 *****

Loss: 68070.824060
Feature norm: 142.634049
Error norm: 133.148415
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4448, 4361) (0.9465, 0.9654, 0.9558)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715527, 0.717159, 0.716315)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #349 *****

Loss: 68070.393222
Feature norm: 142.636167
Error norm: 54.617993
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
 -: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715495, 0.716928, 0.716181)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #350 *****

Loss: 68070.181033
Feature norm: 142.636544
Error norm: 57.915089
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

```

    <: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
    -: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
    |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715495, 0.716928, 0.716181)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #351 *****
Loss: 68069.674832
Feature norm: 142.638784
Error norm: 50.498498
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
    -: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
    |: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715386, 0.716909, 0.716117)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #352 *****
Loss: 68069.529051
Feature norm: 142.640139
Error norm: 125.369144
Active features: 272898
Line search trials: 2
Line search step: 0.161222
Seconds required for this iteration: 0.230
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
    -: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
    |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715440, 0.716928, 0.716154)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #353 *****
Loss: 68069.154562
Feature norm: 142.642826
Error norm: 86.407114
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```


Seconds required for this iteration: 0.122
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715440, 0.716928, 0.716154)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #354 *****
Loss: 68068.677657
Feature norm: 142.647414
Error norm: 44.398793
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #355 *****
Loss: 68068.392717
Feature norm: 142.650172
Error norm: 42.828133
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4452, 4361) (0.9461, 0.9658, 0.9559)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715628, 0.717043, 0.716303)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #356 *****
Loss: 68068.008381
Feature norm: 142.653896
Error norm: 55.870375

Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4452, 4361) (0.9461, 0.9658, 0.9559)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715628, 0.717043, 0.716303)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #357 *****

Loss: 68067.835735
Feature norm: 142.656719
Error norm: 90.344749
Active features: 272898
Line search trials: 2
Line search step: 0.273721
Seconds required for this iteration: 0.225
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4212, 4452, 4361) (0.9461, 0.9658, 0.9559)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715628, 0.717043, 0.716303)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #358 *****

Loss: 68067.492664
Feature norm: 142.659528
Error norm: 48.070728
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #359 *****

Loss: 68067.242386
Feature norm: 142.661276
Error norm: 39.218841
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715440, 0.716928, 0.716154)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #360 *****

Loss: 68066.965691
Feature norm: 142.663141
Error norm: 76.865872
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.121
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715440, 0.716928, 0.716154)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #361 *****

Loss: 68066.849581
Feature norm: 142.667054
Error norm: 181.818000
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.130
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4447, 4361) (0.9467, 0.9654, 0.9559)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13179, 13393, 13461) (0.9840, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715581, 0.717177, 0.716351)
Item accuracy: 19391 / 19989 (0.9701)

Instance accuracy: 699 / 999 (0.6997)

***** Iteration #362 *****

Loss: 68066.476944

Feature norm: 142.666237

Error norm: 67.111521

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.130

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4210, 4448, 4361) (0.9465, 0.9654, 0.9558)

-: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)

|: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)

Macro-average precision, recall, F1: (0.715527, 0.717159, 0.716315)

Item accuracy: 19390 / 19989 (0.9700)

Instance accuracy: 699 / 999 (0.6997)

***** Iteration #363 *****

Loss: 68066.228757

Feature norm: 142.666718

Error norm: 38.775596

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.121

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)

-: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)

|: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)

Macro-average precision, recall, F1: (0.715440, 0.716928, 0.716154)

Item accuracy: 19388 / 19989 (0.9699)

Instance accuracy: 699 / 999 (0.6997)

***** Iteration #364 *****

Loss: 68066.049031

Feature norm: 142.668532

Error norm: 44.195853

Active features: 272898

Line search trials: 1

Line search step: 1.000000

Seconds required for this iteration: 0.118

Performance by label (#match, #model, #ref) (precision, recall, F1):

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)

<: (4210, 4448, 4361) (0.9465, 0.9654, 0.9558)

-: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)

```

|: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715527, 0.717159, 0.716315)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #365 *****
Loss: 68065.846831
Feature norm: 142.672779
Error norm: 77.459612
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4210, 4448, 4361) (0.9465, 0.9654, 0.9558)
  -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
  |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715527, 0.717159, 0.716315)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #366 *****
Loss: 68065.564600
Feature norm: 142.679517
Error norm: 47.352748
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
  -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
  |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #367 *****
Loss: 68065.359930
Feature norm: 142.682168
Error norm: 44.734004
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):

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['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
-: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
|: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #368 *****
Loss: 68064.894809
Feature norm: 142.690248
Error norm: 41.025878
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4450, 4361) (0.9463, 0.9656, 0.9559)
-: (2002, 2147, 2166) (0.9325, 0.9243, 0.9284)
|: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715694, 0.717216, 0.716425)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #369 *****
Loss: 68064.749809
Feature norm: 142.693275
Error norm: 126.057526
Active features: 272898
Line search trials: 2
Line search step: 0.253726
Seconds required for this iteration: 0.223
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4450, 4361) (0.9463, 0.9656, 0.9559)
-: (2002, 2147, 2166) (0.9325, 0.9243, 0.9284)
|: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715694, 0.717216, 0.716425)
Item accuracy: 19391 / 19989 (0.9701)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #370 *****
Loss: 68064.423955
Feature norm: 142.700182
Error norm: 74.757972
Active features: 272898
Line search trials: 1

```

Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #371 *****
Loss: 68064.181600
Feature norm: 142.705740
Error norm: 32.301501
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #372 *****
Loss: 68064.073981
Feature norm: 142.707303
Error norm: 32.387120
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #373 *****
Loss: 68063.918858
Feature norm: 142.709493

Error norm: 50.658451
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #374 *****
Loss: 68063.804447
Feature norm: 142.716065
Error norm: 125.536297
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #375 *****
Loss: 68063.552056
Feature norm: 142.716121
Error norm: 50.208497
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)


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***** Iteration #376 *****
Loss: 68063.397481
Feature norm: 142.718249
Error norm: 31.672530
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
  -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
  |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

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***** Iteration #377 *****
Loss: 68063.316651
Feature norm: 142.720712
Error norm: 34.109032
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
  -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
  |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

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***** Iteration #378 *****
Loss: 68063.256293
Feature norm: 142.723305
Error norm: 81.250721
Active features: 272898
Line search trials: 2
Line search step: 0.291201
Seconds required for this iteration: 0.225
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
  -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
  |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)

```

Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #379 *****

Loss: 68063.128354
Feature norm: 142.726580
Error norm: 65.855095
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
 |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #380 *****

Loss: 68062.795869
Feature norm: 142.736905
Error norm: 34.275831
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2144, 2166) (0.9328, 0.9234, 0.9281)
 |: (13179, 13394, 13461) (0.9839, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715716, 0.717004, 0.716327)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #381 *****

Loss: 68062.611849
Feature norm: 142.743046
Error norm: 34.170252
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)

-: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
|: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #382 *****

Loss: 68062.461469
Feature norm: 142.749758
Error norm: 42.474571
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
-: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
|: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715495, 0.716928, 0.716181)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #383 *****

Loss: 68062.358259
Feature norm: 142.752548
Error norm: 56.375861
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
-: (2000, 2147, 2166) (0.9315, 0.9234, 0.9274)
|: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715440, 0.716928, 0.716154)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 700 / 999 (0.7007)

***** Iteration #384 *****

Loss: 68062.262162
Feature norm: 142.753227
Error norm: 30.264393
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115

Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
-: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
|: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715495, 0.716928, 0.716181)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #385 *****

Loss: 68062.157692
Feature norm: 142.755282
Error norm: 30.046160
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
-: (2000, 2144, 2166) (0.9328, 0.9234, 0.9281)
|: (13179, 13394, 13461) (0.9839, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715716, 0.717004, 0.716327)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #386 *****

Loss: 68062.098487
Feature norm: 142.756886
Error norm: 36.321762
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
-: (2000, 2144, 2166) (0.9328, 0.9234, 0.9281)
|: (13179, 13394, 13461) (0.9839, 0.9791, 0.9815)
Macro-average precision, recall, F1: (0.715716, 0.717004, 0.716327)
Item accuracy: 19390 / 19989 (0.9700)
Instance accuracy: 699 / 999 (0.6997)

***** Iteration #387 *****

Loss: 68061.993822
Feature norm: 142.761548
Error norm: 36.117598
Active features: 272898

Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.118
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2144, 2166) (0.9328, 0.9234, 0.9281)
 |: (13179, 13394, 13461) (0.9839, 0.9791, 0.9815)
 Macro-average precision, recall, F1: (0.715716, 0.717004, 0.716327)
 Item accuracy: 19390 / 19989 (0.9700)
 Instance accuracy: 699 / 999 (0.6997)

***** Iteration #388 *****
 Loss: 68061.815052
 Feature norm: 142.770827
 Error norm: 36.374475
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.117
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4450, 4361) (0.9463, 0.9656, 0.9559)
 -: (2002, 2147, 2166) (0.9325, 0.9243, 0.9284)
 |: (13178, 13392, 13461) (0.9840, 0.9790, 0.9815)
 Macro-average precision, recall, F1: (0.715694, 0.717216, 0.716425)
 Item accuracy: 19391 / 19989 (0.9701)
 Instance accuracy: 698 / 999 (0.6987)

***** Iteration #389 *****
 Loss: 68061.622790
 Feature norm: 142.783567
 Error norm: 45.444694
 Active features: 272898
 Line search trials: 1
 Line search step: 1.000000
 Seconds required for this iteration: 0.117
 Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
 Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
 Item accuracy: 19389 / 19989 (0.9700)
 Instance accuracy: 698 / 999 (0.6987)

***** Iteration #390 *****
 Loss: 68061.489105

Feature norm: 142.791022
Error norm: 36.082361
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #391 *****
Loss: 68061.406160
Feature norm: 142.794592
Error norm: 33.783946
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #392 *****
Loss: 68061.326555
Feature norm: 142.795553
Error norm: 28.222472
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2002, 2149, 2166) (0.9316, 0.9243, 0.9279)
 |: (13177, 13391, 13461) (0.9840, 0.9789, 0.9815)
Macro-average precision, recall, F1: (0.715474, 0.717140, 0.716278)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

```

***** Iteration #393 *****
Loss: 68061.202671
Feature norm: 142.800340
Error norm: 26.309925
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
  -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
  |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)
Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

```

```

***** Iteration #394 *****
Loss: 68061.025583
Feature norm: 142.810934
Error norm: 41.152793
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
  -: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
  |: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715498, 0.716967, 0.716201)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 697 / 999 (0.6977)

```

```

***** Iteration #395 *****
Loss: 68060.974004
Feature norm: 142.814793
Error norm: 42.071557
Active features: 272898
Line search trials: 2
Line search step: 0.256415
Seconds required for this iteration: 0.224
Performance by label (#match, #model, #ref) (precision, recall, F1):
  ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
  <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
  -: (2000, 2145, 2166) (0.9324, 0.9234, 0.9279)
  |: (13178, 13393, 13461) (0.9839, 0.9790, 0.9815)

```

Macro-average precision, recall, F1: (0.715607, 0.716985, 0.716264)
Item accuracy: 19389 / 19989 (0.9700)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #396 *****

Loss: 68060.912199
Feature norm: 142.819336
Error norm: 22.529407
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
 |: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715498, 0.716967, 0.716201)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #397 *****

Loss: 68060.866786
Feature norm: 142.821915
Error norm: 20.818644
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4211, 4451, 4361) (0.9461, 0.9656, 0.9557)
 -: (2000, 2146, 2166) (0.9320, 0.9234, 0.9276)
 |: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715498, 0.716967, 0.716201)
Item accuracy: 19388 / 19989 (0.9699)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #398 *****

Loss: 68060.759265
Feature norm: 142.828672
Error norm: 36.058399
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.118
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)


```

    <: (4210, 4450, 4361) (0.9461, 0.9654, 0.9556)
    -: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
    |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715278, 0.716891, 0.716054)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 697 / 999 (0.6977)

***** Iteration #399 *****
Loss: 68060.660560
Feature norm: 142.839669
Error norm: 53.622593
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
    -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
    |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #400 *****
Loss: 68060.558542
Feature norm: 142.843771
Error norm: 34.938868
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
    ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
    <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
    -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
    |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #401 *****
Loss: 68060.499828
Feature norm: 142.845371
Error norm: 20.903432
Active features: 272898
Line search trials: 1
Line search step: 1.000000

```

Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
 |: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715331, 0.716909, 0.716091)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #402 *****
Loss: 68060.458321
Feature norm: 142.847303
Error norm: 19.080213
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2148, 2166) (0.9311, 0.9234, 0.9272)
 |: (13177, 13392, 13461) (0.9839, 0.9789, 0.9814)
Macro-average precision, recall, F1: (0.715331, 0.716909, 0.716091)
Item accuracy: 19387 / 19989 (0.9699)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #403 *****
Loss: 68060.369869
Feature norm: 142.853299
Error norm: 30.149118
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.117
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
 |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #404 *****
Loss: 68060.272008
Feature norm: 142.863373
Error norm: 39.061599

Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.130
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
 |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #405 *****

Loss: 68060.229720
Feature norm: 142.878866
Error norm: 84.547530
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.119
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
 |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #406 *****

Loss: 68060.141685
Feature norm: 142.875460
Error norm: 26.243049
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
 |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #407 *****

Loss: 68060.111293
Feature norm: 142.875790
Error norm: 20.366081
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.115
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
 |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

***** Iteration #408 *****

Loss: 68060.087704
Feature norm: 142.877666
Error norm: 24.899780
Active features: 272898
Line search trials: 1
Line search step: 1.000000
Seconds required for this iteration: 0.116
Performance by label (#match, #model, #ref) (precision, recall, F1):
 ['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
 <: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
 -: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
 |: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)
Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)
Item accuracy: 19386 / 19989 (0.9698)
Instance accuracy: 698 / 999 (0.6987)

L-BFGS terminated with the stopping criteria

Total seconds required for training: 51.759

End time of the training: 2026-07-25T10:13:48Z

real 0m54.335s
user 0m54.260s
sys 0m0.070s

1.17 Tagging with Trained Model

```
[94]: !time crfsuite tag -m ws_tag3.model /home/ye/aif2/word-seg/data/myPOS/tag/test.  
      ↪crfsuite.txt > /home/ye/aif2/word-seg/data/myPOS/tag/test.tag.out
```

```
real    0m0.058s  
user    0m0.028s  
sys     0m0.030s
```

```
[95]: !head -n 50 /home/ye/aif2/word-seg/data/myPOS/tag/test.tag.out
```

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```

1.19 Check Tagwise Performance

```
[99]: !time crfsuite tag -qt -m ./ws_tag3.model /home/ye/aif2/word-seg/data/myPOS/tag/
      ↪test.crfsuite.txt
```

Performance by label (#match, #model, #ref) (precision, recall, F1):

```

['w', 'y']: (0, 0, 1) (0.0000, 0.0000, 0.0000)
<: (4210, 4449, 4361) (0.9463, 0.9654, 0.9557)
-: (2000, 2149, 2166) (0.9307, 0.9234, 0.9270)
|: (13176, 13391, 13461) (0.9839, 0.9788, 0.9814)

```

Macro-average precision, recall, F1: (0.715222, 0.716891, 0.716028)

Item accuracy: 19386 / 19989 (0.9698)

Instance accuracy: 698 / 999 (0.6987)

Elapsed time: 0.021141 [sec] (47254.2 [instance/sec])

```

real    0m0.044s
user    0m0.022s
sys      0m0.022s

```

1.20 Dump the Trained Model

```
[92]: %pwd
```

```
[92]: '/home/ye/aif2/word-seg/tool/crfsuite/example'
```

```
[93]: !ls ws_tag3.model -hl
```

```
-rw-rw-r-- 1 ye ye 21M Jul 25 17:07 ws_tag3.model
```

```
[101]: !crfsuite dump ./ws_tag3.model | head -n 200
```

```
FILEHEADER = {  
    magic: 1CRF  
    size: 22010616  
    type: FOMC  
    version: 100  
    num_features: 0  
    num_labels: 4  
    num_attrs: 229827  
    off_features: 0x30  
    off_labels: 0x534864  
    off_attrs: 0x5350FD  
    off_labelrefs: 0x1232498  
    off_attrrefs: 0x12324E8  
}
```

```
LABELS = {  
    0: ['w', 'y']  
    1: <  
    2: -  
    3: |  
}
```

```
ATTRIBUTES = {  
    0: w[0]=  
    1: w[1]=  
    2: w[2]=  
    3: w[0] | w[1]= |  
    4: __BOS__  
    5: w[-1]=  
    6: w[0]=  
    7: w[1]=  
    8: w[2]=  
    9: w[-1] | w[0]= |  
    10: w[0] | w[1]= |  
    11: w[-2]=  
    12: w[-1]=
```


13: $w[0] = 6$
 14: $w[1] = 7$
 15: $w[2] = 9$
 16: $w[-1] \mid w[0] = 9 \mid 6$
 17: $w[0] \mid w[1] = 6 \mid 7$
 18: $w[-2] = 9$
 19: $w[-1] = 6$
 20: $w[0] = 7$
 21: $w[1] = 9$
 22: $w[2] = \text{နှစ်}$
 23: $w[-1] \mid w[0] = 6 \mid 7$
 24: $w[0] \mid w[1] = 7 \mid 9$
 25: $w[-2] = 6$
 26: $w[-1] = 7$
 27: $w[0] = 9$
 28: $w[1] = \text{နှစ်}$
 29: $w[2] = \text{ခန့်}$
 30: $w[-1] \mid w[0] = 7 \mid 9$
 31: $w[0] \mid w[1] = 9 \mid \text{နှစ်}$
 32: $w[-2] = 7$
 33: $w[-1] = 9$
 34: $w[0] = \text{နှစ်}$
 35: $w[1] = \text{ခန့်}$
 36: $w[2] = \text{မှန်း}$
 37: $w[-1] \mid w[0] = 9 \mid \text{နှစ်}$
 38: $w[0] \mid w[1] = \text{နှစ်} \mid \text{ခန့်}$
 39: $w[-2] = 9$
 40: $w[-1] = \text{နှစ်}$
 41: $w[0] = \text{ခန့်}$
 42: $w[1] = \text{မှန်း}$
 43: $w[2] = \text{သန်း}$
 44: $w[-1] \mid w[0] = \text{နှစ်} \mid \text{ခန့်}$
 45: $w[0] \mid w[1] = \text{ခန့်} \mid \text{မှန်း}$
 46: $w[-2] = \text{နှစ်}$
 47: $w[-1] = \text{ခန့်}$
 48: $w[0] = \text{မှန်း}$
 49: $w[1] = \text{သန်း}$
 50: $w[2] = \text{ခေါင်}$
 51: $w[-1] \mid w[0] = \text{ခန့်} \mid \text{မှန်း}$
 52: $w[0] \mid w[1] = \text{မှန်း} \mid \text{သန်း}$
 53: $w[-2] = \text{ခန့်}$
 54: $w[-1] = \text{မှန်း}$

55: $w[0] = \text{သန်း}$
 56: $w[1] = \text{ခေါင်}$
 57: $w[2] = \text{စာ}$
 58: $w[-1] \mid w[0] = \text{မှန်း} \mid \text{သန်း}$
 59: $w[0] \mid w[1] = \text{သန်း} \mid \text{ခေါင်}$
 60: $w[-2] = \text{မှန်း}$
 61: $w[-1] = \text{သန်း}$
 62: $w[0] = \text{ခေါင်}$
 63: $w[1] = \text{စာ}$
 64: $w[2] = \text{ရင်း}$
 65: $w[-1] \mid w[0] = \text{သန်း} \mid \text{ခေါင်}$
 66: $w[0] \mid w[1] = \text{ခေါင်} \mid \text{စာ}$
 67: $w[-2] = \text{သန်း}$
 68: $w[-1] = \text{ခေါင်}$
 69: $w[0] = \text{စာ}$
 70: $w[1] = \text{ရင်း}$
 71: $w[2] = \text{အ}$
 72: $w[-1] \mid w[0] = \text{ခေါင်} \mid \text{စာ}$
 73: $w[0] \mid w[1] = \text{စာ} \mid \text{ရင်း}$
 74: $w[-2] = \text{ခေါင်}$
 75: $w[-1] = \text{စာ}$
 76: $w[0] = \text{ရင်း}$
 77: $w[1] = \text{အ}$
 78: $w[2] = \text{ရ}$
 79: $w[-1] \mid w[0] = \text{စာ} \mid \text{ရင်း}$
 80: $w[0] \mid w[1] = \text{ရင်း} \mid \text{အ}$
 81: $w[-2] = \text{စာ}$
 82: $w[-1] = \text{ရင်း}$
 83: $w[0] = \text{အ}$
 84: $w[1] = \text{ရ}$
 85: $w[2] = \text{လူ}$
 86: $w[-1] \mid w[0] = \text{ရင်း} \mid \text{အ}$
 87: $w[0] \mid w[1] = \text{အ} \mid \text{ရ}$
 88: $w[-2] = \text{ရင်း}$
 89: $w[-1] = \text{အ}$
 90: $w[0] = \text{ရ}$
 91: $w[1] = \text{လူ}$
 92: $w[2] = \text{ဦး}$
 93: $w[-1] \mid w[0] = \text{အ} \mid \text{ရ}$
 94: $w[0] \mid w[1] = \text{ရ} \mid \text{လူ}$
 95: $w[-2] = \text{အ}$
 96: $w[-1] = \text{ရ}$
 97: $w[0] = \text{လူ}$

98: $w[1] = \text{ဦး}$
 99: $w[2] = \text{ရေ}$
 100: $w[-1] \mid w[0] = \text{ရ} \mid \text{လူ}$
 101: $w[0] \mid w[1] = \text{လူ} \mid \text{ဦး}$
 102: $w[-2] = \text{ရ}$
 103: $w[-1] = \text{လူ}$
 104: $w[0] = \text{ဦး}$
 105: $w[1] = \text{ရေ}$
 106: $w[2] = \text{၁}$
 107: $w[-1] \mid w[0] = \text{လူ} \mid \text{ဦး}$
 108: $w[0] \mid w[1] = \text{ဦး} \mid \text{ရေ}$
 109: $w[-2] = \text{လူ}$
 110: $w[-1] = \text{ဦး}$
 111: $w[0] = \text{ရေ}$
 112: $w[1] = \text{၁}$
 113: $w[-1] \mid w[0] = \text{ဦး} \mid \text{ရေ}$
 114: $w[0] \mid w[1] = \text{ရေ} \mid \text{၁}$
 115: $w[-2] = \text{ဦး}$
 116: $w[-1] = \text{ရေ}$
 117: $w[2] = \text{၅}$
 118: $w[-1] \mid w[0] = \text{ရေ} \mid \text{၁}$
 119: $w[0] \mid w[1] = \text{၁} \mid \text{၁}$
 120: $w[-2] = \text{ရေ}$
 121: $w[1] = \text{၅}$
 122: $w[2] = \text{၉}$
 123: $w[-1] \mid w[0] = \text{၁} \mid \text{၁}$
 124: $w[0] \mid w[1] = \text{၁} \mid \text{၅}$
 125: $w[0] = \text{၅}$
 126: $w[2] = \text{၃}$
 127: $w[-1] \mid w[0] = \text{၁} \mid \text{၅}$
 128: $w[0] \mid w[1] = \text{၅} \mid \text{၉}$
 129: $w[-1] = \text{၅}$
 130: $w[1] = \text{၃}$
 131: $w[-1] \mid w[0] = \text{၅} \mid \text{၉}$
 132: $w[0] \mid w[1] = \text{၉} \mid \text{၃}$
 133: $w[-2] = \text{၅}$
 134: $w[0] = \text{၃}$
 135: $w[2] = \text{ယောက်}$
 136: $w[-1] \mid w[0] = \text{၉} \mid \text{၃}$
 137: $w[0] \mid w[1] = \text{၃} \mid \text{၁}$
 138: $w[-1] = \text{၃}$
 139: $w[1] = \text{ယောက်}$
 140: $w[2] = \text{ရှိ}$

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141: w[-1] | w[0] = ဂ | ဝ
142: w[0] | w[1] = ဝ | ယောက်
143: w[-2] = ဂ
144: w[0] = ယောက်
145: w[1] = ရှိ
146: w[2] = သည်
147: w[-1] | w[0] = ဝ | ယောက်
148: w[0] | w[1] = ယောက် | ရှိ
149: w[-1] = ယောက်
150: w[0] = ရှိ
151: w[1] = သည်
152: w[-1] | w[0] = ယောက် | ရှိ
153: w[0] | w[1] = ရှိ | သည်
154: w[-2] = ယောက်
155: w[-1] = ရှိ
156: w[0] = သည်
157: w[-1] | w[0] = ရှိ | သည်
158: __EOS__
159: w[1] = တိုင်း
160: w[2] = တွင်
161: w[0] | w[1] = လူ | တိုင်း
162: w[0] = တိုင်း
163: w[1] = တွင်
164: w[2] = သင့်
165: w[-1] | w[0] = လူ | တိုင်း
166: w[0] | w[1] = တိုင်း | တွင်
167: w[-1] = တိုင်း
168: w[0] = တွင်
169: w[1] = သင့်
170: w[2] = မြတ်
171: w[-1] | w[0] = တိုင်း | တွင်
172: w[0] | w[1] = တွင် | သင့်
173: w[-2] = တိုင်း
174: w[-1] = တွင်
175: w[0] = သင့်
176: w[1] = မြတ်

```

```
[102]: !crfsuite dump ./ws_tag3.model > ./ws_tag3.model.txt
```

```
[103]: !tail -n 200 ./ws_tag3.model.txt
```

```

(0) w[0] | w[1] = လျှင် | ရှေ့ --> | : 0.002001
(0) w[-1] | w[0] = လျှင် | ရှေ့ --> | : 0.342502

```

(0) $w[0] | w[1] = \text{ရှေ့} | \text{ခြေ} \rightarrow | : 0.342502$
 (0) $w[-1] | w[0] = \text{ရှေ့} | \text{ခြေ} \rightarrow | : 0.012368$
 (0) $w[0] | w[1] = \text{ရွက်} | \text{လာ} \rightarrow | : 0.005379$
 (0) $w[-1] | w[0] = \text{ရွက်} | \text{လာ} \rightarrow | : 0.000317$
 (0) $w[0] | w[1] = \text{သောက်} | \text{ချောင်း} \rightarrow | : 0.020503$
 (0) $w[-1] | w[0] = \text{သောက်} | \text{ချောင်း} \rightarrow < : 0.099385$
 (0) $w[0] | w[1] = \text{ကဲ} | \text{ဦး} \rightarrow | : 0.057558$
 (0) $w[-1] | w[0] = \text{ကဲ} | \text{ဦး} \rightarrow | : 0.025290$
 (0) $w[0] | w[1] = \text{ကျော်} | \text{တ} \rightarrow | : 0.228098$
 (0) $w[-1] | w[0] = \text{ကျော်} | \text{တ} \rightarrow < : 0.233963$
 (0) $w[0] | w[1] = \text{ဘက်} | \text{ခြုံ} \rightarrow | : 0.114696$
 (0) $w[-1] | w[0] = \text{ဘက်} | \text{ခြုံ} \rightarrow | : 0.037383$
 (0) $w[0] | w[1] = \text{ထား} | \text{ပျို့} \rightarrow | : 0.050724$
 (0) $w[-1] | w[0] = \text{ထား} | \text{ပျို့} \rightarrow | : 0.125101$
 (0) $w[0] | w[1] = \text{ပျို့} | \text{နား} \rightarrow | : 0.125101$
 (0) $w[-1] | w[0] = \text{ပျို့} | \text{နား} \rightarrow | : 0.047008$
 (0) $w[0] | w[1] = \text{နား} | \text{နှစ်} \rightarrow | : 0.047008$
 (0) $w[-1] | w[0] = \text{နား} | \text{နှစ်} \rightarrow | : 0.088119$
 (0) $w[0] | w[1] = \text{အောင်} | \text{ခြုံ} \rightarrow | : 0.213086$
 (0) $w[-1] | w[0] = \text{အောင်} | \text{ခြုံ} \rightarrow | : 0.098216$
 (0) $w[0] | w[1] = \text{ခြုံ} | \text{နော်} \rightarrow | : 0.098216$
 (0) $w[-1] | w[0] = \text{ခြုံ} | \text{နော်} \rightarrow | : 0.021393$
 (0) $w[0] | w[1] = \text{နော်} | \text{ရေ} \rightarrow | : 0.021393$
 (0) $w[-1] | w[0] = \text{နော်} | \text{ရေ} \rightarrow < : 0.039212$
 (0) $w[0] | w[1] = \text{ဤ} | \text{ဥ} \rightarrow | : 0.027485$
 (0) $w[-1] | w[0] = \text{ဤ} | \text{ဥ} \rightarrow < : 0.052937$
 (0) $w[0] | w[1] = \text{တည်း} | \text{ပြီး} \rightarrow | : 0.001094$
 (0) $w[-1] | w[0] = \text{တည်း} | \text{ပြီး} \rightarrow | : 0.004032$
 (0) $w[0] | w[1] = \text{ရွှေ} | \text{စက်} \rightarrow < : 0.070065$
 (0) $w[-1] | w[0] = \text{ရွှေ} | \text{စက်} \rightarrow - : 0.121614$
 (0) $w[0] | w[1] = \text{စက်} | \text{တော်} \rightarrow - : 0.121614$
 (0) $w[-1] | w[0] = \text{စက်} | \text{တော်} \rightarrow | : 0.004100$
 (0) $w[0] | w[1] = \text{ရား} | \text{ခြေ} \rightarrow | : 0.093414$
 (0) $w[-1] | w[0] = \text{ရား} | \text{ခြေ} \rightarrow < : 0.099156$
 (0) $w[0] | w[1] = \text{ခြေ} | \text{တော်} \rightarrow < : 0.099156$
 (0) $w[-1] | w[0] = \text{ခြေ} | \text{တော်} \rightarrow - : 0.185933$
 (0) $w[0] | w[1] = \text{နောင်} | \text{ဗ} \rightarrow | : 0.023179$
 (0) $w[-1] | w[0] = \text{နောင်} | \text{ဗ} \rightarrow < : 0.023529$
 (0) $w[0] | w[1] = \text{၏} | \text{မဏ္ဍိုင်} \rightarrow | : 0.003660$
 (0) $w[-1] | w[0] = \text{၏} | \text{မဏ္ဍိုင်} \rightarrow | : 0.166529$
 (0) $w[0] | w[1] = \text{မဏ္ဍိုင်} | \text{သ} \rightarrow | : 0.166529$

(0) $w[-1] | w[0] = \text{မဏ္ဍိုင်} | \text{သ} \rightarrow <: 0.157215$
 (0) $w[0] | w[1] = \text{မည့်} | \text{ဤ} \rightarrow |: 0.018874$
 (0) $w[-1] | w[0] = \text{မည့်} | \text{ဤ} \rightarrow |: 0.024122$
 (0) $w[0] | w[1] = \text{ဤ} | \text{ရဲ} \rightarrow |: 0.024122$
 (0) $w[-1] | w[0] = \text{ဤ} | \text{ရဲ} \rightarrow <: 0.042284$
 (0) $w[0] | w[1] = \text{ဝင်} | \text{ပုဂ္ဂိုလ်} \rightarrow |: 0.060354$
 (0) $w[-1] | w[0] = \text{ဝင်} | \text{ပုဂ္ဂိုလ်} \rightarrow |: 0.003041$
 (0) $w[0] | w[1] = \text{ကူ} | \text{လိုက်} \rightarrow |: 0.007304$
 (0) $w[-1] | w[0] = \text{ကူ} | \text{လိုက်} \rightarrow <: 0.463897$
 (0) $w[0] | w[1] = \text{ရို} | \text{ဂိုဏ်း} \rightarrow |: 0.003136$
 (0) $w[-1] | w[0] = \text{ရို} | \text{ဂိုဏ်း} \rightarrow <: 0.060354$
 (0) $w[0] | w[1] = \text{နပ်} | \text{လက်} \rightarrow |: 0.033477$
 (0) $w[-1] | w[0] = \text{နပ်} | \text{လက်} \rightarrow <: 0.039189$
 (0) $w[0] | w[1] = \text{ပြင်} | \text{နောင်} \rightarrow |: 0.079752$
 (0) $w[-1] | w[0] = \text{ပြင်} | \text{နောင်} \rightarrow |: 0.000707$
 (0) $w[0] | w[1] = \text{မည့်} | \text{မျိုး} \rightarrow |: 0.003077$
 (0) $w[-1] | w[0] = \text{မည့်} | \text{မျိုး} \rightarrow <: 0.012073$
 (0) $w[0] | w[1] = \text{ကို} | \text{ပျိုး} \rightarrow |: 0.028643$
 (0) $w[-1] | w[0] = \text{ကို} | \text{ပျိုး} \rightarrow |: 0.105071$
 (0) $w[0] | w[1] = \text{ပျိုး} | \text{လိုက်} \rightarrow |: 0.105071$
 (0) $w[-1] | w[0] = \text{ပျိုး} | \text{လိုက်} \rightarrow |: 0.004460$
 (0) $w[0] | w[1] = \text{လို့} | \text{ပြိုင်} \rightarrow |: 0.000585$
 (0) $w[-1] | w[0] = \text{လို့} | \text{ပြိုင်} \rightarrow <: 0.004477$
 (0) $w[0] | w[1] = \text{ပွဲ} | \text{ရပ်} \rightarrow |: 0.046114$
 (0) $w[-1] | w[0] = \text{ပွဲ} | \text{ရပ်} \rightarrow <: 0.106711$
 (0) $w[0] | w[1] = \text{ဟမ္မဒ်} | \text{ကွယ်} \rightarrow |: 0.035705$
 (0) $w[-1] | w[0] = \text{ဟမ္မဒ်} | \text{ကွယ်} \rightarrow <: 0.069235$
 (0) $w[0] | w[1] = \text{ခ} | \text{လီ} \rightarrow <: 0.198667$
 (0) $w[-1] | w[0] = \text{ခ} | \text{လီ} \rightarrow -: 0.165982$
 (0) $w[0] | w[1] = \text{လီ} | \text{ဖာ} \rightarrow -: 0.165982$
 (0) $w[-1] | w[0] = \text{လီ} | \text{ဖာ} \rightarrow -: 0.223739$
 (0) $w[0] | w[1] = \text{ဖာ} | \text{အ} \rightarrow -: 0.125238$
 (0) $w[2] = \text{ဘက္ကရ်} \rightarrow -: 0.120339$
 (0) $w[-1] | w[0] = \text{ဖာ} | \text{အ} \rightarrow -: 0.120339$
 (0) $w[0] | w[1] = \text{အ} | \text{ဘူ} \rightarrow -: 0.120339$
 (0) $w[1] = \text{ဘက္ကရ်} \rightarrow -: 0.016299$
 (0) $w[-1] | w[0] = \text{အ} | \text{ဘူ} \rightarrow -: 0.016299$
 (0) $w[0] | w[1] = \text{ဘူ} | \text{ဘက္ကရ်} \rightarrow -: 0.016299$
 (0) $w[0] = \text{ဘက္ကရ်} \rightarrow |: 0.027257$
 (0) $w[-1] | w[0] = \text{ဘူ} | \text{ဘက္ကရ်} \rightarrow |: 0.027257$
 (0) $w[0] | w[1] = \text{ဘက္ကရ်} | \text{က} \rightarrow |: 0.027257$
 (0) $w[-1] = \text{ဘက္ကရ်} \rightarrow |: 0.004569$

(0) $w[-1] | w[0] = \text{ဘက္ကရီ} | \text{က} \rightarrow | : 0.004569$
 (0) $w[-2] = \text{ဘက္ကရီ} \rightarrow < : 0.007555$
 (0) $w[0] | w[1] = \text{ဖြစ်} | \text{စု} \rightarrow | : 0.008388$
 (0) $w[-1] | w[0] = \text{ဖြစ်} | \text{စု} \rightarrow < : 0.010786$
 (0) $w[2] = \text{အွတ်စ်} \rightarrow - : 0.103069$
 (0) $w[1] = \text{အွတ်စ်} \rightarrow - : 0.098501$
 (0) $w[0] | w[1] = \text{ဖာ} | \text{အွတ်စ်} \rightarrow - : 0.098501$
 (0) $w[0] = \text{အွတ်စ်} \rightarrow - : 0.045981$
 (0) $w[-1] | w[0] = \text{ဖာ} | \text{အွတ်စ်} \rightarrow - : 0.045981$
 (0) $w[0] | w[1] = \text{အွတ်စ်} | \text{မန်း} \rightarrow - : 0.045981$
 (0) $w[-1] = \text{အွတ်စ်} \rightarrow | : 0.084134$
 (0) $w[-1] | w[0] = \text{အွတ်စ်} | \text{မန်း} \rightarrow | : 0.084134$
 (0) $w[-2] = \text{အွတ်စ်} \rightarrow | : 0.143767$
 (0) $w[0] | w[1] = \text{ဆုံး} | \text{ကွန်} \rightarrow | : 0.000392$
 (0) $w[-1] | w[0] = \text{ဆုံး} | \text{ကွန်} \rightarrow < : 0.001350$
 (0) $w[0] | w[1] = \text{ငှင်း} | \text{စက်} \rightarrow | : 0.205127$
 (0) $w[-1] | w[0] = \text{ငှင်း} | \text{စက်} \rightarrow | : 0.036172$
 (0) $w[0] | w[1] = \text{ခါ} | \text{လေး} \rightarrow | : 0.005498$
 (0) $w[-1] | w[0] = \text{ခါ} | \text{လေး} \rightarrow < : 0.311441$
 (0) $w[0] | w[1] = \text{ဖက်} | \text{သွား} \rightarrow - : 0.461423$
 (0) $w[-1] | w[0] = \text{ဖက်} | \text{သွား} \rightarrow | : 0.003381$
 (0) $w[0] | w[1] = \text{ထင်} | \text{ကောင်း} \rightarrow | : 0.171776$
 (0) $w[-1] | w[0] = \text{ထင်} | \text{ကောင်း} \rightarrow | : 0.037144$
 (0) $w[0] | w[1] = \text{ဆား} | \text{ခြစ်} \rightarrow < : 0.123523$
 (0) $w[-1] | w[0] = \text{ဆား} | \text{ခြစ်} \rightarrow - : 0.434820$
 (0) $w[0] | w[1] = \text{လုံး} | \text{မင်း} \rightarrow | : 0.020885$
 (0) $w[-1] | w[0] = \text{လုံး} | \text{မင်း} \rightarrow | : 0.002154$
 (0) $w[0] | w[1] = \text{ဖက်} | \text{လား} \rightarrow | : 0.014830$
 (0) $w[-1] | w[0] = \text{ဖက်} | \text{လား} \rightarrow | : 0.000135$
 (0) $w[0] | w[1] = \text{ညှိ} | \text{အေ} \rightarrow | : 0.030455$
 (0) $w[-1] | w[0] = \text{ညှိ} | \text{အေ} \rightarrow < : 0.082320$
 (0) $w[0] | w[1] = \text{တို့} | \text{ယဉ်} \rightarrow | : 0.000540$
 (0) $w[-1] | w[0] = \text{တို့} | \text{ယဉ်} \rightarrow < : 0.003210$
 (0) $w[0] | w[1] = \text{ငုံ} | \text{ယဉ်} \rightarrow | : 0.011294$
 (0) $w[-1] | w[0] = \text{ငုံ} | \text{ယဉ်} \rightarrow | : 0.072348$
 (0) $w[0] | w[1] = \text{တန်း} | \text{ချင်း} \rightarrow | : 0.198816$
 (0) $w[-1] | w[0] = \text{တန်း} | \text{ချင်း} \rightarrow | : 0.005719$
 (0) $w[0] | w[1] = \text{n} | \text{ပြည်} \rightarrow | : 0.003695$
 (0) $w[-1] | w[0] = \text{n} | \text{ပြည်} \rightarrow < : 0.008992$
 (0) $w[0] | w[1] = \text{လည်} | \text{ပြောင်း} \rightarrow | : 0.010481$
 (0) $w[-1] | w[0] = \text{လည်} | \text{ပြောင်း} \rightarrow < : 0.095882$

(0) $w[0] | w[1] = \text{ရွှေ} | \text{ရင်း} \rightarrow | : 0.047795$
 (0) $w[-1] | w[0] = \text{ရွှေ} | \text{ရင်း} \rightarrow < : 0.058645$
 (0) $w[0] | w[1] = \text{မှ} | \text{မိုက်} \rightarrow | : 0.001728$
 (0) $w[-1] | w[0] = \text{မှ} | \text{မိုက်} \rightarrow < : 0.007265$
 (0) $w[0] | w[1] = \text{၏} | \text{သမ္မ} \rightarrow | : 0.001042$
 (0) $w[-1] | w[0] = \text{၏} | \text{သမ္မ} \rightarrow < : 0.003049$
 (0) $w[0] | w[1] = \text{နှင့်} | \text{ဘုတ်} \rightarrow | : 0.026246$
 (0) $w[-1] | w[0] = \text{နှင့်} | \text{ဘုတ်} \rightarrow | : 0.017939$
 (0) $w[0] | w[1] = \text{ဝင်} | \text{ဥက္ကဋ္ဌ} \rightarrow | : 0.081669$
 (0) $w[-1] | w[0] = \text{ဝင်} | \text{ဥက္ကဋ္ဌ} \rightarrow | : 0.023395$
 (0) $w[0] | w[1] = \text{အင်း} | \text{၁} \rightarrow | : 0.002519$
 (0) $w[-1] | w[0] = \text{အင်း} | \text{၁} \rightarrow < : 0.004513$
 (0) $w[0] | w[1] = \text{ဝ} | \text{လမ်း} \rightarrow | : 0.066940$
 (0) $w[-1] | w[0] = \text{ဝ} | \text{လမ်း} \rightarrow | : 0.006724$
 (0) $w[0] | w[1] = \text{ရဲ့} | \text{ဌာ} \rightarrow | : 0.000330$
 (0) $w[-1] | w[0] = \text{ရဲ့} | \text{ဌာ} \rightarrow < : 0.001171$
 (0) $w[0] | w[1] = \text{ရင်} | \text{ရပ်} \rightarrow | : 0.125018$
 (0) $w[-1] | w[0] = \text{ရင်} | \text{ရပ်} \rightarrow | : 0.021620$
 (0) $w[0] | w[1] = \text{မိ} | \text{တပ်} \rightarrow | : 0.073698$
 (0) $w[-1] | w[0] = \text{မိ} | \text{တပ်} \rightarrow | : 0.006731$
 (0) $w[0] | w[1] = \text{ပေး} | \text{အုပ်} \rightarrow | : 0.055272$
 (0) $w[-1] | w[0] = \text{ပေး} | \text{အုပ်} \rightarrow < : 0.086070$
 (0) $w[0] | w[1] = \text{ချုပ်} | \text{ပိုင်} \rightarrow | : 0.163264$
 (0) $w[-1] | w[0] = \text{ချုပ်} | \text{ပိုင်} \rightarrow | : 0.201396$
 (0) $w[0] | w[1] = \text{ဦး} | \text{ဆန္ဒ} \rightarrow | : 0.000202$
 (0) $w[-1] | w[0] = \text{ဦး} | \text{ဆန္ဒ} \rightarrow < : 0.002019$
 (0) $w[0] | w[1] = \text{သို့} | \text{ယ} \rightarrow | : 0.010375$
 (0) $w[-1] | w[0] = \text{သို့} | \text{ယ} \rightarrow < : 0.012861$
 (0) $w[0] | w[1] = \text{ဖွဲ့} | \text{သာ} \rightarrow | : 0.012748$
 (0) $w[-1] | w[0] = \text{ဖွဲ့} | \text{သာ} \rightarrow < : 0.244811$
 (0) $w[0] | w[1] = \text{စား} | \text{ခန့်} \rightarrow | : 0.006894$
 (0) $w[-1] | w[0] = \text{စား} | \text{ခန့်} \rightarrow < : 0.171166$
 (0) $w[0] | w[1] = \text{လျက်} | \text{မုတ္တ} \rightarrow | : 0.007602$
 (0) $w[-1] | w[0] = \text{လျက်} | \text{မုတ္တ} \rightarrow < : 0.045189$
 (0) $w[0] | w[1] = \text{ဆဲလ်} | \text{ကြောင့်} \rightarrow | : 0.057159$
 (0) $w[-1] | w[0] = \text{ဆဲလ်} | \text{ကြောင့်} \rightarrow | : 0.000092$
 (0) $w[0] | w[1] = \text{လာ} | \text{ဟပ်} \rightarrow | : 0.087682$
 (0) $w[-1] | w[0] = \text{လာ} | \text{ဟပ်} \rightarrow | : 0.139486$
 (0) $w[0] | w[1] = \text{ဟပ်} | \text{ရာ} \rightarrow | : 0.139486$
 (0) $w[-1] | w[0] = \text{ဟပ်} | \text{ရာ} \rightarrow | : 0.004021$
 (0) $w[0] | w[1] = \text{င်} | \text{ကျ} \rightarrow | : 0.123393$
 (0) $w[-1] | w[0] = \text{င်} | \text{ကျ} \rightarrow | : 0.019437$


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(0) w[0]|w[1]=ပြေ|ဖို့ --> |: 0.001746
(0) w[-1]|w[0]=ပြေ|ဖို့ --> |: 0.001063
(0) w[0]|w[1]=စဉ်|ကောင်း --> |: 0.110705
(0) w[-1]|w[0]=စဉ်|ကောင်း --> |: 0.182078
(0) w[0]|w[1]=ကောင်း|ဆွဲ --> |: 0.182078
(0) w[-1]|w[0]=ကောင်း|ဆွဲ --> |: 0.018422
(0) w[0]|w[1]=နက်|ဟာ --> |: 0.018968
(0) w[-1]|w[0]=နက်|ဟာ --> |: 0.002638
(0) w[0]|w[1]=နွေး|သည့် --> |: 0.005286
(0) w[-1]|w[0]=နွေး|သည့် --> |: 0.000131
(0) w[0]|w[1]=ချ|ဘော --> |: 0.072852
(0) w[-1]|w[0]=ချ|ဘော --> <: 0.080975
(0) w[0]|w[1]=ပမ်း|ရင် --> |: 0.038040
(0) w[-1]|w[0]=ပမ်း|ရင် --> |: 0.009633
(0) w[0]|w[1]=__COLON__|၄ --> |: 0.006974
(0) w[-1]|w[0]=__COLON__|၄ --> <: 0.009547
(0) w[0]|w[1]=ဘက်|ရှမ်း --> |: 0.013992
(0) w[-1]|w[0]=ဘက်|ရှမ်း --> |: 0.008630
(0) w[0]|w[1]=လွင်|လား --> |: 0.011020
(0) w[-1]|w[0]=လွင်|လား --> <: 0.129499
(0) w[0]|w[1]=ရှိုး|နှင့် --> |: 0.001564
(0) w[-1]|w[0]=ရှိုး|နှင့် --> |: 0.001830
(0) w[0]|w[1]=မယ့်|ပ --> |: 0.004729
(0) w[-1]|w[0]=မယ့်|ပ --> <: 0.032134
(0) w[0]|w[1]=S|1 --> |: 0.246889
(0) w[-1]|w[0]=S|1 --> <: 0.270271
(0) w[0]|w[1]=3|မှာ --> |: 0.000376
(0) w[-1]|w[0]=3|မှာ --> |: 0.000028
(0) w[0]|w[1]=မှ|ကား --> |: 0.001455
(0) w[-1]|w[0]=မှ|ကား --> <: 0.103079
}

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[]: